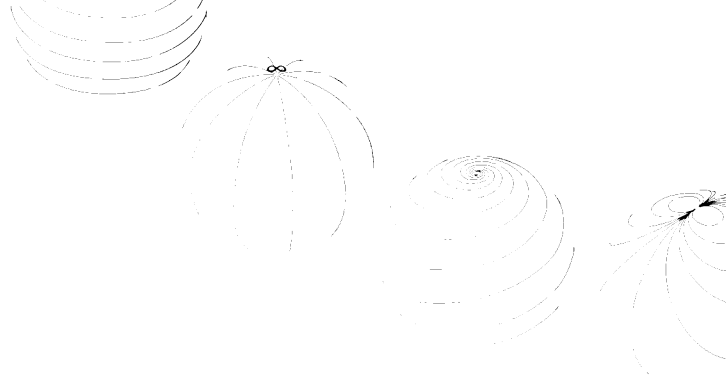




CHAPTER 3

Möbius Transformations and Inversion

3.1 Introduction

3.1.1 Definition and Significance of Möbius Transformations

A Möbius transformation¹ is a mapping of the form

$$M(z) = \frac{az + b}{cz + d}, \quad (3.1)$$

where a, b, c, d are complex constants. These mappings have many beautiful properties, and they find very varied application throughout complex analysis. Despite their apparent simplicity, Möbius transformations lie at the heart of several exciting areas of modern mathematical research. This is due in large part to their intimate and somewhat miraculous connection with the non-Euclidean geometries alluded to in Chapter 1. [This connection is the subject of Chapter 6.] Moreover, these transformations are also intimately connected² with Einstein's Theory of Relativity! This connection has been exploited with remarkable success by Sir Roger Penrose; see [Penrose and Rindler \(1984\)](#).

Two centuries have passed since August Ferdinand Möbius first studied the transformations that now bear his name, but it is fair to say that the rich vein of knowledge which he thereby exposed is still far from being exhausted. For this reason, we shall investigate Möbius transformations in considerably greater depth than is customary.

¹ Also known as a “linear”, “bilinear”, “linear-fractional”, or “homographic” transformation.

² According to [Coxeter \(1967\)](#), this connection was first recognized by H. Liebmann in 1905, the very year that Einstein discovered Special Relativity!

3.1.2 The Connection with Einstein's Theory of Relativity*

Clearly it would be neither appropriate nor feasible for us to explore this connection in detail, but let us at least briefly indicate how Möbius transformations are related to Einstein's Theory of Relativity.

In that theory, the time T and the 3-dimensional Cartesian coordinates (X, Y, Z) of an event are combined into a single *4-vector* (T, X, Y, Z) in 4-dimensional *space-time*. Of course the spatial components of this vector have no absolute significance: rotating the coordinate axes yields different coordinates $(\tilde{X}, \tilde{Y}, \tilde{Z})$ for one and the same point in space. But if two people choose different axes, they will nevertheless agree on the value of $\tilde{X}^2 + \tilde{Y}^2 + \tilde{Z}^2 = X^2 + Y^2 + Z^2$, for this represents the square of the distance to the point.

In contrast to this, we are accustomed to thinking that the *time* component T *does* have an absolute significance. However, Einstein's theory—confirmed by innumerable experiments—tells us that this is wrong. *If two (momentarily coincident) observers are in relative motion, they will disagree about the times at which events occur.* Furthermore, they will no longer agree about the value of $(X^2 + Y^2 + Z^2)$ —this is the famous *Lorentz contraction*. Is there *any* aspect of spacetime that has absolute significance and on which two observers in relative motion must agree? Yes: making a convenient choice of units in which the speed of light is equal to 1, Einstein discovered that both observers will agree on the value of

$$\tilde{T}^2 - (\tilde{X}^2 + \tilde{Y}^2 + \tilde{Z}^2) = T^2 - (X^2 + Y^2 + Z^2).$$

A *Lorentz transformation* $\mathcal{L}$ is a linear transformation of spacetime (a 4×4 matrix) that maps one observer's description (T, X, Y, Z) of an event to another observer's description $(\tilde{T}, \tilde{X}, \tilde{Y}, \tilde{Z})$ of the same event. Put differently, $\mathcal{L}$ is a linear transformation that preserves the quantity $T^2 - (X^2 + Y^2 + Z^2)$, upon which both observers must agree.

Now imagine that the spacetime coordinate origin emits a flash of light—an origin-centred sphere whose radius increases at the speed of light. It turns out that any given $\mathcal{L}$ is completely determined by its effect on the coordinates of the light rays that make up this flash. Here is the next crucial idea: in Ex. 8 we explain how *we may set up a one-to-one correspondence between these light rays and complex numbers*. Thus each Lorentz transformation of spacetime induces a definite mapping of the complex plane. What kinds of complex mappings do we obtain in this way? The miraculous answer turns out to be this:

The complex mappings that correspond to the Lorentz transformations are the Möbius transformations! Conversely, every Möbius transformation of $\mathbb{C}$ yields a unique Lorentz transformation of spacetime. (3.2)

Even among professional physicists, this “miracle” is not as well known as it should be.

The connection exhibited in (3.2) is deep and powerful. Just for starters, it means that any result we establish concerning Möbius transformations will immediately yield a corresponding result in Einstein's Theory of Relativity. Furthermore, these Möbius transformation proofs turn out to be considerably more elegant than direct spacetime proofs.

To really understand the above claims, we strongly recommend that after reading this chapter you consult [Penrose and Rindler \(1984, Ch. 1\)](#), as well as [Shaw \(2006, Ch. 23\)](#) and [Needham \(2021, §6.4\)](#).

3.1.3 Decomposition into Simple Transformations

As a first step towards making sense of (3.1), let us decompose $M(z)$ [exercise] into the following sequence of transformations:

$$\left. \begin{array}{ll} \text{(i)} & z \mapsto z + \frac{d}{c}, \text{ which is a translation;} \\ \text{(ii)} & z \mapsto (1/z); \\ \text{(iii)} & z \mapsto -\frac{(ad-bc)}{c^2}z, \text{ which is an expansion and a rotation;} \\ \text{(iv)} & z \mapsto z + \frac{a}{c}, \text{ which is another translation.} \end{array} \right\} \quad (3.3)$$

Note that if $(ad-bc) = 0$ then $M(z)$ is an uninteresting constant mapping, sending every point z to the same image point (a/c) ; in this exceptional case $M(z)$ is called *singular*. In discussing Möbius transformations we shall therefore always assume that $M(z)$ is *non-singular*, meaning that $(ad-bc) \neq 0$.

Of the four transformations above, only the second one has not yet been investigated. This mapping $z \mapsto (1/z)$ holds the key to understanding Möbius transformations; we shall call it *complex inversion*. The next section examines its many remarkable and powerful properties.

3.2 Inversion

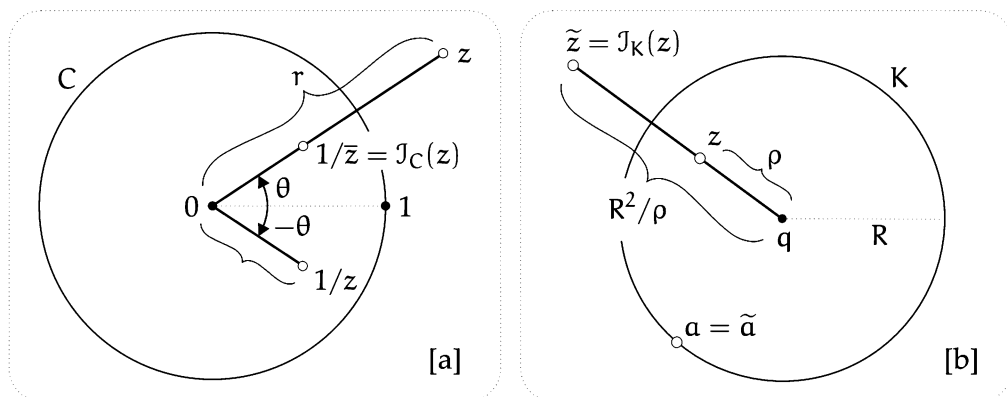
3.2.1 Preliminary Definitions and Facts

The image of $z = r e^{i\theta}$ under complex inversion is $1/(r e^{i\theta}) = (1/r) e^{-i\theta}$: the new length is the reciprocal of the original, and the new angle is the negative of the original. See [3.1a]. Note how a point outside the unit circle C is mapped to a point inside C , and vice versa.

Figure [3.1a] also illustrates a particularly fruitful way of decomposing complex inversion into a two-stage process:

- (i) Send $z = r e^{i\theta}$ to the point that is in the same direction as z but that has reciprocal length, namely the point $(1/r) e^{i\theta} = (1/\bar{z})$.
- (ii) Apply complex conjugation (i.e., reflection in the real axis), which sends $(1/\bar{z})$ to $\overline{(1/\bar{z})} = (1/z)$.

Check for yourself that the order in which we apply these mappings is immaterial.



[3.1] [a] Geometric inversion in the unit circle C : $z \mapsto J_C(z) \equiv (1/\bar{z})$ leaves the points of C fixed, and swaps the interior and exterior. When composed with conjugation, it yields complex inversion, $z \mapsto (1/z)$. [b] Definition of geometric inversion $z \mapsto J_K(z)$ in a general circle K .

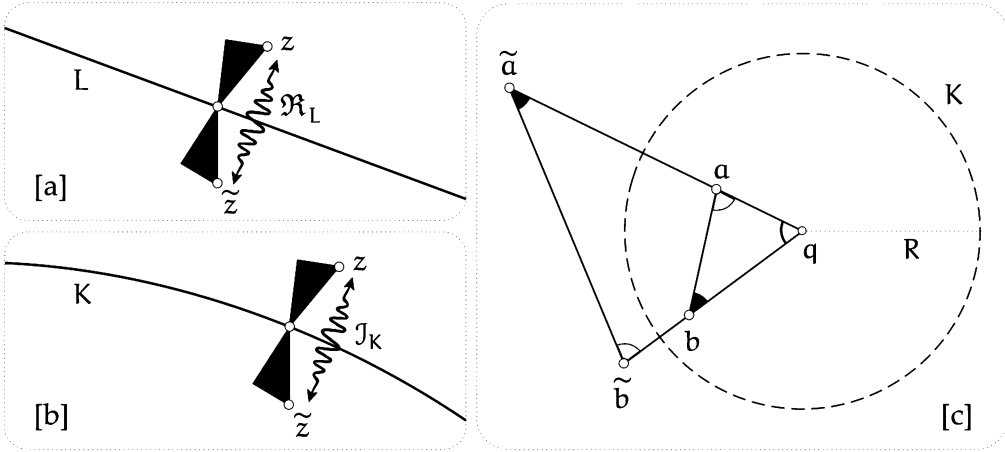
While stage (ii) is geometrically trivial, we shall see that the mapping in stage (i) is filled with surprises; it is called³ *geometric inversion*, or simply *inversion*. Clearly, the unit circle C plays a special role for this mapping: the inversion interchanges the interior and exterior of C , while each point *on* C remains fixed (i.e., is mapped to itself). For this reason we write the mapping as $z \mapsto J_C(z) = (1/\bar{z})$, and we call J_C (a little more precisely than before) “inversion in C ”.

This added precision in terminology is important because, as illustrated in [3.1b], there is a natural way of generalizing J_C to inversion in an *arbitrary* circle K (say with centre q and radius R). Clearly, this “inversion in K ”, written $z \mapsto \tilde{z} = J_K(z)$, should be such that the interior and exterior of K are interchanged, while each point on K remains fixed. If ρ is the distance from q to z , then *we define* $\tilde{z} = J_K(z)$ *to be the point in the same direction from* q *as* z , *and at distance* (R^2/ρ) *from* q . [Check for yourself that this definition does indeed perform as advertised.]

As usual, we invite you to use a computer to verify empirically the many results we shall derive concerning inversion. However, in the case of this particular mapping, you can also construct (fairly easily) a *mechanical* instrument that will carry out the mapping for you; see Ex. 2.

Although we shall not need it for a while, it is easy enough to obtain a *formula* for $J_K(z)$. Because the connecting complex numbers from q to z and to $\tilde{z}$ both have the

³ In older works it is often called “transformation by reciprocal radii”.



[3.2] [a] Ordinary reflection in a line. [b] If the radius of K is large, so that it starts to look straight, then geometric inversion $z \mapsto \tilde{z} = J_K(z)$ looks very much like ordinary reflection. More precisely, as z approaches K , then $J_K(z)$ is ultimately its reflection in the tangent to K . [c] Under inversion in K , the triangle aqb is similar to the triangle $\tilde{a}\tilde{b}\tilde{q}$.

same direction, and their lengths are ρ and (R^2/ρ) , it follows that $(\tilde{z}-q)(\overline{z-q}) = R^2$. Solving for $\tilde{z}$,

$$J_K(z) = \frac{R^2}{\bar{z} - \bar{q}} + q = \frac{q\bar{z} + (R^2 - |q|^2)}{\bar{z} - \bar{q}}. \quad (3.4)$$

For example, if we put $q = 0$ and $R = 1$, then we recover $J_C(z) = (1/\bar{z})$.

There is a very interesting similarity (which will deepen as we go on) between inversion $J_K(z)$ in a circle K and reflection $\mathfrak{R}_L(z)$ in a line L . See [3.2a] and [3.2b]. First, L divides the plane into two pieces, or “components”, which are interchanged by $\mathfrak{R}_L(z)$; second, each point on the boundary between the components remains fixed; third, $\mathfrak{R}_L(z)$ is *involutory* or *self-inverse*, meaning that $\mathfrak{R}_L \circ \mathfrak{R}_L$ is the identity mapping, leaving every point fixed. To put this last property differently, consider a point z and its reflection $\tilde{z} = \mathfrak{R}_L(z)$ in L . Such a pair are said to be “mirror images”, or to be “symmetric with respect to L ”. The involutory property says that the reflection causes such a pair of points to *swap places*.

Check for yourself that $J_K(z)$ shares all three of these properties. Furthermore, the black triangle in [3.2b] illustrates the fact that if K is large then the effect of J_K on a small shape close to K looks very much like ordinary reflection. [We will explain this later, but you might like to check this empirically using a computer.] For these reasons, and others still to come, $J_K(z)$ is often also called *reflection in a circle*, and the pair of points z and $\tilde{z} = J_K(z)$ are said to be *symmetric with respect to K* .

We end this subsection with two simple properties of inversion, the first of which will serve as the springboard for the investigations that follow. Let us use the symbol $[cd]$ to stand for the distance $|c - d|$ between two points c and d . We hope that

no confusion will arise from this, the square brackets serving as a reminder that $[cd]$ is *not* the product of the complex numbers c and d .

In [3.2c], a and b are two arbitrary points, and $\tilde{a} = \mathcal{I}_K(a)$ and $\tilde{b} = \mathcal{I}_K(b)$ are their images under inversion in K . By definition, $[qa][q\tilde{a}] = R^2 = [qb][q\tilde{b}]$, and so

$$[qa]/[qb] = [q\tilde{b}]/[q\tilde{a}].$$

Noting the common angle $\angle aqb = \angle \tilde{a}q\tilde{b}$, we deduce that

$$\begin{aligned} &\text{If inversion in a circle centred at } q \text{ maps two points } a \text{ and } b \text{ to } \tilde{a} \text{ and } \tilde{b}, \\ &\text{then the triangles } aqb \text{ and } \tilde{b}q\tilde{a} \text{ are similar.} \end{aligned} \quad (3.5)$$

Lastly, let us find the relationship between the separation $[ab]$ of two points, and the separation $[\tilde{a}\tilde{b}]$ of their images under inversion. Using (3.5),

$$[\tilde{a}\tilde{b}]/[ab] = [q\tilde{b}]/[qa] = R^2/[qa][qb],$$

and so the separation of the image points is given by

$$[\tilde{a}\tilde{b}] = \left(\frac{R^2}{[qa][qb]} \right) [ab]. \quad (3.6)$$

3.2.2 Preservation of Circles

Let us examine the effect of $\mathcal{I}_K$ on lines and then on circles. If a line L passes through the centre q of K , then clearly $\mathcal{I}_K$ maps L to itself, which we may write as $\mathcal{I}_K(L) = L$. Of course we don't mean that each point of L remains fixed, for $\mathcal{I}_K$ interchanges the portions of L interior and exterior to K ; the only *points* of L that remain fixed are the two places where it intersects K .

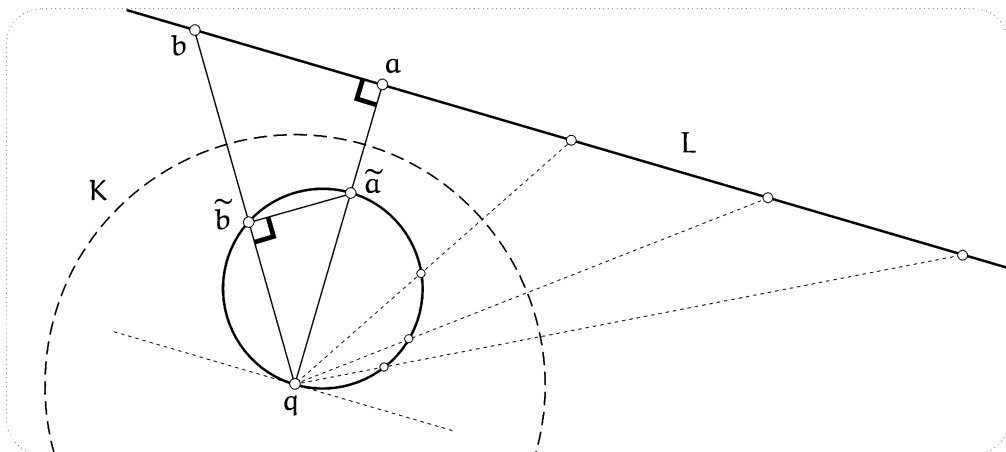
Matters become much more interesting when we consider a general line L that does not pass through q . Figure [3.3] provides the surprising answer:

$$\begin{aligned} &\text{If a line } L \text{ does not pass through the centre } q \text{ of } K, \text{ then inversion in } K \\ &\text{maps } L \text{ to a circle that passes through } q. \end{aligned} \quad (3.7)$$

Here b is an arbitrary point on L , while a is the intersection of L with the perpendicular line through q . By virtue of (3.5), $\angle q\tilde{b}\tilde{a} = \angle qab = (\pi/2)$, so $\tilde{b}$ lies on the circle having the line-segment $q\tilde{a}$ as diameter. Done. Notice, incidentally, that the tangent at q of the image circle is parallel to L .

Note that (3.7) makes no mention of the radius R of K . You may therefore be concerned that in [3.3] we have chosen R so that K does not intersect L ; what happens if K *does* intersect L ? Check for yourself that, while the picture looks somewhat different in this case, the geometric argument above continues to apply without any modification.

We now give a less direct, but more instructive way of understanding why (3.7) does not depend on the size of K . We will show that if the result holds for one



[3.3] The geometric inverse of a line is a circle through the centre of inversion. By virtue of [3.2c], the illustrated right triangles are similar, so we see that the geometric inverse of a line L is a circle that passes through the centre of the inversion, and that its tangent there is parallel to L .

circle K_1 (radius R_1) centred at q , then it will hold for any other circle K_2 (radius R_2) centred at q .

Let z be an arbitrary point, and let $\tilde{z}_1 = J_{K_1}(z)$ and $\tilde{z}_2 = J_{K_2}(z)$. Obviously $\tilde{z}_1$ and $\tilde{z}_2$ are both in the same direction from q as z , and you can easily check that the ratio of their distances from q is independent of the location of z :

$$[q\tilde{z}_2]/[q\tilde{z}_1] = (R_2/R_1)^2 \equiv k, \text{ say.}$$

Thus

$$J_{K_2} = \mathcal{D}_q^k \circ J_{K_1}, \quad (3.8)$$

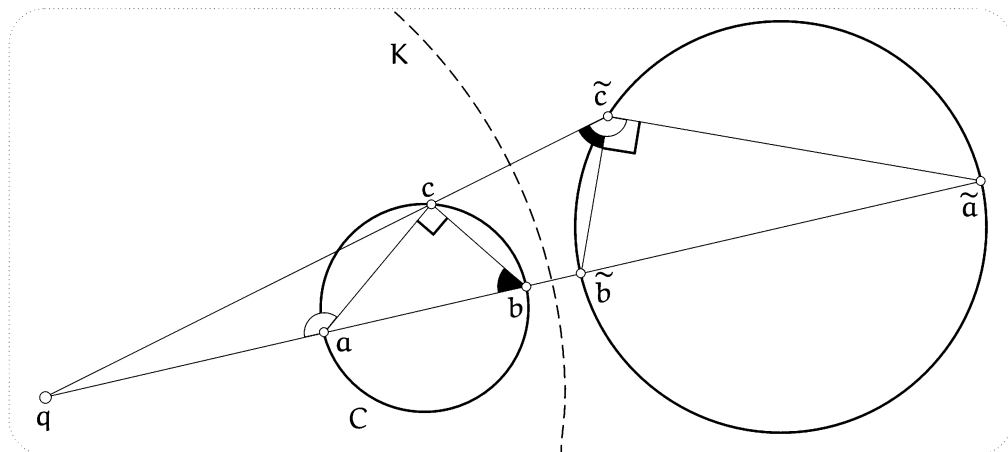
where the “central dilation” $\mathcal{D}_q^k$ [see p. 45] is an expansion (centred at q) of the plane by a factor of k . It follows [exercise] that if (3.7) holds for K_1 then it also holds for K_2 .

Look again at [3.3]. Since J_K is involutory, it simply *swaps* the line and circle, and so the image of any circle through q is a line not passing through q . But what happens to a general circle C that does not pass through q ? Initially, suppose that C does not contain q in its interior. Figure [3.4] provides the beautiful answer:

If a circle C does not pass through the centre q of K , then inversion in K maps C to another circle not passing through q . (3.9)

This fundamental result is often described by saying that inversion “preserves circles”.

It follows from (3.8) that if (3.9) is true for one choice of K , then it will be true for any choice of K . We may therefore conveniently choose K so that C lies inside



[3.4] Geometric inversion maps circles to circles! *This beautiful and important fact is readily confirmed by first observing that [3.2c] implies that the two grey angles are equal to each other, and, likewise, the two black angles are equal to each other. But this means that the right angle at c , being the difference of the grey and black angles, implies the illustrated right angle at $\tilde{c}$. Done!*

it, as illustrated. Here a and b are the ends of a diameter of C , and they therefore subtend a right angle at a general point c on C . To understand (3.9), first use (3.5) to check that both the shaded angles are equal, and that both the black angles are equal. Next look at the triangle abc , and observe that the external shaded angle at a is the sum of the two illustrated internal angles: the right angle at c and the black angle at b . It follows that $\angle \tilde{a} \tilde{c} \tilde{b} = (\pi/2)$, and hence that $\tilde{a}$ and $\tilde{b}$ are the ends of a diameter of a circle through $\tilde{c}$. Thus we have demonstrated (3.9) in the case where C does not contain q . We leave it to you to check that the same line of reasoning establishes the result in the case where C does contain q .

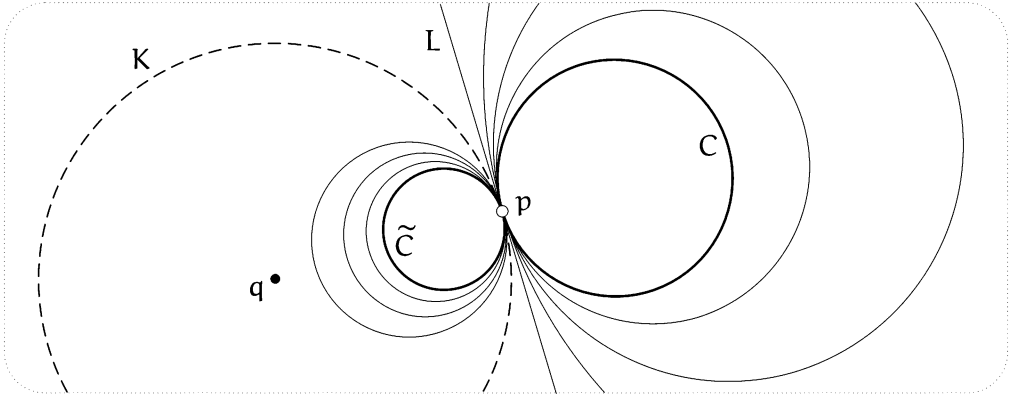
The result (3.7) is in fact a special limiting case of (3.9). Figure [3.5] shows a line L , the point p on L closest to the centre q of the inversion, and a circle C tangent to L at p . As its radius tends to infinity, C tends to L , and the image circle $\tilde{C} = \mathcal{I}_K(C)$ tends to a circle through q .

Later we will be able to give a much cleaner way of seeing that (3.7) and (3.9) are two aspects of a single result.

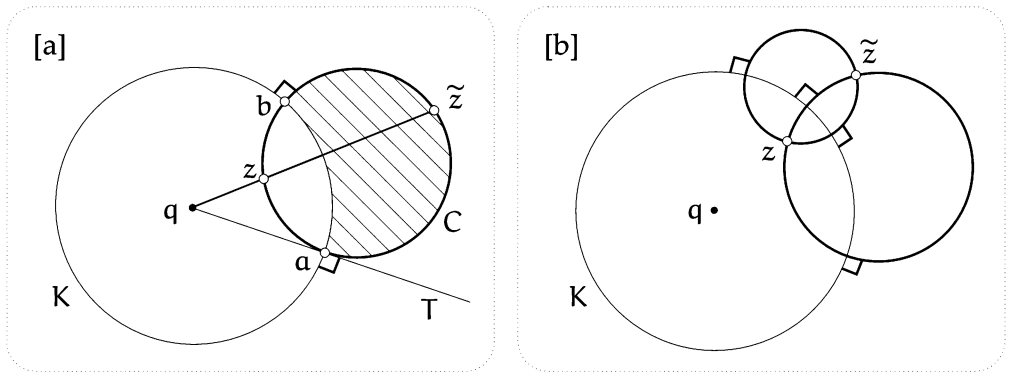
3.2.3 Constructing Inverse Points Using Orthogonal Circles

Consider [3.6a]. The circle C cuts the circle of inversion K at right angles at a and b . In other words, the tangent T to C at a (for example) passes through q . Under inversion in K , a and b remain fixed, and T is mapped into itself. Thus the image of C must be a circle that again passes through a and b and that is again orthogonal to K . But clearly there is only *one* circle with these properties, namely C itself. Thus,

$$\text{Under inversion in } K, \text{ every circle orthogonal to } K \text{ is mapped to itself.} \quad (3.10)$$



[3.5] Inversion of a line is the limit of inversion of a circle. Here we see that [3.3] is in fact merely a special limiting case of the general result shown in [3.4]. As the radius of C tends to infinity, so that C becomes the line L , the image circle $\tilde{C} = \mathcal{I}_K(C)$ becomes a circle through q .

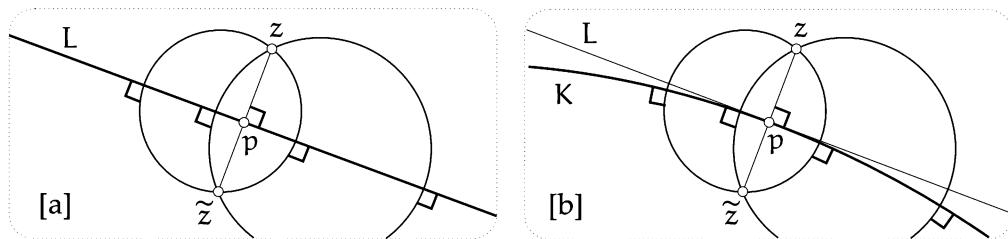


[3.6] Constructing the geometric inverse using orthogonal circles. [a] If C is orthogonal to K , then it is mapped to itself, because the image must be another circle that again passes through the fixed points a and b , and the tangent T at a is mapped into itself. Thus it can only be C itself. [b] It follows that the geometric inverse $\tilde{z}$ of a point z may be constructed as the second intersection point of any two circles through z that are orthogonal to K .

Figure [3.6a] illustrates two immediate consequences of this result. First, the disc bounded by C is also mapped to itself, the shaded and hatched regions into which K divides it being swapped by the inversion. Second, a line from q through a point z on C intersects C for the second time at the inverse point $\tilde{z}$.

Another consequence (the key result of this subsection) is the geometric construction shown in [3.6b], the verification of which is left to you.

The inverse $\tilde{z}$ of z in K is the second intersection point of any two circles that pass through z and are orthogonal to K .



[3.7] Inversion (reflection) in a line is the limit of inversion in a circle. **[a]** The reflection $\tilde{z}$ of z in the line L can be constructed as the second intersection point of any two circles through z that are orthogonal to L . **[b]** The same construction for geometric inversion in the circle K . As the radius of K tends to infinity, this becomes reflection in the tangent L .

Note that the construction of $\tilde{z}$ in [3.6a] is the special limiting case in which the radius of one of the circles tends to infinity, and so becomes a line through q . For other, less important, geometric constructions of inversion, see Ex. 1.

The previously mentioned analogy between inversion in K and reflection in a line L now deepens, for the reflection $\tilde{z} = \mathfrak{R}_L(z)$ of z in L can be obtained using *precisely the same construction*; see [3.7a]. Note that the line-segment joining z and $\tilde{z}$ is orthogonal to L , and that its intersection p with L is equidistant from z and $\tilde{z}$: $[pz]/[p\tilde{z}] = 1$.

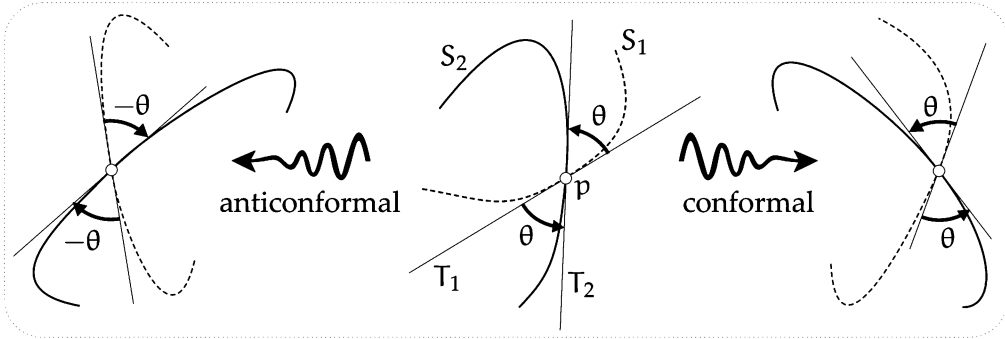
As illustrated in [3.7b], the segment of L in the vicinity of p can be approximated by an arc of a large circle K tangent to L at p . Here $\tilde{z} = \mathcal{J}_K(z)$ is the image under inversion in K of the same point z as before. As you can see, there is virtually no difference between the two figures. More precisely, as the radius of K tends to infinity, inversion in K *becomes* reflection in L . In particular, $[pz]/[p\tilde{z}]$ tends to unity, or equivalently, $[p\tilde{z}]$ is “ultimately equal” to $[pz]$. We can now understand what was happening in figure [3.2b].

We can also check this result algebraically. First, though, observe that from the geometric point of view it is sufficient to demonstrate the result for a single choice of the line L and a single point p on it. Let us therefore choose L to be the real axis, and let p be the origin. The circle K of radius R centred at $q = iR$ is therefore tangent to L at p . Using (3.4), we obtain [exercise]

$$\mathcal{J}_K(z) = \frac{\bar{z}}{1 - (i\bar{z}/R)}.$$

Thus as R tends to infinity we find that $\mathcal{J}_K(z)$ is ultimately equal to $\mathfrak{R}_L(z) = \bar{z}$, as was to be shown.

Here is another way of looking at the result. Instead of making K larger and larger, let z move closer and closer to an arbitrary point p on a circle K of fixed size. As z approaches p from any direction, $\mathcal{J}_K(z)$ is ultimately equal to $\mathfrak{R}_T(z)$, where T is the tangent to K at p .



[3.8] Conformal and anticonformal mappings preserve and reverse angles, respectively.

Again, we can also get this algebraically using the above equation. If R is fixed and $|z| < R$, then [exercise]

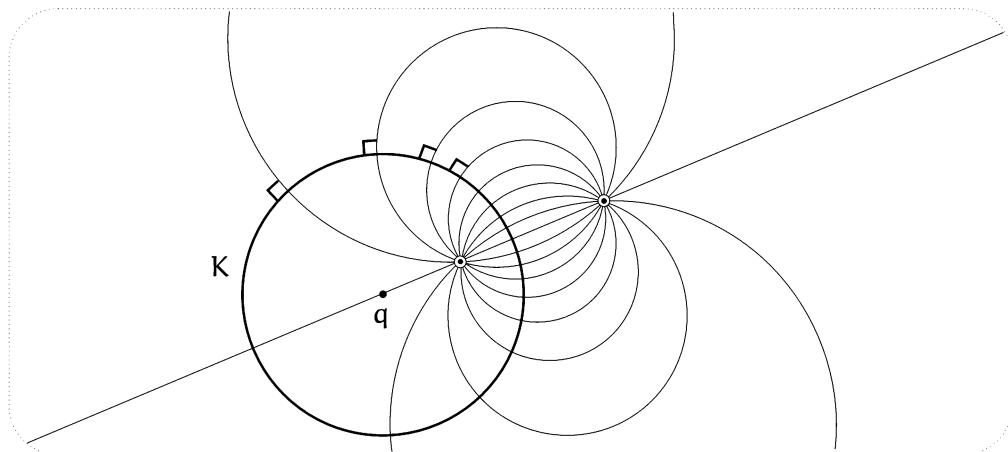
$$J_K(z) = \bar{z} + \frac{i\bar{z}^2}{R} - \frac{\bar{z}^3}{R^2} + \dots$$

Thus as z approaches $p = 0$, $J_K(z)$ is ultimately equal to $\mathfrak{R}_L(z) = \bar{z}$, which is reflection in the tangent to K at p .

3.2.4 Preservation of Angles

Let us begin by discussing what is meant by “preservation of angles”. In the centre of [3.8] are two curves S_1 and S_2 intersecting at a point p . Provided these curves are sufficiently smooth at p , then, as illustrated, we may draw their tangent lines T_1 and T_2 at p . We now define the “angle between S_1 and S_2 ” at p to be the acute angle θ from T_1 to T_2 . Thus this angle θ has a *sign* attached to it: the angle between S_2 and S_1 is *minus* the illustrated angle between S_1 and S_2 . If we now apply a sufficiently smooth transformation to the curves, then the image curves will again possess tangents at the image of p , and so there will be a well-defined angle between these image curves.

If the angle between the image curves is the same as the angle between the original curves through p , then we say that the transformation has “preserved” the angle at p . It is perfectly possible that the transformation preserves the angle between one pair of curves through p , but not *every* pair through p . However, if the transformation *does* preserve the angle between *every* pair of curves through p , then we say that it is *conformal* at p . We stress that this means that both the magnitude *and the sign* of the angles are preserved; see the right of [3.8]. If every angle at p is instead mapped to an angle of equal magnitude but opposite sign, then we say that the mapping is *anticonformal* at p ; see the left of [3.8]. If the mapping is conformal at every point in the region where it is defined, then we call it a *conformal mapping*; if it is instead anticonformal at every point, then we call it an *anticonformal mapping*. Finally, if a mapping is known to preserve the *magnitude* of angles, but we



[3.9] Through any given point z not on K , there is precisely one circle orthogonal to K that passes through z in any given direction.

are unable to say whether or not it preserves their sense, then we call it an *isogonal mapping*.

It is easy enough to think of concrete mappings that are either conformal or anticonformal. For example, a translation $z \mapsto (z + c)$ is conformal, as is a rotation and expansion of the plane given by $z \mapsto az$. On the other hand, $z \mapsto \bar{z}$ is anticonformal, as is any reflection in a line. The analogy between such a reflection and inversion in a circle now gets even deeper, for

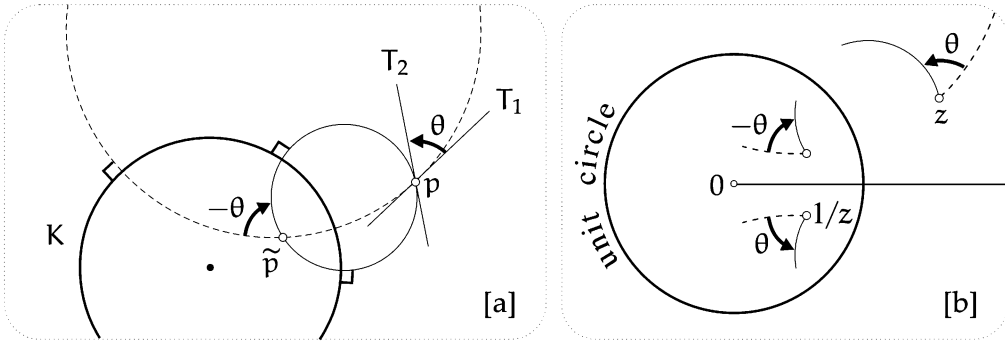
Inversion in a circle is an anticonformal mapping.

To see this, first look at [3.9]. This illustrates the fact that given any point z not on K , there is precisely one circle orthogonal to K that passes through z in any given direction. [Given the point and the direction, can you think how to *construct* this circle?]

As in [3.8], suppose that two curves S_1 and S_2 intersect at p , and that their tangents there are T_1 and T_2 , the angle between them being θ . To find out what happens to this angle under inversion in K , let us replace S_1 and S_2 with the unique circles orthogonal to K that pass through p in the same directions as directions S_1 and S_2 , i.e., circles whose tangents at p are T_1 and T_2 . See [3.10a]. Since inversion in K maps each of these circles to themselves, the new angle at $\tilde{p} = J_K(p)$ is $-\theta$. Done.

Figure [3.10b] illustrates the effect of $z \mapsto (1/z)$ on angles. Since this mapping is equivalent to reflection (i.e., inversion) in the unit circle followed by reflection in the real axis (both of which are anticonformal), we see that their composition reverses the angle twice, restoring it to its original value:

Complex inversion, $z \mapsto (1/z)$, is conformal.



[3.10] Geometric inversion is anticonformal, so complex inversion is conformal.

[a] When two circles intersect, the magnitudes of both angles of intersection are the same, by symmetry. But, taking account of the sense of the angles, it follows that geometric inversion is anticonformal. **[b]** Since complex conjugation is also anticonformal, its composition with geometric inversion reverses angles twice. Therefore, complex inversion, $z \mapsto (1/z)$, is conformal.

By the same reasoning, it follows more generally that

The composition of an even number of reflections (in lines or circles) is a conformal mapping, while the composition of an odd number of such reflections is an anticonformal mapping.

3.2.5 Preservation of Symmetry

Consider [3.11a], which shows two points a and b that are symmetric with respect to a line L . If reflection in a line M maps a to $\tilde{a}$, b to $\tilde{b}$, and L to $\tilde{L}$, then clearly the image points $\tilde{a}$ and $\tilde{b}$ are again symmetric with respect to the image line $\tilde{L}$. In brief, reflection in lines “preserves symmetry” with respect to lines.

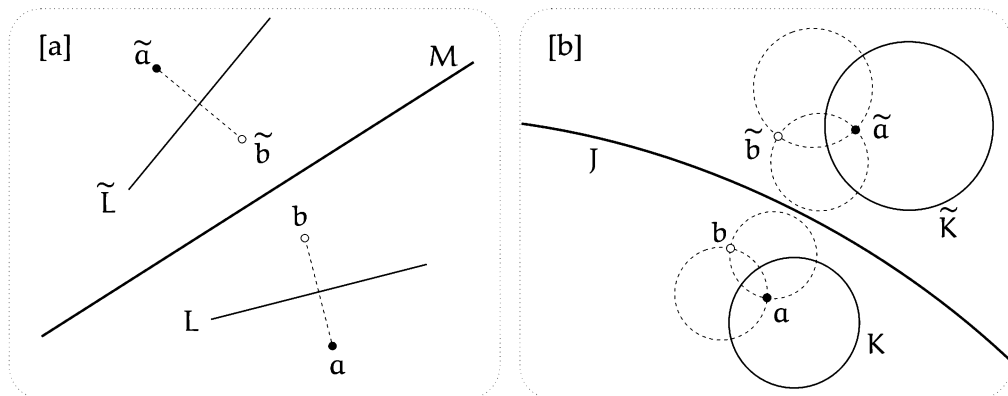
We now show that reflection in circles also preserves symmetry with respect to circles:

If a and b are symmetric with respect to a circle K , then their images $\tilde{a}$ and $\tilde{b}$ under inversion in any circle J are again symmetric with respect to the image $\tilde{K}$ of K .

To understand this, first note that, since inversion is anticonformal, (3.10) is just a special case of the following more general result:

Inversion maps any pair of orthogonal circles to another pair of orthogonal circles.

Of course if one of the circles passes through the centre of inversion then its image will be a line. However, if we think of lines as merely being circles of infinite radius then the result is true without qualification.



[3.11] Geometric inversion preserves symmetry. **[a]** If a and b are symmetric with respect to a line L , then their reflections $\tilde{a}$ and $\tilde{b}$ in a line M are still symmetric with respect to the reflection $\tilde{L}$ of L : we say that symmetry is preserved. **[b]** Since geometric inversion preserves circles and orthogonality, we see that if a and b are symmetric with respect to a circle K , then their “reflections” $\tilde{a}$ and $\tilde{b}$ in a circle J are still symmetric with respect to the reflection $\tilde{K}$ of K : here, too, symmetry is preserved!

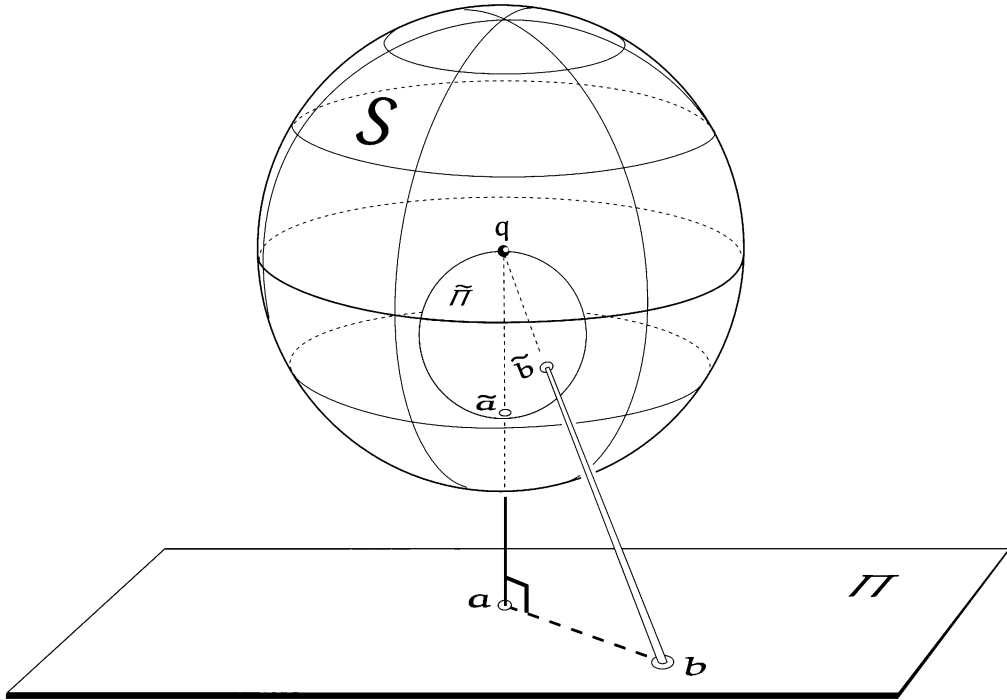
The preservation of symmetry result is now easily understood. See [3.11b]. Since the two dashed circles through a and b are orthogonal to K , their images under inversion in J are likewise orthogonal to $\tilde{K}$, and they therefore intersect in a pair of points that are symmetric with respect to $\tilde{K}$.

3.2.6 Inversion in a Sphere

Inversion $\mathcal{I}_S$ of three-dimensional space in a sphere S (radius R and centre q) is defined in the obvious way: if p is a point in space at distance ρ from q , then $\mathcal{I}_S(p)$ is the point in the same direction from q as p , and at distance (R^2/ρ) from q . We should explain that this is not generalization for its own sake; soon we will see how this three-dimensional inversion sheds new light on two-dimensional inversion in $\mathbb{C}$.

Without any additional work, we may immediately generalize most of the above results on inversion in circles to results on inversion in spheres. For example, reconsider [3.3]. If we rotate this figure (in space) about the line through q and a , then we obtain [3.12], in which the circle of inversion K has swept out a sphere of inversion S , and the line has swept out a plane Π . Thus we have the following result:

Under inversion in a sphere centred at q , a plane Π that does not contain q is mapped to a sphere that contains q and whose tangent plane there is parallel to Π . Conversely, a sphere containing q is mapped to a plane that is parallel to the tangent plane of that sphere at q . (3.11)



[3.12] Inversion in a sphere sends a plane to a sphere through the centre of inversion. Rotating [3.3] (in space) about the line through q and a , we see that inversion in the sphere S maps the plane Π to a sphere that contains q and whose tangent plane there is parallel to Π .

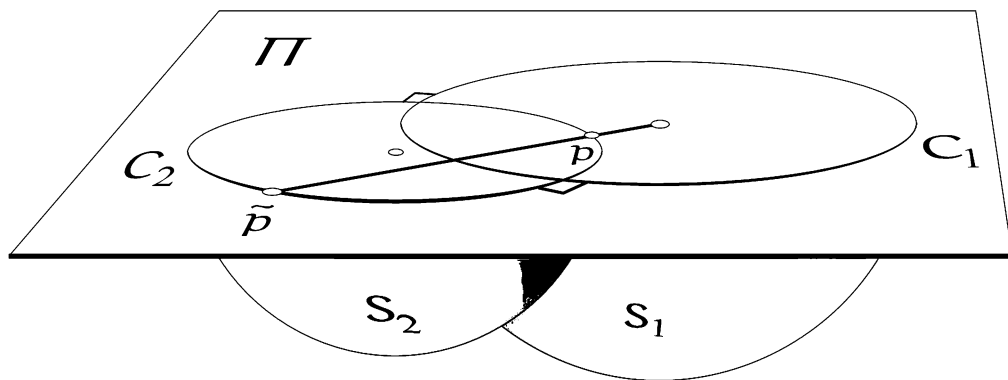
By the same token, if we rotate figure [3.4] about the line through q and a , then we find that

Under inversion in a sphere, the image of a sphere that does not contain the centre of inversion is another sphere that does not contain the centre of inversion.

This result immediately tells us what will happen to a circle in space under inversion in a sphere, for such a circle may be thought of as the intersection of two spheres. Thus we easily deduce [exercise] the following result:

Under inversion in a sphere, the image of a circle C that does not pass through the centre q of inversion is another circle that does not pass through q . If C does pass through q then the image is a line parallel to the tangent of C at q . (3.12)

The close connection between inversion in a circle and reflection in a line also persists: reflection in a plane is a limiting case of inversion in a sphere. For this reason, inversion in a sphere is also called “reflection in a sphere”. Of



[3.13] Let S_1 and S_2 be intersecting spheres, and let C_1 and C_2 be the great circles in which these spheres intersect any plane Π passing through their centres. Then S_1 and S_2 are orthogonal if and only if C_1 and C_2 are orthogonal. Also illustrated is the fact that if we restrict attention to Π then geometric inversion of space in S_1 induces two-dimensional inversion in the circle C_1 .

particular importance is the fact that such three-dimensional reflections again preserve symmetry:

Let K be a plane or sphere, and let a and b be symmetric points with respect to K . Under a three-dimensional reflection in any plane or sphere, the images of a and b are again symmetric with respect to the image of K . (3.13)

We now describe the steps leading to this result; they are closely analogous to the steps leading to the two-dimensional preservation of symmetry result.

If we rotate figure [3.6a] about the line joining the centres of K and C , we deduce that

Under inversion in a sphere K , every sphere orthogonal to K is mapped to itself. (3.14)

When we say that spheres are “orthogonal” we mean that their tangent planes are orthogonal at each point of their circle of intersection. However, in order to be able to easily draw on previous results, let us rephrase this three-dimensional description in two-dimensional terms:

Let S_1 and S_2 be intersecting spheres, and let C_1 and C_2 be the great circles in which these spheres intersect any plane Π passing through their centres. Then S_1 and S_2 are orthogonal if and only if C_1 and C_2 are orthogonal.

See [3.13]. This figure is also intended to help you see that if we restrict attention to Π then the three-dimensional inversion in S_1 is identical to the two-dimensional inversion in C_1 . This way of viewing inversion in spheres allows us to quickly generalize earlier results.

For example, referring back to [3.6b], we find—make sure you see this—that if p lies in Π then $\tilde{p} = \mathcal{I}_{S_1}(p)$ may be constructed as the second intersection point of any two circles like C_2 that (i) lie in Π , (ii) are orthogonal to C_1 , and (iii) pass through p .

Next, suppose that S_1 and S_2 in [3.13] are subjected to inversion in a third sphere K . Choose Π to be the unique plane passing through the centres of S_1 , S_2 , K , and let C be the great circle in which K intersects Π . Since $\mathcal{I}_C$ maps C_1 and C_2 to orthogonal circles, we deduce [exercise] that (3.14) is a special case of the following result:

$$\text{Orthogonal spheres invert to orthogonal spheres.} \quad (3.15)$$

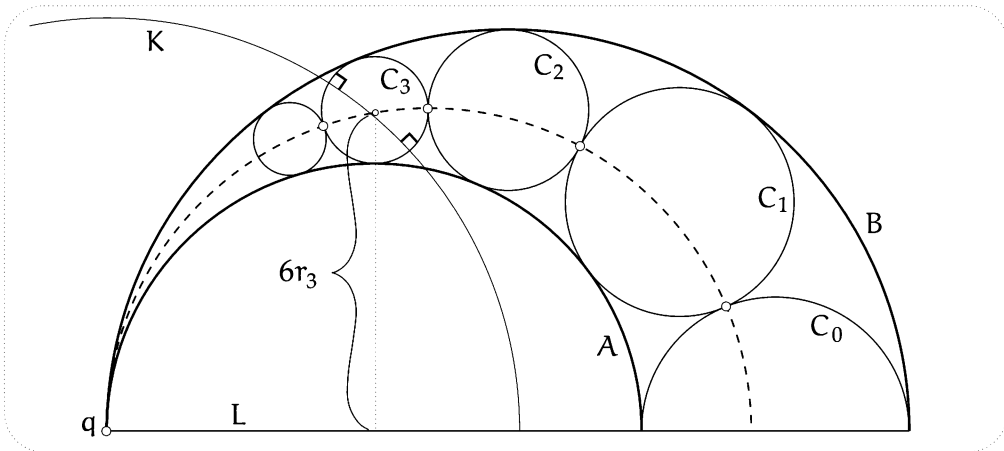
Here we are considering a plane to be a limiting case of a sphere.

Putting these facts together, you should now be able to see the truth of (3.13).

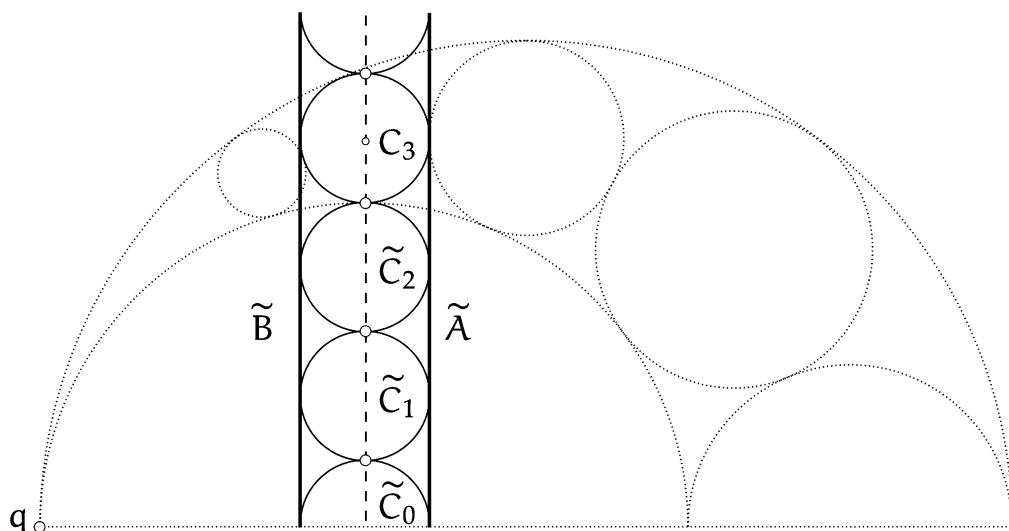
3.3 Three Illustrative Applications of Inversion

3.3.1 A Problem on Touching Circles

For our first problem, consider [3.14], in which we imagine that we are given two circles A and B that touch at q . As illustrated, we now construct the circle C_0 that touches A and B and whose centre lies on the horizontal line L through the centres of A and B . Finally, we construct the chain of circles C_1 , C_2 , etc., such that C_{n+1} touches C_n , A , and B .



[3.14] Given any two circles A and B that touch at q , as shown, construct touching circles, C_i , as shown. Then, very surprisingly, (1) the points of contact of the chain C_i all lie on a circle that also touches A and B at q , and (2) the height of the centre of C_n above L is $2n$ times the radius of C_n , as illustrated for $n = 3$.



[3.15] Referring to the previous figure, consider the illustrated (unique) circle K centred at q that cuts C_3 at right angles. Geometric inversion in K maps C_3 to itself, and it maps A and B to parallel vertical lines. The truth of both of the mysterious, bulleted facts below suddenly becomes obvious!

The figure illustrates two remarkable facts about this chain of circles:

- The points of contact of the chain C_0, C_1, C_2 , etc., all lie on a circle [dashed] touching A and B at q .
- If the radius of C_n is r_n , then the height above L of the centre of C_n is $2nr_n$. The figure illustrates this for C_3 .

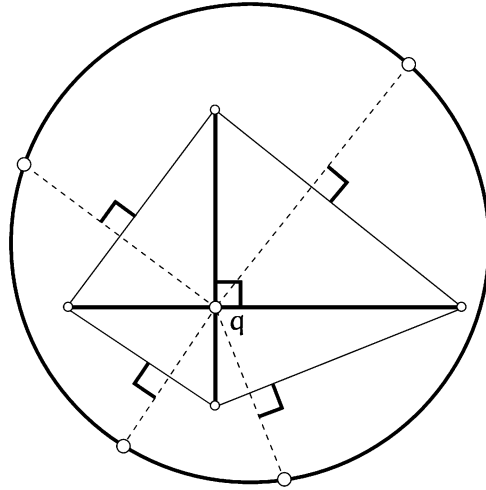
Before reading further, see if you can prove either of these results using conventional geometric methods.

Inversion allows us to demonstrate both these results in a single elegant swoop. In [3.14], we have drawn the unique circle K centred at q that cuts C_3 at right angles. Thus inversion in K will map C_3 to itself, and it will map A and B to parallel vertical lines; see [3.15]. Check for yourself that the stated results are immediate consequences of this figure.

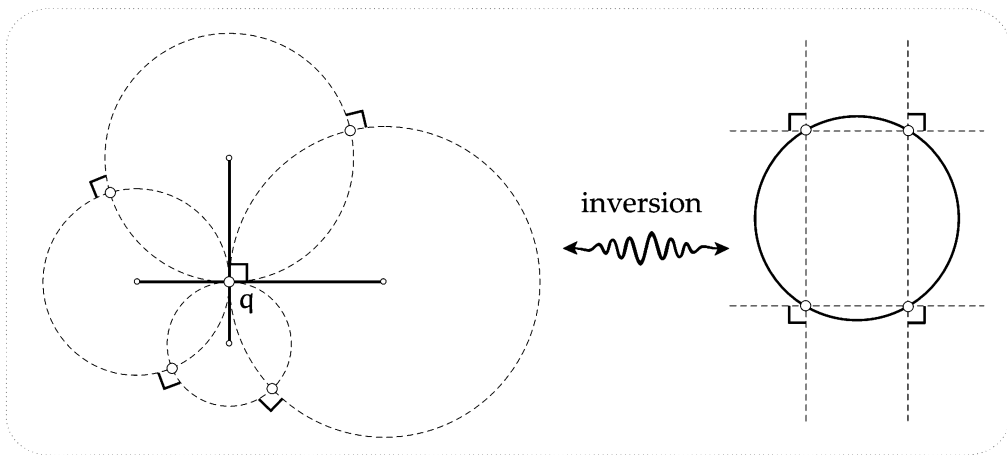
3.3.2 A Curious Property of Quadrilaterals with Orthogonal Diagonals

Figure [3.16] shows a shaded quadrilateral whose diagonals intersect orthogonally at q . If we now reflect q in each of the edges of the quadrilateral, then we obtain four new points. Very surprisingly, *these four points lie on a circle*⁴. As with the previous problem, see if you can prove this by ordinary means.

⁴ I am grateful to my friend Paul Zeitz for challenging me with this problem, which appeared in the USA Mathematical Olympiad.



[3.16] A curious property of quadrilaterals with orthogonal diagonals. Consider a quadrilateral whose diagonals intersect orthogonally at q . If we reflect q in each of the edges of the quadrilateral, then we obtain four new points. Very surprisingly, these four points lie on a circle!



[3.17] Geometric explanation of the previous figure. First, on the left, we use the construction in [3.7a] to represent the reflection of q in an edge as the second intersection point of any two circles through q whose centres lie on that edge. Here we have chosen the centres of these circles to be the vertices of the quadrilateral. Shown on the right is the result of performing an inversion in any circle centred at q . The orthogonal circles become orthogonal lines, intersecting in a rectangle, so these four points lie on a circle. But this means that the original four points were also on a circle. Done!

To demonstrate the result using inversion, we first use the construction in [3.7a] to represent the reflection of q in an edge as the second intersection point of any two circles through q whose centres lie on that edge. More precisely, let us choose the centres of these circles to be the *vertices* of the quadrilateral; see the LHS of [3.17].

Note that, because the diagonals are orthogonal, a pair of these circles centred at the ends of an edge will intersect orthogonally both at q and at the reflection of q in that edge.

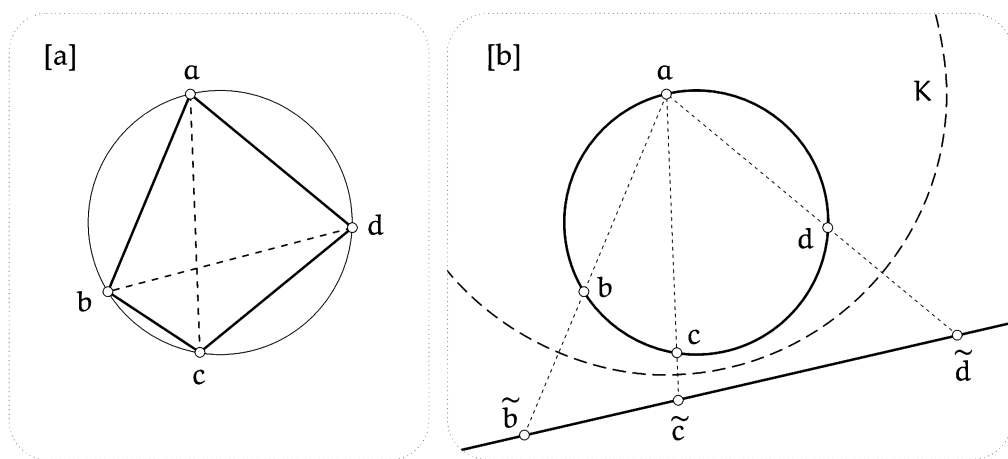
It follows that if we now apply an inversion in any circle centred at q , then a pair of such orthogonal circles through q will be mapped to a pair of orthogonal lines (parallel to the diagonals of the original quadrilateral); see the RHS of [3.17]. Thus the images of the four reflections of q are the vertices of a rectangle, and they therefore lie on a circle. The desired result follows immediately. Why?

3.3.3 Ptolemy's Theorem

Figure [3.18a] shows a quadrilateral $abcd$ inscribed in a circle. Ptolemy (c. CE 125) discovered the beautiful fact that *the sum of the product of the opposite sides is the product of the diagonals*. In symbols,

$$[ad][bc] + [ab][cd] = [ac][bd].$$

We note that for Ptolemy this was not merely interesting, it was a crucial tool for doing *astronomy*! See Ex. 9. His original proof (which is reproduced in most geometry texts) is elegant and simple, but it is very difficult to discover on one's own. On the other hand, once one has become comfortable with inversion, the following proof is almost mechanical.



[3.18] [a] A cyclic quadrilateral is subject to **Ptolemy's Theorem**: $[ad][bc] + [ab][cd] = [ac][bd]$. **[b]** Here is the result of inverting figure [a] in any circle K centred at a . Clearly, $[\tilde{b}\tilde{c}] + [\tilde{c}\tilde{d}] = [\tilde{b}\tilde{d}]$. But (3.6) dictates how the separations of these inverted points are related to the separations of the original points, and Ptolemy's Theorem then follows immediately!

Inverting figure [3.18a] in a circle K centred at one of the vertices (say a), we obtain [3.18b], in which

$$[\tilde{b}\tilde{c}] + [\tilde{c}\tilde{d}] = [\tilde{b}\tilde{d}].$$

Recalling that (3.6) tells us how the separation of two inverted points is related to the separation of the original points, we deduce that

$$\frac{[bc]}{[ab][ac]} + \frac{[cd]}{[ac][ad]} = \frac{[bd]}{[ab][ad]}.$$

Multiplying both sides by $([ab][ac][ad])$, we deduce Ptolemy's Theorem.

3.4 The Riemann Sphere

3.4.1 The Point at Infinity

In discussing inversion we saw that results about lines could always be understood as special limiting cases of results about circles, simply by letting the radius tend to infinity. This limiting process is nevertheless tiresome and clumsy; how much better it would be if lines could literally be described as circles of infinite radius.

Here is another, related inconvenience. Inversion in the unit circle is a one-to-one mapping of the plane to itself that swaps pairs of points. The same is true of the mapping $z \mapsto (1/z)$. However, there are exceptions: no image point is presently associated with $z = 0$, nor is 0 to be found among the image points.

To resolve both these difficulties, note that as z moves further and further away from the origin, $(1/z)$ moves closer and closer to 0. Thus as z travels to ever greater distances (in any direction), it is as though it were approaching a single *point at infinity*, written ∞ , whose image is 0. Thus, by definition, this point ∞ satisfies the following equations:

$$\frac{1}{\infty} = 0, \quad \frac{1}{0} = \infty.$$

The addition of this single point at infinity turns the complex plane into the so-called *extended complex plane*. Thus we may now say, without qualification, that $z \mapsto (1/z)$ is a one-to-one mapping of the extended plane to itself.

If a curve passes through $z = 0$ then (by definition) the image curve under $z \mapsto (1/z)$ will be a curve through the point at infinity. Conversely, if the image curve passes through 0 then the original curve passed through the point ∞ . Since $z \mapsto (1/z)$ swaps a circle through 0 with a line, we may now say that a line is just a circle that happens to pass through the point at infinity, and (without further qualification) inversion in a "circle" sends "circles" to "circles".

This is all very tidy, but it leaves one feeling none the wiser. We are accustomed to using the symbol ∞ only in conjunction with a limiting process, not as a thing in its own right; how are we to grasp its new meaning as a definite point that is infinitely far away?

3.4.2 Stereographic Projection

Riemann's profoundly beautiful answer to this question was to interpret complex numbers as points on a *sphere* Σ , instead of as points in a plane. Throughout the following discussion, imagine the complex plane positioned *horizontally* in space. In order to be definite about which way up the plane is, suppose that when we look down on $\mathbb{C}$ from above, a *positive* (i.e., counterclockwise) rotation of $(\pi/2)$ carries 1 to i . Now let Σ be the sphere centred at the origin of $\mathbb{C}$, and let it have unit radius so its "equator" coincides with the unit circle⁵.

We now seek to set up a correspondence between points on Σ and points in $\mathbb{C}$. If we think of Σ as the surface of the Earth, then this is the ancient problem of how to draw a geographical map. In an atlas you will find many different ways of drawing maps, the reason for the variety being that no single map can faithfully represent every aspect of a curved⁶ surface on a flat piece of paper. Although distortions of some kind are inevitably introduced, different maps can "preserve" or "faithfully represent" some (but not all) features of the curved surface. For example, a map can preserve angles at the expense of distorting areas.

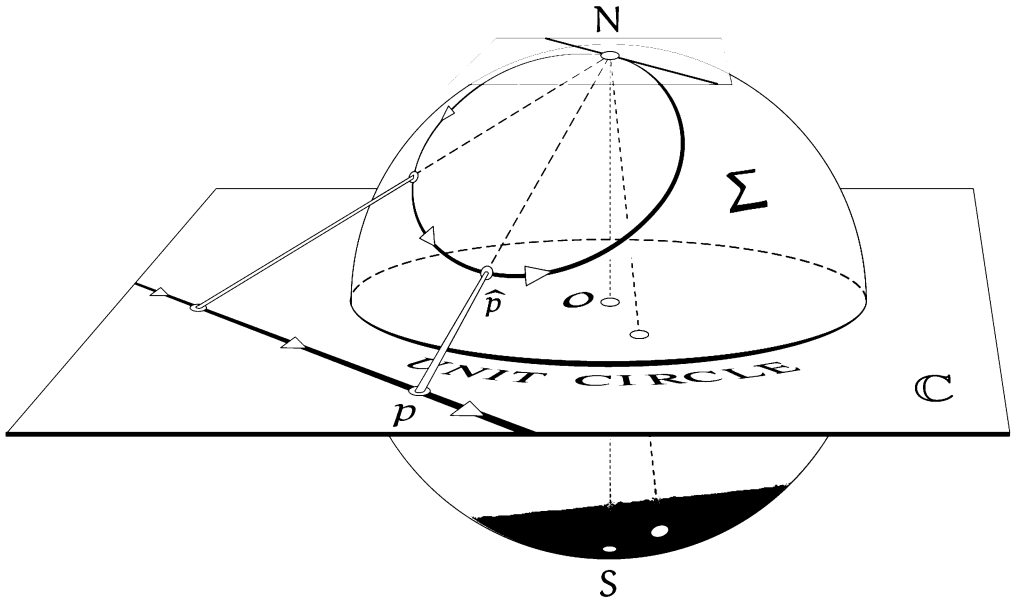
Ptolemy (c. CE 125) was the first to construct such a map, which he used to plot the positions of heavenly bodies on the "celestial sphere". His method is called *stereographic projection*, and we will soon see how perfectly it is adapted to our needs. Figure [3.19] illustrates its definition. From the north pole N of the sphere Σ , draw the line through the point p in $\mathbb{C}$; the *stereographic image* of p on Σ is the point $\hat{p}$ where this line intersects Σ . Since this gives us a one-to-one correspondence between points in $\mathbb{C}$ and points on Σ , let us also say that p is the stereographic image of $\hat{p}$. No confusion should arise from this, the context making it clear whether we are mapping $\mathbb{C}$ to Σ , or vice versa.

Note the following immediate facts: (i) the interior of the unit circle is mapped to the southern hemisphere of Σ , and in particular 0 is mapped to the south pole, S ; (ii) each point on the unit circle is mapped to itself, now viewed as lying on the equator of Σ ; (iii) the exterior of the unit circle is mapped to the northern hemisphere of Σ , *except* that N is not the image of any finite point in the plane.

However, it is clear that as p moves further and further away from the origin (in any direction), $\hat{p}$ moves closer and closer to N . This strongly suggests that N is *the stereographic image of the point at infinity*. Thus stereographic projection establishes a one-to-one correspondence between every point of the *extended* complex plane and every point of Σ . Instead of merely speaking of a "correspondence" between

⁵ Some works instead define Σ to be tangent to the complex plane at its south pole.

⁶ This concept of "curvature" will be defined more precisely in Chapter 6.



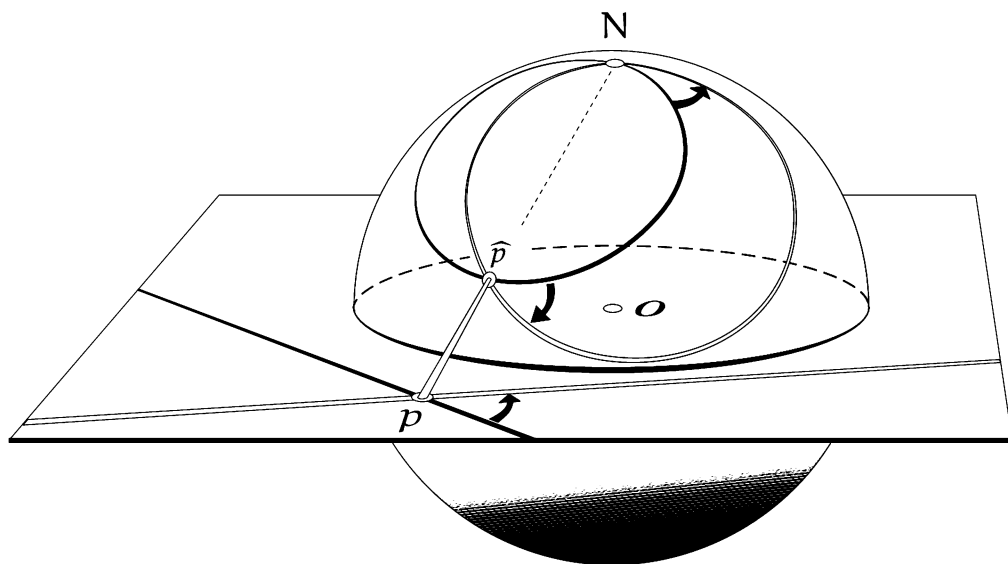
[3.19] Stereographic Projection maps the surface of the (clear glass) sphere to its equatorial plane. Standing at the north pole, N , fire a laser beam at a point $\hat{p}$ on the surface of the sphere: it goes on to hit the plane at p , called the stereographic projection (or image) of $\hat{p}$. Conversely, $\hat{p}$ is symmetrically called the stereographic projection (or image) of p . If p moves along a line in the plane, the laser beam sweeps out a plane, so the beam cuts the sphere in a circle that passes through N , and its tangent there is parallel to the line in the plane. This is because the equatorial plane and the tangent plane to the sphere at N are parallel—both are horizontal—so the plane swept out by the laser beam intersects these parallel planes in parallel lines.

complex numbers and points of Σ , we can imagine that the points of Σ are the complex numbers. For example, $S = 0$ and $N = \infty$. Once stereographic projection has been used to label each point of Σ with a complex number, Σ is called the *Riemann sphere*.

We have already discussed the fact that a line in $\mathbb{C}$ may be viewed as a circle passing through the point at infinity. The Riemann sphere now transforms this abstract idea into a literal fact:

The stereographic image of a line in the plane is a circle on Σ passing through $N = \infty$. (3.16)

To see this, observe that as p moves along the line shown in [3.19], the line connecting N to p sweeps out a plane through N . Thus $\hat{p}$ moves along the intersection of this plane with Σ , which is a circle passing through N . Done. In addition, note that *the tangent to this circle at N is parallel to the original line*. This is explained in the caption.



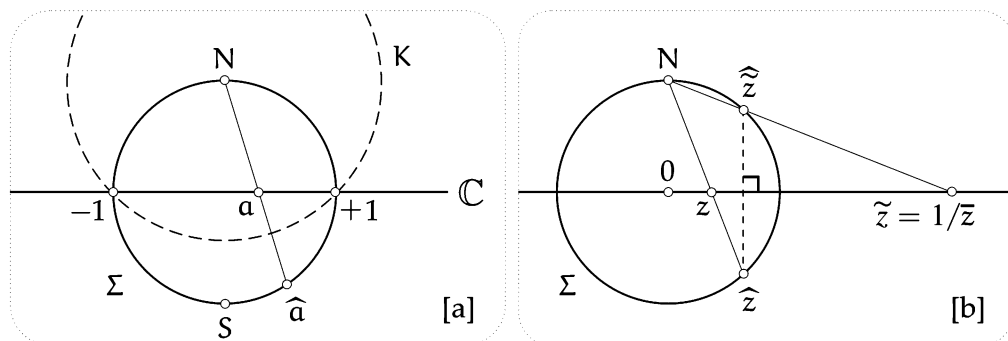
[3.20] Stereographic Projection is conformal. *If two lines intersect in the plane, their stereographic projections on the sphere are two circles that intersect at the same angle at N , for the tangents to these circles at N are parallel to the lines in the plane. But, by symmetry, the angle between these circles at $\hat{p}$ is the same as their angle at N , which we have just seen is the same as the angle between the lines in the plane. Done!*

From this last fact it follows that *stereographic projection preserves angles*. Consider [3.20], which shows two lines intersecting at p , together with their circular, stereographic projections. By symmetry, the magnitude of the angle of intersection between the circles is the same at their two intersection points, $\hat{p}$ and N . Since their tangents at N are parallel to the original lines in the plane, it follows that the illustrated angles at p and $\hat{p}$ are of equal *magnitude*. But before we can say that stereographic projection is “conformal”, we must assign a *sense* to the angle on the sphere.

According to our convention, the illustrated angle at p (from the black curve to the white one) is *positive*, i.e., it is counterclockwise when viewed from above the plane. From the perspective from which we have drawn [3.20], the angle at $\hat{p}$ is *negative*, i.e., clockwise. However, if we were looking at this angle from *inside* the sphere then it would be positive. Thus

If we define the sense of an angle on Σ by its appearance to an observer inside Σ , then stereographic projection is conformal. (3.17)

[HISTORICAL NOTE: It is quite remarkable that although stereographic projection had been well known since Ptolemy first put it to practical use around 125 CE, its beautiful and fundamentally important *conformality* was not discovered



[3.21] [a] Two-dimensional stereographic projection of the Riemann sphere Σ to the plane $\mathbb{C}$ is merely the restriction to Σ of the action of *three-dimensional geometric inversion in the sphere K of radius $\sqrt{2}$* (here shown in cross section) centred at N that cuts through Σ along its unit-circle equator in $\mathbb{C}$. [b] Inversion of $\mathbb{C}$ in the unit circle induces a reflection of the Riemann sphere in its equatorial plane, $\mathbb{C}$. If we compose this with conjugation, which reflects Σ in the vertical plane through the real axis, then we obtain complex inversion $z \mapsto (1/z)$. But reflection across two perpendicular planes is equivalent to a rotation of space about their line of intersection through angle π . Thus, **the mapping $z \mapsto (1/z)$ in $\mathbb{C}$ induces a rotation of the Riemann sphere about the real axis through an angle of π .**

for another 1500 years! This was first done around 1590 by Thomas Harriot.⁷ In Chapter 6 we shall meet Harriot again, for in 1603 he made another profoundly important discovery about the geometry of the sphere, this time relating the angles in a spherical triangle to the triangle's area—(6.9) on page 317.]

Clearly, any origin-centred circle in the plane is mapped to a horizontal circle on Σ , but what happens to a general circle? The startling answer is that *it too is mapped to a circle on the Riemann sphere!* This is quite difficult to see if we stick to our original definition of stereographic projection, but it suddenly becomes obvious if we change our point of view. Look again at [3.12], and observe how closely it resembles the definition of stereographic projection.

To make the connection precise, let K be the sphere centred at the north pole N of Σ that intersects Σ along its equator (the unit circle of $\mathbb{C}$). Figure [3.21a] shows a vertical cross section (through N and the real axis), of K , Σ , and $\mathbb{C}$. The full three-dimensional picture is obtained by rotating this figure about the line through N and S . We now see that

If K is the sphere of radius $\sqrt{2}$ centred at N , then stereographic projection is the restriction to $\mathbb{C}$ or Σ of inversion in K .

⁷ See Stillwell (2010, §16.2). For a short sketch of Harriot's life, see Stillwell (2010, §17.7). The first full biography of this very remarkable, unsung hero of mathematics and science—born within a mile of Stephen Hawking's place of birth (and mine!)—has at last been published: Arianrhod (2019).

In other words, if α is a point of $\mathbb{C}$ and $\hat{\alpha}$ is its stereographic projection on Σ , then $\hat{\alpha} = \mathcal{I}_K(\alpha)$ and $\alpha = \mathcal{I}_K(\hat{\alpha})$.

Appealing to our earlier work on inversion in spheres, (3.12) confirms our claim that

Stereographic projection preserves circles.

Note that (3.16) could also have been derived from (3.12) in this way.

3.4.3 Transferring Complex Functions to the Sphere

Stereographic projection enables us to transfer the action of any complex function to the Riemann sphere. Given a complex mapping $z \mapsto w = f(z)$ of $\mathbb{C}$ to itself, we obtain a corresponding mapping $\hat{z} \mapsto \hat{w}$ of Σ to itself, where $\hat{z}$ and $\hat{w}$ are the stereographic images of z and w . We shall say that $z \mapsto w$ *induces* the mapping $\hat{z} \mapsto \hat{w}$ of Σ .

For example, consider what happens if we transfer $f(z) = \bar{z}$ to Σ . Clearly [exercise],

Complex conjugation in $\mathbb{C}$ induces a reflection of the Riemann sphere in the vertical plane passing through the real axis.

For our next example, consider $z \mapsto \tilde{z} = (1/\bar{z})$, which is inversion in the unit circle C . Figure [3.21b] shows a vertical cross section of Σ taken through N and the point z in $\mathbb{C}$. This figure also illustrates the very surprising result of transferring this inversion to Σ :

Inversion of $\mathbb{C}$ in the unit circle induces a reflection of the Riemann sphere in its equatorial plane, $\mathbb{C}$. (3.18)

Here is an elegant way of seeing this. First note that not only are the pair of points z and $\tilde{z}$ symmetric (in the two-dimensional sense) with respect to C , but they are also symmetric (in the three-dimensional sense) with respect to the sphere Σ . Now apply the three-dimensional preservation of symmetry result (3.13). Since z and $\tilde{z}$ are symmetric with respect to Σ , their stereographic images $\hat{z} = \mathcal{I}_K(z)$ and $\hat{\tilde{z}} = \mathcal{I}_K(\tilde{z})$ will be symmetric with respect to $\mathcal{I}_K(\Sigma)$. But $\mathcal{I}_K(\Sigma) = \mathbb{C}$. Done! A more elementary (but less illuminating) derivation may be found in Ex. 6.

By combining the above results, we can now find the effect of complex inversion on the Riemann sphere. In $\mathbb{C}$, we know that $z \mapsto (1/\bar{z})$ is equivalent to inversion in the unit circle, followed by complex conjugation. The induced mapping on Σ is therefore the composition of *two reflections in perpendicular planes through the real axis*—one horizontal, the other vertical. However, it is not hard to see (perhaps with the aid of an orange) that the net effect of successively reflecting Σ in *any*

two perpendicular planes through the real axis is a *rotation* of Σ about the real axis through angle π . Thus we have shown that

The mapping $z \mapsto (1/z)$ in $\mathbb{C}$ induces a rotation of the Riemann sphere about the real axis through an angle of π . (3.19)

Recall that the point ∞ was originally defined by the property that it be swapped with 0 under complex inversion, $z \mapsto (1/z)$. The result (3.19) vividly illustrates the correctness of identifying N with the point at infinity, for the point 0 in $\mathbb{C}$ corresponds to the south pole S of Σ , and the rotation of π about the real axis does indeed swap S with N .

3.4.4 Behaviour of Functions at Infinity

Suppose two curves in $\mathbb{C}$ extend to arbitrarily large distances from the origin. Abstractly, one would say that they meet at the point at infinity. On Σ this becomes a literal intersection at N , and if each of the curves arrives at N in a well defined direction, then one can even assign an “intersection angle at ∞ ”. For example, [3.20] illustrates that if two lines in $\mathbb{C}$ intersect at a finite point and contain an angle α there, then they intersect for a second time at ∞ and they contain an angle $-\alpha$ at that point.

Transferring a complex function to the Riemann sphere enables one to examine its behaviour “at infinity” exactly as one would at any other point. In particular, one can look to see if the function preserves the angle between any two curves passing through ∞ . For example, the result (3.19) shows that complex inversion does preserve such angles at N , and it is therefore said to be “conformal at infinity”. By the same token, this rotation of Σ will also preserve the angle between two curves that pass through the singularity $z = 0$ of $z \mapsto (1/z)$, so complex inversion is conformal there too. In brief, *complex inversion is conformal throughout the extended complex plane*.

In this chapter we have found it convenient to depict $z \mapsto w$ as a mapping of $\mathbb{C}$ to *itself*, and in the above example we have likewise interpreted the induced mapping $\hat{z} \mapsto \hat{w}$ as sending points on the sphere to other points on the *same* sphere. However, it is often better to revert to the convention of the previous chapter, whereby the mapping sends points in the z -plane to image points residing in a second copy of $\mathbb{C}$, the w -plane. In the same spirit, the induced mapping $\hat{z} \mapsto \hat{w}$ may be viewed as mapping points in one sphere (the z -sphere) to points in a second sphere (the w -sphere). We illustrate this with an example.

Consider $z \mapsto w = z^n$, where n is a positive integer. The top half of [3.22] illustrates the effect of the mapping (in the case $n = 2$) on a grid of small “squares” abutting the unit circle and two rays containing an angle θ . Very mysteriously, the images of these “squares” in the w -plane are *again* almost square. In the next chapter we will show that this is just one consequence of a more basic mystery,

namely, that $z \mapsto w = z^n$ is *conformal*. Indeed, we will show that if a mapping is conformal, then any infinitesimal shape is mapped to a *similar* infinitesimal shape.

Since stereographic projection is known to be conformal, we would therefore anticipate that when we transfer the grid from the z -plane to the z -sphere, the result will again be a grid of “squares”. That this does indeed happen can be seen at the bottom left of [3.22]; the bottom right of [3.22] illustrates the same phenomenon as we pass from the image grid in the w -plane to the image grid on the w -sphere. Quite generally, any conformal mapping of $\mathbb{C}$ will induce a conformal mapping of Σ that will (as one consequence) map a grid of infinitesimal squares to another grid of infinitesimal squares.

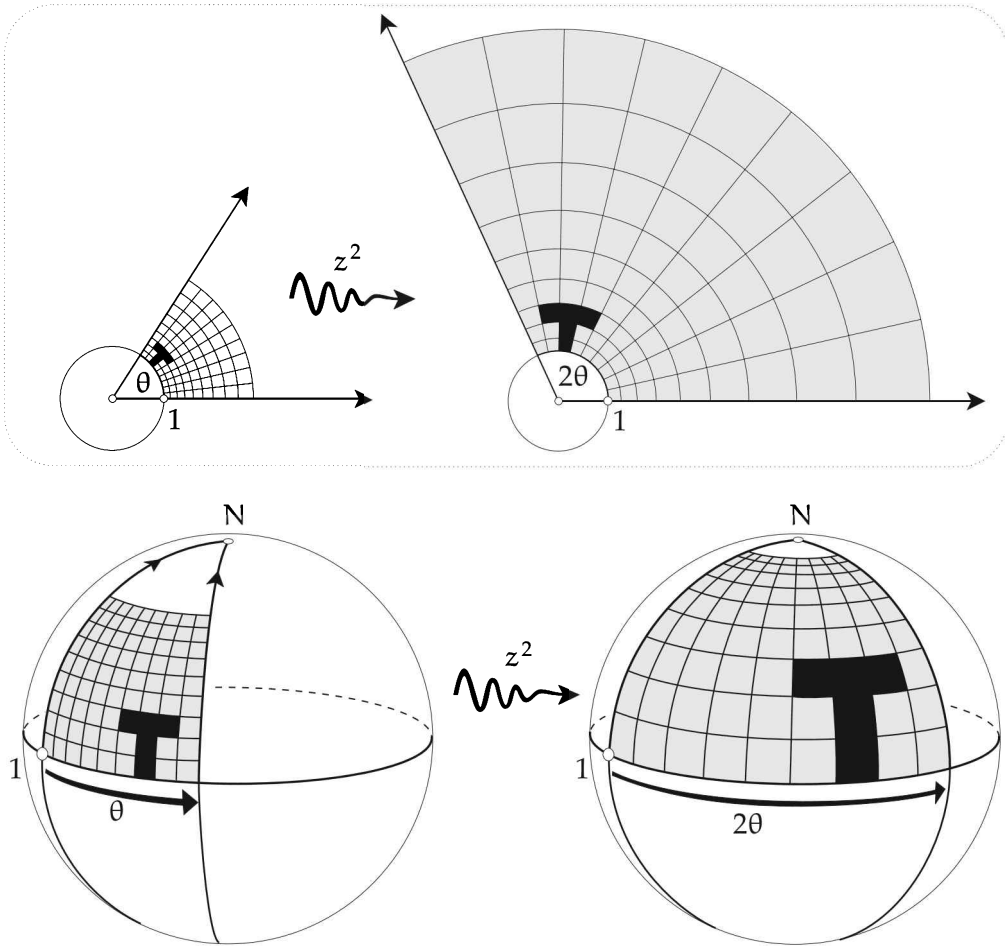
Figure [3.22] not only manifests the conformality of $z \mapsto w = z^2$, but it also illustrates that there exist points at which this conformality *breaks down*. Clearly, the angle θ at the origin is *doubled*; more generally, $z \mapsto w = z^n$ multiplies angles at 0 by n . Quite generally, if the conformality of an otherwise conformal mapping breaks down at a particular point p , then p is called a *critical point* of the mapping. Thus we may say that 0 is a critical point of $z \mapsto w = z^n$.

If we restrict ourselves to $\mathbb{C}$ then this is the *only* critical point of this mapping. However, if we look at the induced mapping of Σ , then the figure makes it clear that in the *extended* complex plane there is a second critical point at infinity: angles there are multiplied by n , just as they were at 0. Thus, more precisely than before, the claim is that $z \mapsto w = z^n$ is a conformal mapping whose only critical points are 0 and ∞ .

Next, we discuss how the behaviour of a complex mapping at infinity may be investigated *algebraically*. Complex inversion rotates Σ so that a neighbourhood of $N = \infty$ becomes a neighbourhood of $S = 0$. Thus to examine behaviour near infinity we may first apply complex inversion and then examine the neighbourhood of the origin. Algebraically, this means that to study $f(z)$ at infinity we should study $F(z) \equiv f(1/z)$ at the origin. For example, $f(z)$ is conformal at infinity if and only if $F(z)$ is conformal at the origin.

For example, if $f(z) = (z + 1)^3/(z^5 - z)$, then $F(z) = z^2(1 + z)^3/(1 - z^4)$, which has a double root at 0. Thus instead of merely saying that $f(z)$ “dies away to zero like $(1/z^2)$ as z tends to infinity”, we can now say (more precisely) that $f(z)$ has a double root at $z = \infty$.

This process can also be used to extend the concept of a branch point of a multi-function to the point at infinity. For example, if $f(z) = \log(z)$ then $F(z) = -\log(z)$. Thus $f(z)$ not only has a logarithmic branch point at $z = 0$, it also has one at $z = \infty$.



[3.22] The conformality of $z \mapsto z^n$. In the next chapter we will begin to address the remarkable and mysterious fact that $z \mapsto z^n$ is conformal, here illustrated in the case $n = 2$ [TOP]. We will also see that conformality suffices to explain the illustrated fact that a grid of small squares is ultimately mapped to a grid of squares, in the limit that they vanish. Note, however, that it is not conformal at 0: the angles between the rays are doubled. We see this too at the south pole of the Riemann sphere, and at the north pole, corresponding to ∞ . Since stereographic projection has been proven to be conformal, it follows that the conformality of $z \mapsto z^2$ is transferred to the induced mapping of the Riemann sphere [BOTTOM], and therefore it, too, ultimately maps squares to squares.

3.4.5 Stereographic Formulae*

In this subsection we derive explicit formulae connecting the coordinates of a point z in $\mathbb{C}$ and its stereographic projection $\hat{z}$ on Σ . These formulae will prove useful in investigating non-Euclidean geometry, but if you don't plan to study Chapter 6 then you should feel free to skip this subsection.

To begin with, let us describe z with Cartesian coordinates: $z = x + iy$. Similarly, let (X, Y, Z) be the Cartesian coordinates of $\hat{z}$ on Σ ; here the X - and Y -axes are chosen to coincide with the x - and y -axes of $\mathbb{C}$, so that the positive Z -axis passes through N . To make yourself comfortable with these coordinates, check the following facts: the equation of Σ is $X^2 + Y^2 + Z^2 = 1$, the coordinates of N are $(0, 0, 1)$, and similarly $S = (0, 0, -1)$, $1 = (1, 0, 0)$, $i = (0, 1, 0)$, etc.

Now let us find the formula for the stereographic projection $z = x + iy$ of the point $\hat{z}$ on Σ in terms of the coordinates (X, Y, Z) of $\hat{z}$. Let $z' = X + iY$ be the foot of the perpendicular from $\hat{z}$ to $\mathbb{C}$. Clearly, the desired point z is in the same direction as z' , so

$$z = \frac{|z|}{|z'|} z'.$$

Now look at [3.23a], which shows the vertical cross section of Σ and $\mathbb{C}$ taken through N and $\hat{z}$; note that this vertical plane necessarily also contains z' and z . From the similarity of the illustrated right triangles with hypotenuses $N\hat{z}$ and Nz , we immediately deduce [exercise] that

$$\frac{|z|}{|z'|} = \frac{1}{1 - Z},$$

and so we obtain our first stereographic formula:

$$x + iy = \frac{X + iY}{1 - Z}. \quad (3.20)$$

Let us now invert this formula to find the coordinates of $\hat{z}$ in terms of those of z . Since [exercise]

$$|z|^2 = \frac{1 + Z}{1 - Z},$$

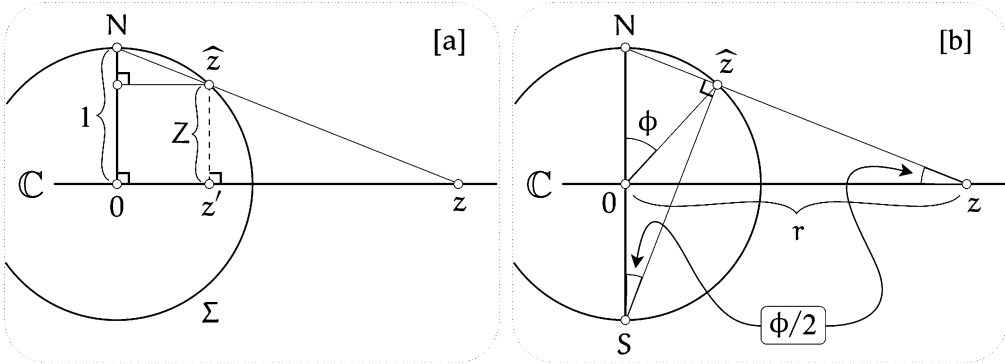
we obtain [exercise]

$$X + iY = \frac{2z}{1 + |z|^2} = \frac{2x + i2y}{1 + x^2 + y^2}, \quad \text{and} \quad Z = \frac{|z|^2 - 1}{|z|^2 + 1}. \quad (3.21)$$

Although it is often useful to describe the points of Σ with the three coordinates (X, Y, Z) , this is certainly unnatural, for the sphere is intrinsically *two* dimensional. If we instead describe $\hat{z}$ with the more natural (two-dimensional) spherical polar coordinates (ϕ, θ) then we obtain a particularly neat stereographic formula.

First recall⁸ that θ measures angle around the Z -axis, with $\theta = 0$ being assigned to the vertical half-plane through the positive X -axis: thus for a point z in $\mathbb{C}$, the angle θ is simply the usual angle from the positive real axis to z . The definition of ϕ is illustrated in [3.23b]—it is the angle subtended at the centre of Σ by the points N and $\hat{z}$: for example, the equator corresponds to $\phi = (\pi/2)$. By convention, $0 \leq \phi \leq \pi$.

⁸ This is the American convention; in my native England the roles of θ and ϕ are the reverse of those stated here.



[3.23] Two stereographic formulae. [a] From the similarity of the illustrated triangles, we immediately obtain a Cartesian formula for $z = x + iy = \frac{x+iy}{1-z}$ in terms of the Cartesian coordinates (X, Y, Z) of the point $\hat{z}$ on the sphere. This can be inverted to find $\hat{z}$ in terms of z ; see (3.21). [b] If we instead use geographical coordinates (θ, ϕ) to describe $\hat{z}$ on the sphere, the figure shows that we obtain a particularly elegant and useful polar formula: $z = \cot(\phi/2) e^{i\theta}$.

If z is the stereographic projection of the point $\hat{z}$ having coordinates (ϕ, θ) , then clearly $z = r e^{i\theta}$, and so it only remains to find r as a function of ϕ . From [3.23b] it is clear [exercise] that the triangles $N\hat{z}S$ and $N0z$ are similar, and because the angle $\angle NS\hat{z} = (\phi/2)$, it follows [exercise] that $r = \cot(\phi/2)$. Thus our new stereographic formula is

$$z = \cot(\phi/2) e^{i\theta}. \quad (3.22)$$

We will now illustrate this formula with two applications. In Ex. 8 we also show how this formula may be used to establish a beautiful alternative interpretation of stereographic projection, due to Sir Roger Penrose.

As our first application, let us rederive the result (3.19). As above, let $\hat{z}$ be a general point of Σ having coordinates (ϕ, θ) , and let $\tilde{\hat{z}}$ be the point to which it is carried when we rotate Σ by π about the real axis. Check for yourself (perhaps with the aid of an orange) that the coordinates of $\tilde{\hat{z}}$ are $(\pi - \phi, -\theta)$. Thus if $\tilde{z}$ is the stereographic image of $\tilde{\hat{z}}$, then

$$\tilde{z} = \cot \left[\frac{\pi}{2} - \frac{\phi}{2} \right] e^{-i\theta} = \frac{1}{\cot(\phi/2)} e^{-i\theta} = \frac{1}{z},$$

as was to be shown.

For our second application, recall that if two points on a sphere are diametrically opposite each other (such as the north and south poles) then they are said to be *antipodal*. Let us show that

If $\hat{p}$ and $\hat{q}$ are antipodal points of Σ , then their stereographic projections p and q are related by the following formula:

$$q = -(1/\bar{p}).$$

Put differently, $q = -\mathcal{I}_C(p)$, where C is the unit circle. Note that the relationship between p and q is actually symmetrical (as clearly it should be): $p = -(1/\bar{q})$. To verify (3.23), first check for yourself that if $\hat{p}$ has coordinates (ϕ, θ) then $\hat{q}$ has coordinates $(\pi - \phi, \pi + \theta)$. The remainder of the proof is almost identical to the previous calculation. For an elementary geometric proof, see Ex. 6.

3.5 Möbius Transformations: Basic Results

3.5.1 Preservation of Circles, Angles, and Symmetry

From (3.3) we know that a general Möbius transformation $M(z) = \frac{az+b}{cz+d}$ can be decomposed into the following sequence of more elementary transformations: a translation, complex inversion, a rotation, an expansion, and a second translation. Since each of these transformations preserves circles, angles, and symmetry, we immediately deduce the following fundamental results:

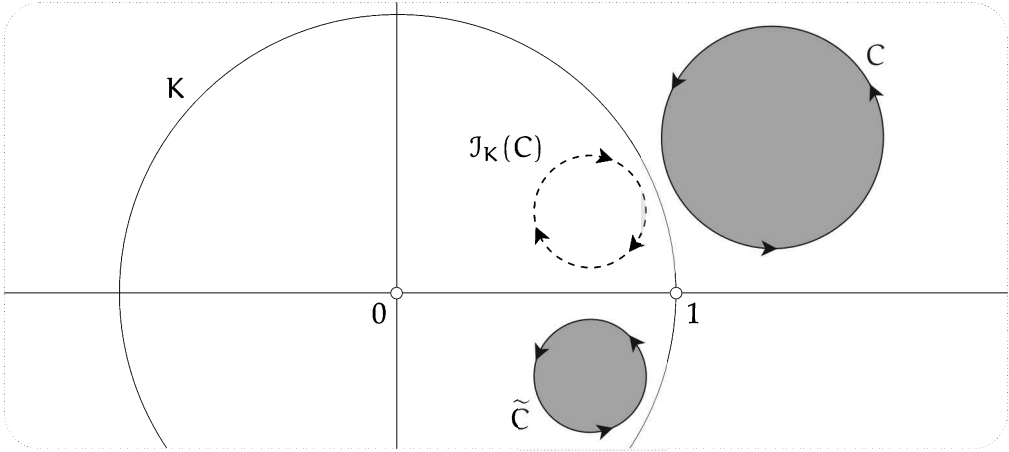
- Möbius transformations map circles to circles.⁹
- Möbius transformations are conformal.
- If two points are symmetric with respect to a circle, then their images under a Möbius transformation are symmetric with respect to the image circle. This is called the “Symmetry Principle”.

We know that a circle C will map to a circle—of course lines are now included as “circles”—but what will happen to the *disc* bounded by C ? First we give a useful way of thinking about this disc. Imagine yourself walking round C moving *counterclockwise*; your motion gives C what is called a *positive sense* or *orientation*. Of the two regions into which this positively oriented circle divides the plane, the disc may now be identified as the one lying to your *left*.

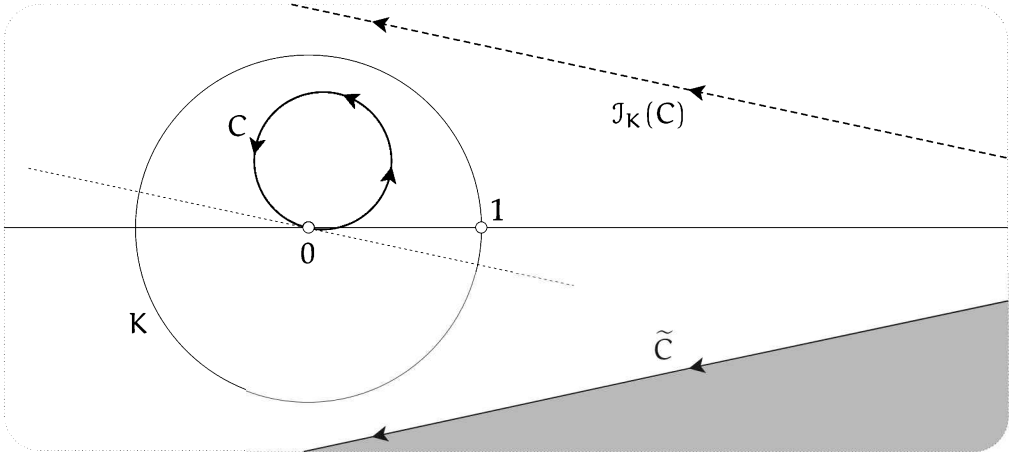
Now consider the effect of the four transformations in (3.3) on the disc and on the positively oriented circle bounding it. Translations, rotations, and expansions all preserve the orientation of C and map the interior of C to the interior of the image $\tilde{C}$ of C . However, the effect of complex inversion on C depends on whether or not C contains the origin. If C does not contain the origin, then $\tilde{C}$ has the same orientation as C , and the interior of C is mapped to the interior of $\tilde{C}$. This is easily understood by looking at [3.24].

If C does contain the origin then $\tilde{C}$ has the opposite orientation and the interior of C is mapped to the exterior of $\tilde{C}$. If C passes through the origin then its interior is mapped to the half-plane lying to the left of the oriented line $\tilde{C}$. See [3.25].

⁹ Remarkably, Carathéodory (1937) proved that this property actually *characterizes* the Möbius transformations: they alone have this property!



[3.24] A Möbius transformation maps an oriented circle C to an oriented circle $\tilde{C}$ in such a way that the region to the left of C is mapped to the region to the left of $\tilde{C}$.



[3.25] A limiting form of the previous figure has the image disc expand to become a half-plane, but it is still true that the region to the left of C is mapped to the region to the left of $\tilde{C}$. In the next chapter we will see that this is a more universal phenomenon, having little to do with the specific geometry of Möbius transformations.

To summarize,

A Möbius transformation maps an oriented circle C to an oriented circle $\tilde{C}$ in such a way that the region to the left of C is mapped to the region to the left of $\tilde{C}$. (3.24)

3.5.2 Non-Uniqueness of the Coefficients

To specify a particular Möbius transformation $M(z) = \frac{az+b}{cz+d}$ it seems that we need to specify the *four* complex numbers a , b , c , and d , which we call the *coefficients* of

the Möbius transformation. In geometric terms, this would mean that to specify a particular Möbius transformation we would need to know the images of any four distinct points. This is wrong.

If k is an arbitrary (non-zero) complex number then

$$\frac{az + b}{cz + d} = M(z) = \frac{kaz + kb}{kcz + kd}.$$

In other words, multiplying the coefficients by k yields one and the same mapping, and so only the *ratios* of the coefficients matter. Since *three* complex numbers are sufficient to pin down the mapping— (a/b) , (b/c) , (c/d) , for example—we conjecture (and later prove) that

*There exists a unique Möbius transformation sending any three points
to any other three points.* (3.25)

In the course of gradually establishing this one result we shall be led to further important properties of Möbius transformations.

If you read the last section of Chapter 1, then (3.25) may be ringing a bell: the similarity transformations needed to do Euclidean geometry are also determined by their effect on three points. Indeed, we saw in that chapter that such similarities can be expressed as complex functions of the form $f(z) = az + b$, and so they actually *are* Möbius transformations, albeit of a particularly simple kind. However, for such a similarity to exist, the image points must form a triangle that is similar to the triangle formed by the original points. But in the case of Möbius transformations there is no such restriction, and this opens the way to more flexible, non-Euclidean geometries in which Möbius transformations play the role of the “motions”. This is the subject of Chapter 6.

Let us make a further remark on the non-uniqueness of the coefficients of a Möbius transformation. Recall from the beginning of this chapter that the interesting Möbius transformations are the non-singular ones, for which $(ad - bc) \neq 0$. For if $(ad - bc) = 0$ then $M(z) = \frac{az+b}{cz+d}$ crushes the entire plane down to the single point (a/c) . If M is non-singular, then we may multiply its coefficients by $k = \pm 1/\sqrt{ad - bc}$, in which case the *new* coefficients satisfy

$$(ad - bc) = 1;$$

the Möbius transformation is then said to be *normalized*. When investigating the properties of a *general* Möbius transformation, it turns out to be very convenient to work with this normalized form. However, when doing calculations with *specific* Möbius transformations, it is usually best *not* to normalize them.

3.5.3 The Group Property

In addition to preserving circles, angles, and symmetry, the mapping

$$z \mapsto w = M(z) = \frac{az + b}{cz + d} \quad (ad - bc) \neq 0$$

is also *one-to-one* and *onto*. This means that if we are given any point w in the w -plane, there is one (and only one) point z in the z -plane that is mapped to w . We can show this by explicitly finding the inverse transformation $w \mapsto z = M^{-1}(w)$. Solving the above equation for z in terms of w , we find [exercise] that M^{-1} is *also* a Möbius transformation:

$$M^{-1}(z) = \frac{dz - b}{-cz + a}. \quad (3.26)$$

Note that if M is normalized, then this formula for M^{-1} is *automatically* normalized as well.

If we look at the induced mapping on the Riemann sphere, then we find that a Möbius transformation actually establishes a one-to-one correspondence between points of the complete z -sphere and points of the complete w -sphere, including their points at infinity. Indeed you may easily convince yourself that

$$M(\infty) = (a/c) \quad \text{and} \quad M(-d/c) = \infty.$$

Using (3.26), you may check for yourself that $M^{-1}(a/c) = \infty$ and $M^{-1}(\infty) = -(d/c)$.

Next, consider the composition $M \equiv (M_2 \circ M_1)$ of two Möbius transformations,

$$M_2(z) = \frac{a_2z + b_2}{c_2z + d_2} \quad \text{and} \quad M_1(z) = \frac{a_1z + b_1}{c_1z + d_1}.$$

A simple calculation [exercise] shows that M is *also* a Möbius transformation:

$$M(z) = (M_2 \circ M_1)(z) = \frac{(a_2a_1 + b_2c_1)z + (a_2b_1 + b_2d_1)}{(c_2a_1 + d_2c_1)z + (c_2b_1 + d_2d_1)}. \quad (3.27)$$

It is clear geometrically that if M_1 and M_2 are non-singular, then so is M . This is certainly not obvious algebraically, but later in this section we shall introduce a new algebraic approach that does make it obvious.

If you have studied “groups”, or if you read the final section of Chapter 1, then you will realize that we have now established the following: *The set of non-singular Möbius transformations forms a group under composition.* For, (i) the identity mapping $\mathcal{E}(z) = z$ belongs to the set; (ii) the composition of two members of the set yields a third member of the set; (iii) every member of the set possesses an inverse that also lies in the set.

3.5.4 Fixed Points

As another step towards establishing (3.25), let us show that if a Möbius transformation exists mapping three given points to three other given points, then it is *unique*. To this end, we now introduce the extremely important concept of the *fixed points* of a Möbius transformation. Quite generally, p is called a fixed point of a mapping f if $f(p) = p$, in which case one may also say that p is “mapped to itself”,

or that it “remains fixed”. Note that under the identity mapping, $z \mapsto \mathcal{E}(z) = z$, every point is a fixed point.

By definition, then, the fixed points of a general Möbius transformation $M(z)$ are the solutions of

$$z = M(z) = \frac{az + b}{cz + d}.$$

Since this is merely a quadratic in disguise, we deduce that

With the exception of the identity mapping, a Möbius transformation has at most two fixed points.

From the above result it follows that if a Möbius transformation is known to have more than two fixed points, then it must be the identity. This enables us to establish the uniqueness part of (3.25). Suppose that M and N are two Möbius transformations that both map the three given points (say q, r, s) to the three given image points. Since $(N^{-1} \circ M)$ is a Möbius transformation that has q, r , and s as fixed points, we deduce that it must be the identity mapping, and so $N = M$. Done.

We now describe the fixed points explicitly. If $M(z)$ is *normalized*, then the two fixed points ξ_+, ξ_- are given by [exercise]

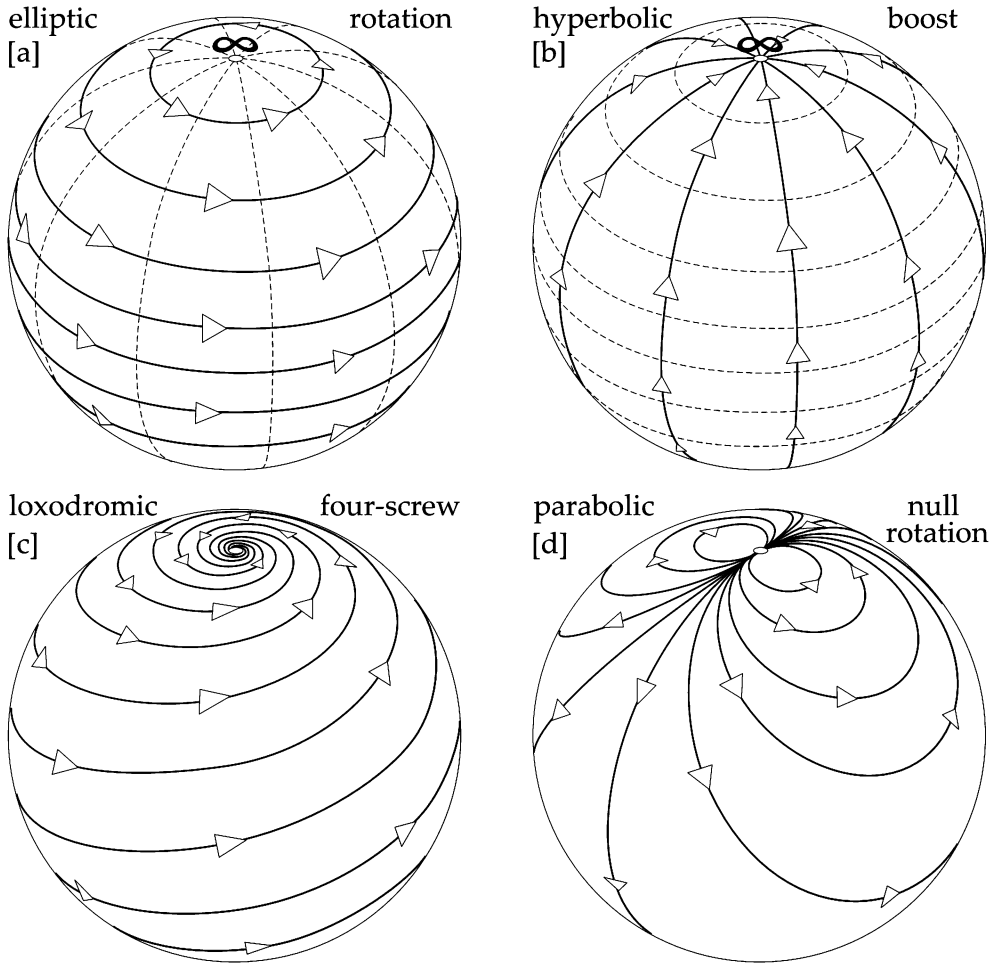
$$\xi_{\pm} = \frac{(a - d) \pm \sqrt{(a + d)^2 - 4}}{2c}. \quad (3.28)$$

In the exceptional case where $(a + d) = \pm 2$, the two fixed points $\xi_{\pm}$ coalesce into the single fixed point $\xi = (a - d)/2c$. In this case the Möbius transformation is called *parabolic*.

3.5.5 Fixed Points at Infinity

Let us now briefly outline how the fixed point can be used to classify the Möbius transformations into just four achetypes. The full mathematical details are worked out later, in Section 3.7. Recall our earlier observation (3.2), that, miraculously, the Möbius transformations exactly correspond to the symmetries (called *Lorentz transformations*) of Minkowski and Einstein’s spacetime. Thus this classification of the Möbius transformations is important for relativity theory, too! We cannot explore this in detail here, but see Needham (2021, §6.4) for the technical details of the spacetime interpretation via Lorentz transformations

Provided $c \neq 0$ then the fixed points both lie in the finite plane; we now discuss the fact that if $c = 0$ then at least one fixed point is at infinity. If $c = 0$ then the Möbius transformation takes the form $M(z) = Az + B$, which represents, as we have mentioned, the most general “direct” (i.e., conformal) similarity transformation of the plane. If we write $A = \rho e^{i\alpha}$ then this may be viewed as the composition of an



[3.26] Classification of Möbius and Lorentz transformations. Each of the four types of transformation has two names, depending on whether it is viewed as acting on $\mathbb{C}$ [name on the left] or on spacetime [name on the right]. The mathematical classification depends on analysing the fixed points, and this is done in detail in Section 3.7. For the technical details of the spacetime interpretation via Lorentz transformations, see [Needham \(2021, §6.4\)](#).

origin-centred rotation of α , an origin-centred expansion by ρ , and finally a translation of B . Let us visualize each of these three transformations on the Riemann sphere.

With $\alpha > 0$, figure [3.26a] illustrates that the rotation $z \mapsto e^{i\alpha}z$ in $\mathbb{C}$ induces an equal rotation of Σ about the vertical axis through its centre. Horizontal circles on Σ rotate (in the direction of the arrows) into themselves and are therefore called *invariant curves* of the transformation. This figure makes it vividly clear that the fixed points of such a rotation are 0 and ∞ . Note also that the (great) circles

through these fixed points (which are orthogonal to the invariant circles) are permuted among themselves. This pure rotation is the simplest, archetypal example of a so-called *elliptic* Möbius transformation.

With $\rho > 1$, figure [3.26b] illustrates the induced transformation on Σ corresponding to the origin-centred expansion of $\mathbb{C}$, $z \mapsto \rho z$. If $\rho < 1$ then we have a contraction of $\mathbb{C}$, and points on Σ move due South instead of due North. Again it is clear that the fixed points are 0 and ∞ , but the roles of the two families of curves in [3.26a] are now reversed: the invariant curves are the great circles through the fixed points at the poles, and the orthogonal horizontal circles are permuted among themselves. This pure expansion is the simplest, archetypal example of a so-called *hyperbolic* Möbius transformation.

Figure [3.26c] shows the combined effect of the rotation and expansion in [3.26a] and [3.26b]. Here the invariant curves are the illustrated “spirals”; however, the two families of circles in [3.26a] (or [3.26b]) are both invariant as a whole, in the sense that the members of each family are permuted among themselves. This rotation and expansion is the archetypal *loxodromic* Möbius transformation, of which the elliptic and hyperbolic transformations are particularly important special cases.

Finally, [3.26d] illustrates a translation. Since the invariant curves in $\mathbb{C}$ are the family of parallel lines in the direction of the translation, the invariant curves on Σ are the family of circles whose common tangent at ∞ is parallel to the invariant lines in $\mathbb{C}$. Since ∞ is the only fixed point, a pure translation is an example of a parabolic Möbius transformation.

Note the following consequence of the above discussion:

A Möbius transformation has a fixed point at ∞ if and only if it is a similarity, $M(z) = (az + b)$. Furthermore, ∞ is the sole fixed point if and only if $M(z)$ is a translation, $M(z) = (z + b)$. (3.29)

Later we will use this to show that each Möbius transformation is equivalent, in a certain sense, to one (and only one) of the four types shown in [3.26].

3.5.6 The Cross-Ratio

Returning to (3.25), we have already established that if we can find a Möbius transformation M that maps three given points q, r, s to three other given points $\tilde{q}, \tilde{r}, \tilde{s}$, then M is unique. It thus remains to show that such an M always exists.

To see this, first let us arbitrarily choose three points q', r', s' , *once and for all*. Next, suppose we can write down a Möbius transformation mapping three arbitrary points q, r, s to these particular three points, q', r', s' ; let $M_{qrs}(z)$ denote this Möbius transformation. In exactly the same way we could also write down $M_{\tilde{q}\tilde{r}\tilde{s}}(z)$. By virtue of the group property, it is now easy to see that

$$M = M_{\tilde{q}\tilde{r}\tilde{s}}^{-1} \circ M_{qrs}$$

is a Möbius transformation mapping q, r, s to q', r', s' and thence to $\tilde{q}, \tilde{r}, \tilde{s}$, as was desired.

Now the real trick is to choose q', r', s' in such a way as to make it easy to write down $M_{qrs}(z)$. We don't like to pull rabbits out of hats, but try $q' = 0, r' = 1$, and $s' = \infty$. Along with this special choice comes a special, standard notation: *the unique Möbius transformation mapping three given points q, r, s to $0, 1, \infty$ (respectively) is written $[z, q, r, s]$.*

In order to map q to $q' = 0$ and s to $s' = \infty$, the numerator and denominator of $[z, q, r, s]$ must be proportional to $(z - q)$ and $(z - s)$, respectively. Thus $[z, q, r, s] = k \left(\frac{z - q}{z - s} \right)$, where k is a constant. Finally, since $k \left(\frac{r - q}{r - s} \right) = [r, q, r, s] \equiv 1$, we deduce that

$$[z, q, r, s] = \frac{(z - q)(r - s)}{(z - s)(r - q)}.$$

This is not quite so rabbit-like as it appears. Two hundred years prior to Möbius' investigations, Girard Desargues had discovered the importance of the expression $[z, q, r, s]$ within the subject of projective geometry, where it was christened the *cross-ratio* of z, q, r, s (in this order¹⁰). Its significance in that context is briefly explained in Ex. 14, but the reader is urged to consult [Stillwell \(2010\)](#) for greater detail and background. We can now restate (3.25) in a more explicit form:

The unique Möbius transformation $z \mapsto w = M(z)$ sending three points q, r, s to any other three points $\tilde{q}, \tilde{r}, \tilde{s}$ is given by

$$\frac{(w - \tilde{q})(\tilde{r} - \tilde{s})}{(w - \tilde{s})(\tilde{r} - \tilde{q})} = [w, \tilde{q}, \tilde{r}, \tilde{s}] = [z, q, r, s] = \frac{(z - q)(r - s)}{(z - s)(r - q)}. \quad (3.30)$$

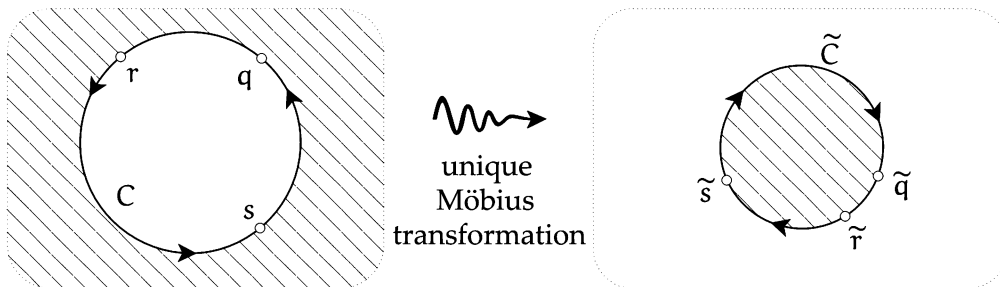
Although we have not done so, in any concrete case one could easily go on to solve this equation for w , thereby obtaining an explicit formula for $w = M(z)$.

The result (3.30) may be rephrased in various helpful ways. For example, if a Möbius transformation maps four points p, q, r, s to $\tilde{p}, \tilde{q}, \tilde{r}, \tilde{s}$ (respectively) then the cross-ratio is invariant: $[\tilde{p}, \tilde{q}, \tilde{r}, \tilde{s}] = [p, q, r, s]$. Conversely, p, q, r, s can be mapped to $\tilde{p}, \tilde{q}, \tilde{r}, \tilde{s}$ by a Möbius transformation if their cross-ratios are equal.

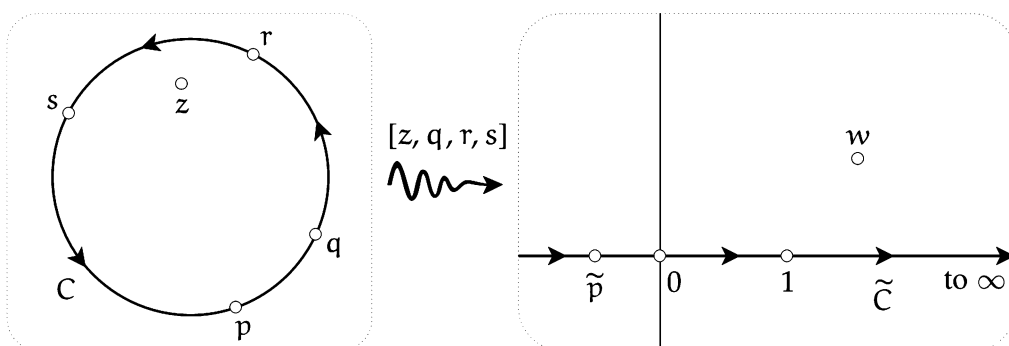
Recalling (3.24), we also obtain the following:

Let C be the unique circle through the points q, r, s in the z -plane, oriented so that these points succeed one another in the stated order. Likewise, let $\tilde{C}$ be the unique oriented circle through $\tilde{q}, \tilde{r}, \tilde{s}$ in the w -plane. Then the Möbius transformation given by (3.30) maps C to $\tilde{C}$, and it maps the region lying to the left of C to the region lying to the left of $\tilde{C}$. (3.31)

¹⁰ Different orders yield different values; see Ex. 16. Unfortunately, there is no firm convention as to which of these values is "the" cross-ratio. For example, our definition agrees with [Carathéodory \(1964\)](#), [Penrose and Rindler \(1984\)](#), and [Jones and Singerman \(1987\)](#), but it is different from the equally common definition of [Ahlfors \(1979\)](#).



[3.27] *There exists a unique Möbius transformation that maps three ordered points on the circle C to three other ordered points on the circle $\tilde{C}$, and the region to the left of C is mapped to the region to the left of $\tilde{C}$.*



[3.28] Geometric interpretation of the cross-ratio. *The cross-ratio $w = [z, q, r, s]$ can be vividly pictured as the image of z under the unique Möbius transformation that maps the oriented circle C through q, r, s to the real axis in such a way that these three points map to $0, 1, \infty$. Also illustrated is the fact that a point p lies on the circle C through q, r, s if and only if $\text{Im } [p, q, r, s] = 0$.*

This is illustrated in [3.27].

This in turn gives us a more vivid picture of the cross-ratio: $w = [z, q, r, s]$ is the image of z under the unique Möbius transformation that maps the oriented circle C through q, r, s to the real axis in such a way that these three points map to $0, 1, \infty$. If q, r, s induce a positive orientation on C then the interior of C is mapped to the upper half-plane; if they induce a negative orientation, then the image is the lower half-plane. This is illustrated in [3.28], from which we immediately deduce a neat equation for the circle C :

A point p lies on the circle C through q, r, s if and only if

$$\text{Im } [p, q, r, s] = 0. \quad (3.32)$$

Furthermore, if q, r, s induce a positive orientation on C (as in [3.28]), then p lies inside C if and only if $\text{Im}[p, q, r, s] > 0$. If the orientation of C is negative, then the inequality is reversed.

For a more elementary proof of (3.32), see Ex. 15.

3.6 Möbius Transformations as Matrices*

3.6.1 Empirical Evidence of a Link with Linear Algebra

As you were reading about the group property of Möbius transformations, you may well have experienced *déjà vu*, for the results we obtained were remarkably reminiscent of the behaviour of matrices in *linear algebra*. Before explaining the *reason* for this connection between Möbius transformations and linear algebra, let us be more explicit about the empirical evidence for believing that such a connection exists.

We begin by associating with every Möbius transformation $M(z)$ a corresponding 2×2 matrix $[M]$:

$$M(z) = \frac{az + b}{cz + d} \quad \longleftrightarrow \quad [M] = \begin{bmatrix} a & b \\ c & d \end{bmatrix}.$$

Since the coefficients of the Möbius transformation are not unique, neither is the corresponding matrix: if k is any non-zero constant, then the matrix $k[M]$ corresponds to the same Möbius transformation as $[M]$. However, if $[M]$ is normalized by imposing $(ad - bc) = 1$, then there are just two possible matrices associated with a given Möbius transformation: if one is called $[M]$, the other is $-[M]$; in other words, the matrix is determined “uniquely up to sign”. This apparently trivial fact turns out to have deep significance in both mathematics and physics; see [Penrose and Rindler \(1984, Ch. 1\)](#) and [Penrose \(2005\)](#).

At this point there exists a strong possibility of confusion, so we issue the following **WARNING**: In linear algebra we are—or should be!—accustomed to thinking of a real 2×2 matrix as representing a linear transformation of $\mathbb{R}^2$. For example, $\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ represents a rotation of the plane through $(\pi/2)$. That is, when we apply it to a vector $\begin{pmatrix} x \\ y \end{pmatrix}$ in $\mathbb{R}^2$, we obtain

$$\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} -y \\ x \end{pmatrix} = \left\{ \begin{pmatrix} x \\ y \end{pmatrix} \text{ rotated by } (\pi/2) \right\}.$$

In stark contrast, the matrix $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$ corresponding to a Möbius transformation generally has *complex* numbers as its entries, and so it *cannot* be interpreted as a linear transformation of $\mathbb{R}^2$. Even if the entries *are* real, it must not be thought of in

this way. For example, the matrix $\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ corresponds to the Möbius transformation $M(z) = -(1/z)$, which is certainly not a linear transformation of $\mathbb{C}$. To avoid confusion, we will adopt the following notational convention: *We use (ROUND) brackets for a real matrix corresponding to a linear transformation of $\mathbb{R}^2$ or of $\mathbb{C}$, and we use [SQUARE] brackets for a (generally) complex matrix corresponding to a Möbius transformation of $\mathbb{C}$.*

Despite this warning, we have the following striking parallels between the behaviour of Möbius transformations and the matrices that represent them:

- The identity Möbius transformation $\mathcal{E}(z) = z$ corresponds to the familiar identity matrix, $[\mathcal{E}] = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$.
- The Möbius transformation $M(z)$ with matrix $[M] = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ possesses an inverse if and only if the *matrix* possesses an inverse. For recall that $[M]$ is non-singular if and only if its determinant $\det[M] = (ad - bc)$ is non-zero.
- If we look at (3.26), we see that the matrix of the inverse Möbius transformation $M^{-1}(z)$ is the same as the inverse matrix $[M]^{-1}$. To put this succinctly,

$$[M^{-1}] = [M]^{-1}.$$

- In linear algebra we compose two linear transformations by multiplying their matrices; indeed, this is the origin of the multiplication rule. If we multiply the matrices $[M_2]$ and $[M_1]$ corresponding to the two Möbius transformations $M_2(z)$ and $M_1(z)$, then we obtain

$$\begin{bmatrix} a_2 & b_2 \\ c_2 & d_2 \end{bmatrix} \begin{bmatrix} a_1 & b_1 \\ c_1 & d_1 \end{bmatrix} = \begin{bmatrix} a_2 a_1 + b_2 c_1 & a_2 b_1 + b_2 d_1 \\ c_2 a_1 + d_2 c_1 & c_2 b_1 + d_2 d_1 \end{bmatrix}.$$

But look at (3.27)! This is simply the matrix of the composite Möbius transformation $(M_2 \circ M_1)(z)$. Thus *multiplication of Möbius matrices corresponds to composition of Möbius transformations*:

$$[M_2] [M_1] = [M_2 \circ M_1].$$

3.6.2 The Explanation: Homogeneous Coordinates

Clearly this cannot all be coincidence, but what is really going on here?! The answer is simple, yet subtle. To see it we must first describe the complex plane with a

completely new kind of coordinate system. Instead of expressing $z = x + iy$ in terms of two real numbers, we write it as the *ratio of two complex numbers*, z_1 and z_2 :

$$z = \frac{z_1}{z_2}.$$

The ordered pair of complex numbers $[z_1, z_2]$ are called *homogeneous coordinates* of z . In order that this ratio be well defined we demand that $[z_1, z_2] \neq [0, 0]$. To each ordered pair $[z_1 \text{ arbitrary}, z_2 \neq 0]$ there corresponds precisely one point $z = (z_1/z_2)$, but to each point z there corresponds an infinite set of homogeneous coordinates, $[kz_1, kz_2] = k[z_1, z_2]$, where k is an arbitrary non-zero complex number.

What about a pair of the form $[z_1, 0]$? By holding z_1 fixed as z_2 tends to 0, it is clear that $[z_1, 0]$ must be identified with the point at infinity. Thus the totality of pairs $[z_1, z_2]$ provide coordinates for the *extended* complex plane. The introduction of homogeneous coordinates thereby accomplishes for algebra what the Riemann sphere accomplishes for geometry—it does away with the exceptional role of ∞ .

Just as we use the symbol $\mathbb{R}^2$ to denote the set of pairs (x, y) of *real* numbers, so we use the symbol $\mathbb{C}^2$ to denote the set of pairs $[z_1, z_2]$ of *complex* numbers. To highlight the distinction between $\mathbb{R}^2$ and $\mathbb{C}^2$, we use conventional round brackets when writing down an element (x, y) of $\mathbb{R}^2$, but we use square brackets for an element $[z_1, z_2]$ of $\mathbb{C}^2$.

Just as a linear transformation of $\mathbb{R}^2$ is represented by a real 2×2 matrix, so a linear transformation of $\mathbb{C}^2$ is represented by a complex 2×2 matrix:

$$\begin{bmatrix} z_1 \\ z_2 \end{bmatrix} \longmapsto \begin{bmatrix} w_1 \\ w_2 \end{bmatrix} = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} z_1 \\ z_2 \end{bmatrix} = \begin{bmatrix} a z_1 + b z_2 \\ c z_1 + d z_2 \end{bmatrix}.$$

But if $[z_1, z_2]$ and $[w_1, w_2]$ are thought of as the homogeneous coordinates in $\mathbb{C}^2$ of the point $z = (z_1/z_2)$ in $\mathbb{C}$ and its image point $w = (w_1/w_2)$, then the above linear transformation of $\mathbb{C}^2$ induces the following (non-linear) transformation of $\mathbb{C}$:

$$z = \frac{z_1}{z_2} \longmapsto w = \frac{w_1}{w_2} = \frac{a z_1 + b z_2}{c z_1 + d z_2} = \frac{a (z_1/z_2) + b}{c (z_1/z_2) + d} = \frac{az + b}{cz + d}.$$

This is none other than the most general Möbius transformation!

We have thus explained why Möbius transformations in $\mathbb{C}$ behave so much like linear transformations—they *are* linear transformations, only they act on the homogeneous coordinates in $\mathbb{C}^2$, rather than directly on the points of $\mathbb{C}$ itself.

As with the cross-ratio, homogeneous coordinates first arose in projective geometry, and for this reason they are often also called *projective* coordinates. See [Stillwell \(2010\)](#) for greater detail on the history of the idea. We cannot move on without mentioning that in recent times these homogeneous coordinates have provided the key to great conceptual advances (and powerful new computational techniques) in Einstein's Theory of Relativity. This pioneering body of work on *2-spinors* is due to

Sir Roger Penrose. See [Penrose and Rindler \(1984, Ch. 1\)](#), [Penrose \(2005\)](#), and [Shaw \(2006, Ch. 23\)](#).

3.6.3 Eigenvectors and Eigenvalues*

The above representation of Möbius transformations as matrices provides an elegant and practical method of doing concrete calculations. More significantly, however, it also means that in developing the theory of Möbius transformations we suddenly have access to a whole range of new ideas and techniques taken from linear algebra.

We begin with something very simple. We previously remarked that while it is geometrically obvious that the composition of two non-singular Möbius transformations is again non-singular, it is far from obvious algebraically. Our new point of view rectifies this, for recall the following elementary property of determinants:

$$\det\{[M_2] [M_1]\} = \det[M_2] \det[M_1].$$

Thus if $\det[M_2] \neq 0$ and $\det[M_1] \neq 0$, then $\det\{[M_2] [M_1]\} \neq 0$, as was to be shown. This also sheds further light on the virtue of working with normalized Möbius transformations. For if $\det[M_2] = 1$ and $\det[M_1] = 1$, then $\det\{[M_2] [M_1]\} = 1$. Thus the set of normalized 2×2 matrices form a *group*—a “subgroup” of the full group of non-singular matrices.

For our second example, consider the *eigenvectors* of a linear transformation $[M] = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ of $\mathbb{C}^2$. By definition, an eigenvector is a vector $\mathfrak{z} = \begin{bmatrix} \mathfrak{z}_1 \\ \mathfrak{z}_2 \end{bmatrix}$ whose “direction” is unaltered by the transformation, in the sense that its image is simply a multiple $\lambda \mathfrak{z}$ of the original; this multiple λ is called the *eigenvalue* of the eigenvector. In other words, an eigenvector satisfies the equation

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} \mathfrak{z}_1 \\ \mathfrak{z}_2 \end{bmatrix} = \lambda \begin{bmatrix} \mathfrak{z}_1 \\ \mathfrak{z}_2 \end{bmatrix}.$$

In terms of the corresponding Möbius transformation in $\mathbb{C}$, this means that $z = (\mathfrak{z}_1/\mathfrak{z}_2)$ is mapped to $M(z) = (\lambda \mathfrak{z}_1/\lambda \mathfrak{z}_2) = z$, and so

$$z = (\mathfrak{z}_1/\mathfrak{z}_2) \text{ is a fixed point of } M(z) \text{ if and only if } \mathfrak{z} = \begin{bmatrix} \mathfrak{z}_1 \\ \mathfrak{z}_2 \end{bmatrix} \text{ is an eigenvector of } [M]. \quad (3.33)$$

Note that one immediate benefit of this approach is that there is no longer any real distinction between a finite fixed point and a fixed point at ∞ , for the latter merely corresponds to an eigenvector of the form $\begin{bmatrix} \mathfrak{z}_1 \\ 0 \end{bmatrix}$. For example, consider how

elegantly we may rederive the fact that ∞ is a fixed point if and only if $M(z)$ is a similarity transformation. If ∞ is a fixed point then

$$\lambda \begin{bmatrix} z_1 \\ 0 \end{bmatrix} = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} z_1 \\ 0 \end{bmatrix} = \begin{bmatrix} a z_1 \\ c z_1 \end{bmatrix}.$$

Thus $c = 0$, $\lambda = a$, and $M(z) = (a/d)z + (b/d)$.

Recall that if the matrix $[M]$ represents the Möbius transformation $M(z)$, then so does the matrix $k[M]$ obtained by multiplying the entries by k . The fact that eigenvectors carry geometric information about $M(z)$ shows up in the fact that they are independent of the choice of k . Indeed, if $\mathbf{z}$ is an eigenvector of $[M]$ (with eigenvalue λ) then it is also an eigenvector of $k[M]$, but with eigenvalue $k\lambda$:

$$\{k[M]\}\mathbf{z} = k\lambda\mathbf{z}.$$

Since the eigenvalue *does* depend on the arbitrary choice of k , it appears that its value can have no bearing on the geometric nature of the mapping $M(z)$. Very surprisingly, however, if $[M]$ is *normalized* then the exact opposite is true! In the next section we will show that *the eigenvalues of the normalized matrix $[M]$ completely determine the geometric nature of the corresponding Möbius transformation $M(z)$* . In anticipation of this result, let us investigate the eigenvalues further.

Recall the fact that the eigenvalues of $[M]$ are the solutions of the so-called *characteristic equation*, $\det\{[M] - \lambda[\mathcal{E}]\} = 0$, where $[\mathcal{E}]$ is the identity matrix $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$. Using the fact that $[M]$ is normalized, we find [exercise] that the characteristic equation is

$$\lambda^2 - (a + d)\lambda + 1 = 0,$$

which (for later use) may be written as

$$\lambda + \frac{1}{\lambda} = a + d. \quad (3.34)$$

The first thing we notice about this equation is that there are typically two eigenvalues, λ_1 and λ_2 , and they are determined solely by the value of $(a + d)$. By inspecting the coefficients of the quadratic we immediately deduce that

$$\lambda_1 \lambda_2 = 1 \quad \text{and} \quad \lambda_1 + \lambda_2 = (a + d). \quad (3.35)$$

Thus if we know λ_1 , then $\lambda_2 = (1/\lambda_1)$. We emphasize this point because it is not obvious when we simply write down the formula for the eigenvalues:

$$\lambda_1, \lambda_2 = \frac{1}{2} \left\{ (a + d) \pm \sqrt{(a + d)^2 - 4} \right\}.$$

Aficionados of linear algebra will recognize (3.35) as a special case of the following general result on the eigenvalues $\lambda_1, \lambda_2, \dots, \lambda_n$ of any $n \times n$ matrix N :

$$\lambda_1 \lambda_2 \dots \lambda_n = \det N \quad \text{and} \quad \lambda_1 + \lambda_2 + \dots + \lambda_n = \operatorname{tr} N,$$

where $\operatorname{tr} N \equiv$ (the sum of the diagonal elements of N) is called the *trace* of N . For future use, recall the following nice property of the trace function: *If N and P are both $n \times n$ matrices, then*

$$\operatorname{tr}\{NP\} = \operatorname{tr}\{PN\}. \quad (3.36)$$

In the case of 2×2 matrices (which is all that we shall ever need) this is easily verified by a direct calculation [exercise].

3.6.4 Rotations of the Sphere as Möbius Transformations*

This subsection is optional because its main result is only needed in Chapter 6. Furthermore, in that chapter we shall treat the same result in a much better and simpler way; the only purpose of this subsection is to further illustrate the connections that exist between Möbius transformations and linear algebra.

Let us investigate what it might mean to say that two vectors $\mathbf{p}$ and $\mathbf{q}$ in $\mathbb{C}^2$ are “orthogonal”. Two vectors $\mathbf{p}$ and $\mathbf{q}$ in $\mathbb{R}^2$ are orthogonal if and only if their dot product vanishes:

$$\mathbf{p} \cdot \mathbf{q} = \begin{pmatrix} p_1 \\ p_2 \end{pmatrix} \cdot \begin{pmatrix} q_1 \\ q_2 \end{pmatrix} = p_1 q_1 + p_2 q_2 = 0.$$

Thus it would seem natural to say that $\mathbf{p}$ and $\mathbf{q}$ are “orthogonal” if $\mathbf{p} \cdot \mathbf{q} = 0$. This will not do. In particular, whereas we would like the dot product of any nonzero vector with itself to be a positive real number, we find that $\begin{bmatrix} 1 \\ i \end{bmatrix} \cdot \begin{bmatrix} 1 \\ i \end{bmatrix} = 0$, for example. As it stands, the dot product is not suitable for use in $\mathbb{C}^2$.

The standard solution to this difficulty is to generalize the dot product $\mathbf{p} \cdot \mathbf{q}$ to the so-called *inner product*, $\langle \mathbf{p}, \mathbf{q} \rangle \equiv \overline{\mathbf{p}} \cdot \mathbf{q}$:

$$\langle \mathbf{p}, \mathbf{q} \rangle = \left\langle \begin{bmatrix} p_1 \\ p_2 \end{bmatrix}, \begin{bmatrix} q_1 \\ q_2 \end{bmatrix} \right\rangle = \overline{p_1} q_1 + \overline{p_2} q_2.$$

We cannot go into all the reasons why this is the “right” generalization, but observe that it shares the following desirable properties of the dot product:

$$\begin{aligned} \langle \mathbf{p}, \mathbf{p} \rangle &\geq 0 \quad \text{and} \quad \langle \mathbf{p}, \mathbf{p} \rangle = 0 \text{ if and only if } p_1 = 0 = p_2; \\ \langle \mathbf{p} + \mathbf{q}, \mathbf{r} \rangle &= \langle \mathbf{p}, \mathbf{r} \rangle + \langle \mathbf{q}, \mathbf{r} \rangle \quad \text{and} \quad \langle \mathbf{r}, \mathbf{p} + \mathbf{q} \rangle = \langle \mathbf{r}, \mathbf{p} \rangle + \langle \mathbf{r}, \mathbf{q} \rangle. \end{aligned}$$

Note, however, that it is not commutative: $\langle \mathbf{q}, \mathbf{p} \rangle = \overline{\langle \mathbf{p}, \mathbf{q} \rangle}$.

We now agree that $\mathbf{p}$ and $\mathbf{q}$ are “orthogonal” if and only if

$$\langle \mathbf{p}, \mathbf{q} \rangle = \overline{p_1} q_1 + \overline{p_2} q_2 = 0.$$

What does this “orthogonality” mean in terms of the points $p = (p_1/p_2)$ and $q = (q_1/q_2)$ whose homogeneous coordinate vectors are $\mathbf{p}$ and $\mathbf{q}$? The answer is surprising. As you may easily check, the above equation says that $q = -(1/\overline{p})$, and so from (3.23) we deduce that

Two vectors in $\mathbb{C}^2$ are orthogonal if and only if they are the homogeneous coordinates of antipodal points on the Riemann sphere.

Suppose we could find a linear transformation $[R]$ of $\mathbb{C}^2$ that were analogous to a rotation—what transformation of the Riemann sphere Σ would be induced by the corresponding Möbius transformation $R(z)$? By “analogous to a rotation”, we mean that $[R]$ preserves the inner product:

$$\langle [R]\mathbf{p}, [R]\mathbf{q} \rangle = \langle \mathbf{p}, \mathbf{q} \rangle. \quad (3.37)$$

In particular, $[R]$ maps every pair of orthogonal vectors to another such pair, and $R(z)$ therefore maps every pair of antipodal points on Σ to another such pair. We shall not attempt a real proof, but since the transformation of Σ is also known to be continuous and conformal¹¹, it can only be a *rotation* of Σ .

The desired invariance of the inner product (3.37) may be neatly rephrased using an operation called the *conjugate transpose*, denoted by a superscript $*$. This operation takes the complex conjugate of each element in a matrix and then interchanges the rows and columns:

$$\mathbf{p}^* = \begin{bmatrix} p_1 \\ p_2 \end{bmatrix}^* = [\overline{p_1}, \overline{p_2}] \quad \text{and} \quad [R]^* = \begin{bmatrix} a & b \\ c & d \end{bmatrix}^* = \begin{bmatrix} \overline{a} & \overline{c} \\ \overline{b} & \overline{d} \end{bmatrix}.$$

Since the inner product can now be expressed in terms of ordinary matrix multiplication as $\langle \mathbf{p}, \mathbf{q} \rangle = \mathbf{p}^* \mathbf{q}$, and since [exercise] $\{[R]\mathbf{p}\}^* = \mathbf{p}^* [R]^*$, we find that (3.37) takes the form

$$\mathbf{p}^* \{[R]^* [R]\} \mathbf{q} = \mathbf{p}^* \mathbf{q}.$$

Clearly this is satisfied if

$$[R]^* [R] = [\mathcal{E}], \quad (3.38)$$

and in linear algebra it is shown that this is also a *necessary* condition.

Matrices satisfying equation (3.38) are extremely important in both mathematics and physics—they are called *unitary matrices*. In the present case of normalized 2×2

¹¹ If it were not continuous then it could, for example, exchange points on two antipodal patches of Σ while leaving the remainder fixed. If it were continuous but *anticonformal*, then it could map each point to its antipodal point, or to its reflection in a plane through the centre of Σ .

matrices, we can easily find the most general unitary matrix $[R]$ by re-expressing (3.38) as $[R]^* = [R]^{-1}$:

$$\begin{bmatrix} \bar{a} & \bar{c} \\ \bar{b} & \bar{d} \end{bmatrix} = \begin{bmatrix} d & -b \\ -c & a \end{bmatrix} \implies [R] = \begin{bmatrix} a & b \\ -\bar{b} & \bar{a} \end{bmatrix}.$$

Although we have left some unsatisfactory gaps in the above reasoning, we have nevertheless arrived at an important truth: *The most general rotation of the Riemann sphere can be expressed as a Möbius transformation of the form*

$$R(z) = \frac{az + b}{-\bar{b}z + \bar{a}}. \quad (3.39)$$

This was first discovered by Gauss, around 1819.

3.7 Visualization and Classification*

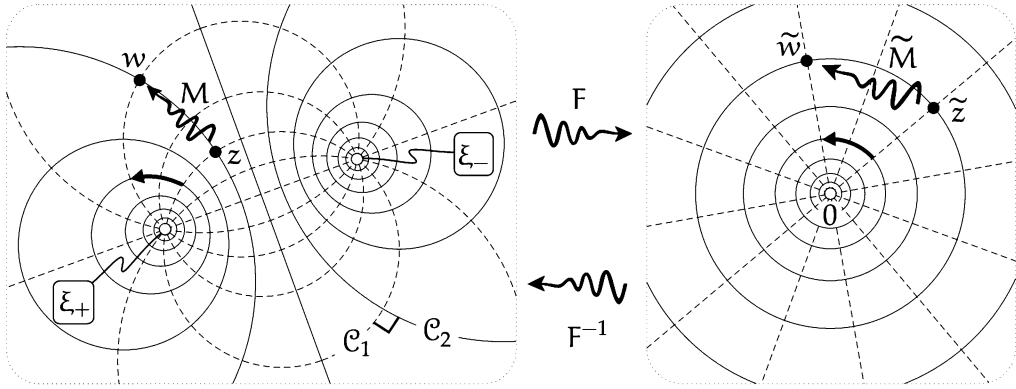
3.7.1 The Main Idea

Although the decomposition (3.3) of a general Möbius transformation $M(z)$ has proved valuable in obtaining results, it makes $M(z)$ appear much more complicated than it is. In this section we will reveal this hidden simplicity by examining the fixed points in greater detail; this will enable us to visualize Möbius transformations in a particularly vivid way. In the process we will clarify our earlier remark that Möbius transformations can be classified into four types, each $M(z)$ being “equivalent” to one (and only one) of the four types of transformation illustrated in [3.26]. The lovely idea behind this classification scheme is due to Felix Klein.

To begin with, suppose that $M(z)$ has two distinct fixed points, ξ_+ and ξ_- . Now look at the LHS of [3.29], and in particular at the family $\mathcal{C}_1$ of circles [shown dashed] passing through the fixed points. If we think of $M(z)$ as a mapping $z \mapsto w = M(z)$ of this figure to itself, then *each member of $\mathcal{C}_1$ is mapped to another member of $\mathcal{C}_1$* . Why?

Still with reference to the LHS of [3.29], suppose that p [not shown] is an arbitrary point on the line through ξ_+ and ξ_- , but lying outside the segment connecting the fixed points. If K is the circle of radius $\sqrt{[p\xi_+][p\xi_-]}$ centred at p , then ξ_+ and ξ_- are symmetric with respect to K . Thus K cuts each member of $\mathcal{C}_1$ at right angles (cf. [3.9]). By varying the position of p we thus obtain a family $\mathcal{C}_2$ of circles [shown solid] such that *ξ_+ and ξ_- are symmetric with respect to each member of $\mathcal{C}_2$, and each member of $\mathcal{C}_2$ is orthogonal to each member of $\mathcal{C}_1$* .

Now we come to the main idea: to the LHS of [3.29] we apply a Möbius transformation $F(z)$ that sends one fixed point (say ξ_+) to 0, and the other fixed point (ξ_-) to ∞ . The



[3.29] The geometric idea behind the classification of the Möbius transformations. Given a Möbius transformation $M(z)$ with two fixed points, ξ_+ and ξ_- , two families of circles naturally arise: the dashed family $\mathcal{C}_1$ passing through the fixed points, and the orthogonal family $\mathcal{C}_2$, such that ξ_+ and ξ_- are symmetric with respect to each member of $\mathcal{C}_2$. Note that $M(z)$ must map members of $\mathcal{C}_1$ amongst themselves, and likewise members of $\mathcal{C}_2$ are mapped amongst themselves. The key idea is to now apply a Möbius transformation $F(z) = \frac{z - \xi_+}{z - \xi_-}$ that sends one fixed point to the origin (south pole of the Riemann sphere) and the other to ∞ (the north pole). Thus, on the right, the circles $\mathcal{C}_1$ become rays, and the circles $\mathcal{C}_2$ become origin-centred circles. Then the induced mapping $\tilde{M} = F \circ M \circ F^{-1}$ clearly has fixed points 0 and ∞ , and must therefore take the form $\tilde{M}(\tilde{z}) = m\tilde{z}$, where $m = \rho e^{i\alpha}$ is called the multiplier; in the illustrated case, $\rho = 1$ and $\alpha = (\pi/3)$. This multiplier m completely characterizes the original Möbius transformation, M , and therefore serves to classify it. In this manner, we obtain the first three archetypes shown in [3.26].

RHS of [3.29] shows the image of the LHS under such a Möbius transformation, the simplest example of which is

$$F(z) = \frac{z - \xi_+}{z - \xi_-}.$$

[Note that we have not bothered to write this in normalized form.] Since F is a Möbius transformation, it must map the members of $\mathcal{C}_1$ to the circles passing through 0 and ∞ , i.e., to lines through the origin [shown dashed]. Furthermore, since F is conformal, two such lines must contain the same angle at 0 as the corresponding $\mathcal{C}_1$ circles do at ξ_+ . We have tried to make this easy to see in our picture by drawing $\mathcal{C}_1$ circles passing through ξ_+ in evenly spaced directions, each one making an angle of $(\pi/6)$ with the next.

As an aside, observe that we now have a second, simpler explanation of the existence of the family $\mathcal{C}_2$ of circles orthogonal to $\mathcal{C}_1$. Since the illustrated set of origin-centred circles are orthogonal to lines through 0, their images under F^{-1} must be circles orthogonal to each member of $\mathcal{C}_1$.

Next, let $\tilde{z} = F(z)$ and $\tilde{w} = F(w)$ be the images under F of z and $w = M(z)$. We may now think of F as carrying the original Möbius transformation $z \mapsto w = M(z)$

on the left over to a transformation $\widetilde{M}$ on the right, namely $\widetilde{z} \mapsto \widetilde{w} = \widetilde{M}(\widetilde{z})$. More explicitly,

$$\widetilde{w} = F(w) = F(M[z]) = F(M[F^{-1}(\widetilde{z})]),$$

and so

$$\widetilde{M} = F \circ M \circ F^{-1}. \quad (3.40)$$

Since $\widetilde{M}$ is the composition of three Möbius transformations, it is itself a Möbius transformation. Furthermore, it follows immediately from the construction that the fixed points of $\widetilde{M}$ are 0 and ∞ . But we have already seen that if a Möbius transformation leaves these points fixed, it can only be of the form

$$\widetilde{M}(\widetilde{z}) = m \widetilde{z},$$

where $m = \rho e^{i\alpha}$ is simply a complex number. Geometrically, $\widetilde{M}$ is just a rotation by α combined with an expansion by ρ .

This complex number m not only constitutes a complete description of the mapping $\widetilde{M}$ but, as we will see shortly, it also completely characterizes the geometric nature of the original Möbius transformation M . The number m is called the *multiplier* of $M(z)$.

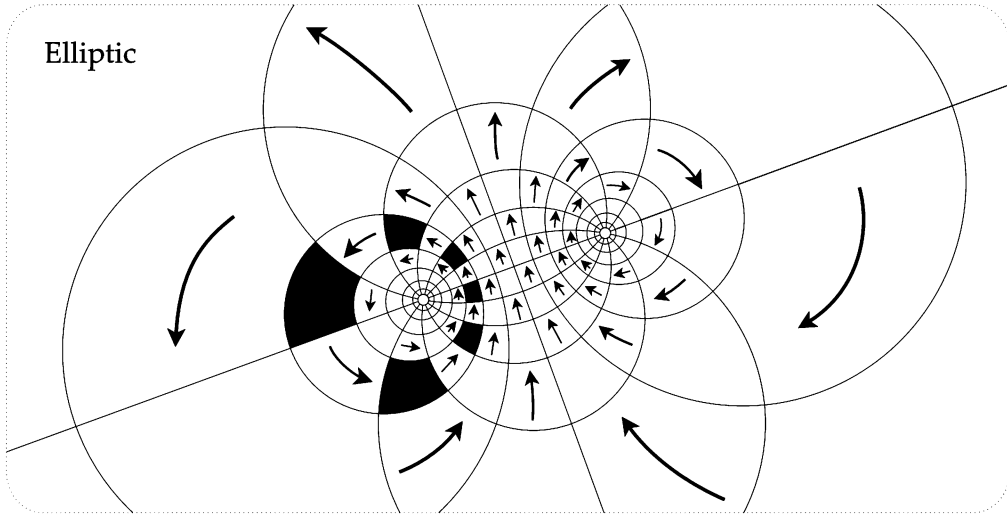
3.7.2 Elliptic, Hyperbolic, and Loxodromic Transformations

Before reading on, refresh your memory of the classification (shown in [3.26a,b,c]) of Möbius transformations of the form $\widetilde{M}(\widetilde{z}) = m \widetilde{z}$.

We call $M(z)$ an *elliptic Möbius transformation* if $\widetilde{M}$ is elliptic, meaning that the latter is a pure rotation corresponding to $m = e^{i\alpha}$. Since $\widetilde{M}$ is a rotation if and only if it maps each origin-centred circle to itself, $M(z)$ is elliptic if and only if it maps each $\mathcal{C}_2$ circle to itself. With $\alpha = (\pi/3)$, the RHS of [3.29] illustrates the effect of $\widetilde{M}$ on the point $\widetilde{z}$. On the LHS you can see the corresponding, unambiguous effect of M : it moves z along its $\mathcal{C}_2$ circle till it lies on the $\mathcal{C}_1$ circle making angle $(\pi/3)$ with the original $\mathcal{C}_1$ through z .

Figure [3.30]¹² is intended to give a more vivid impression of this same elliptic transformation. Each shaded “rectangle” is mapped by $M(z)$ to the next one in the direction of the arrows—some of these regions have been filled with black to emphasize this. This figure may be viewed as typical, with one exception. Because we have chosen $\alpha = (\pi/3)$, six successive applications of M yield the identity, and one therefore says that M has *period* 6. More generally, if $\alpha = (m/n)2\pi$, where (m/n) is a fraction reduced to lowest terms, then M has period n . Of course this is not typical. In general $(\alpha/2\pi)$ will be irrational, and no matter how many times we apply M we will never obtain the identity.

¹² Shading inspired by Ford (1929).

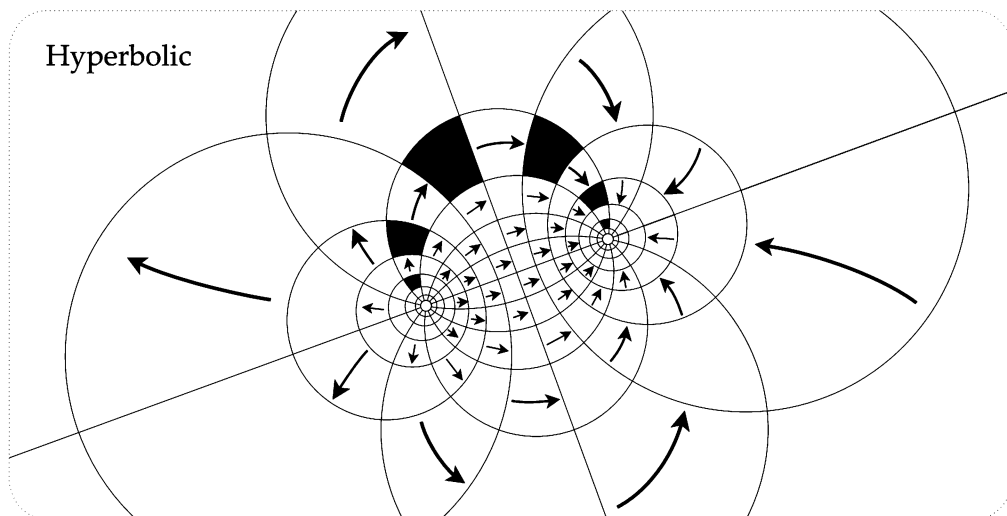


[3.30] An Elliptic Möbius transformation has multiplier $m = e^{i\alpha}$, so that $\widetilde{M}$ is a pure rotation. See [3.26a]. Each shaded “rectangle” is mapped by $M(z)$ to the next one in the direction of the arrows—some of these regions have been filled with black to emphasize this. This figure may be viewed as typical, with one exception. Because we have chosen $\alpha = (\pi/3)$, six successive applications of M yield the identity, and one therefore says that M has period 6.

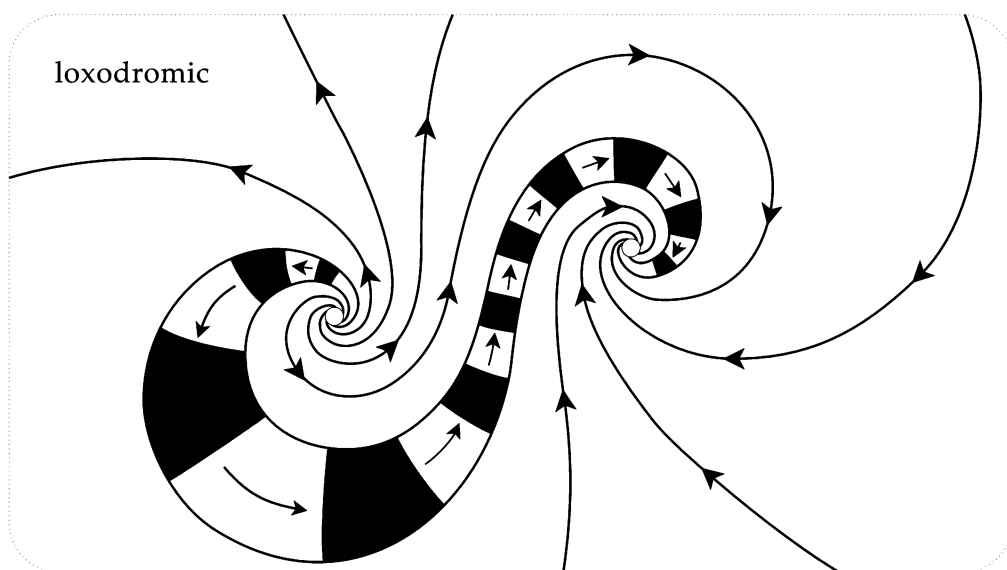
We call $M(z)$ a *hyperbolic Möbius transformation* if $\widetilde{M}$ is hyperbolic, meaning that the latter is a pure expansion corresponding to $m = \rho \neq 1$. Since $\widetilde{M}$ is an expansion if and only if it maps each line through the origin to itself, $M(z)$ is hyperbolic if and only if it maps each $\mathcal{C}_1$ circle to itself. Figure [3.31] illustrates such a transformation with $\rho > 1$. Note that if we repeatedly apply this mapping then any shape (such as the small black square near ξ_+) is repelled away from ξ_+ , eventually being sucked into ξ_- . In this case ξ_+ is called the *repulsive fixed point* and ξ_- is called the *attractive fixed point*; if $m = \rho < 1$ then the roles of ξ_+ and ξ_- are reversed.

Finally, if $m = \rho e^{i\alpha}$ has a general value, and $\widetilde{M}$ is the composition of both a rotation and an expansion, then M is called a *loxodromic Möbius transformation*. In this case neither the $\mathcal{C}_1$ circles, nor the $\mathcal{C}_2$ circles are invariant. The curves that are invariant are illustrated in [3.32], which also shows the effect of successive applications of M to a small square near ξ_+ . In studying this figure, you may find it helpful to note that

The loxodromic Möbius transformation with fixed points $\xi_{\pm}$ and multiplier $m = \rho e^{i\alpha}$ is the composition (in either order) of (i) the elliptic Möbius transformation with multiplier $m = e^{i\alpha}$ and fixed points $\xi_{\pm}$; (ii) the hyperbolic Möbius transformation with multiplier $m = \rho$ and fixed points $\xi_{\pm}$. (3.41)



[3.31] A Hyperbolic Möbius transformation has multiplier $m = \rho \neq 1$, so that $\tilde{M}$ is a pure expansion. See [3.26b]. Iterating the mapping pushes everything away from the repulsive fixed point and sucks everything towards the attractive fixed point.



[3.32] A Loxodromic Möbius transformation has general multiplier $m = \rho e^{i\alpha}$, with $\rho \neq 1$ and $\alpha \neq 0$, so that $\tilde{M}$ expands by ρ and rotates by α . See [3.26c]. Iterating the mapping pushes everything away from the repulsive fixed point and sucks everything towards the attractive fixed point, moving along the illustrated invariant curves.

Just as in the case of a hyperbolic transformation, note that one fixed point is repulsive while the other is attractive. In this figure we have taken $\alpha > 0$ and $\rho > 1$; how would it look if α were negative, or if ρ were less than one?

3.7.3 Local Geometric Interpretation of the Multiplier

In [3.29] we *arbitrarily* elected to send ξ_+ to 0, rather than ξ_- . In this sense our definition of m is clearly ambiguous. How would the new value of m be related to the old one if we were to instead send ξ_- to 0?

Note that (3.40) may be expressed as $(F \circ M) = (\widetilde{M} \circ F)$. Writing $w = M(z)$, and recalling the definition of F , we therefore have

$$\frac{w - \xi_+}{w - \xi_-} = m \left(\frac{z - \xi_+}{z - \xi_-} \right). \quad (3.42)$$

[This formula is often called the *normal form* of the Möbius transformation.] Interchanging ξ_+ and ξ_- in this formula is equivalent to sending ξ_- to 0 and ξ_+ to ∞ , in which case we obtain

$$\frac{w - \xi_+}{w - \xi_-} = \frac{1}{m} \left(\frac{z - \xi_+}{z - \xi_-} \right).$$

Thus the multiplier has changed from m to $(1/m)$, and both of these values can lay equal claim to being called “the” multiplier. Let us therefore refine our language and call the number m occurring in (3.42) the multiplier *associated with* ξ_+ ; we will sometimes write it as m_+ to emphasize this. In these terms, we have just shown that the *multipliers associated with the two fixed points are the reciprocals of one another*. Let us try to understand this more geometrically.

Reconsider [3.29], in which the multiplier associated with ξ_+ is $m = e^{i(\pi/3)}$. We now seek to interpret m directly in terms of [3.30], without the assistance of the RHS of [3.29]. The closer we are to ξ_+ , the more closely do the members of $\mathcal{C}_2$ resemble tiny concentric circles centred at ξ_+ . This is easy to understand: (A) as we examine smaller and smaller neighbourhoods of ξ_+ , the $\mathcal{C}_1$ circles look more and more like their tangent lines at ξ_+ ; (B) by definition, each $\mathcal{C}_2$ cuts every $\mathcal{C}_1$ circle orthogonally.

From these remarks, it is now clear that the *local* effect of M (in an infinitesimal neighbourhood of ξ_+) is a rotation centred at ξ_+ through angle $(\pi/3)$ —this is the meaning of the multiplier $m_+ = e^{i(\pi/3)}$ associated with ξ_+ . Of course exactly the same reasoning applies to the infinitesimal neighbourhood of ξ_- , but we see from [3.30] that the positive rotation at ξ_+ forces an *equal and opposite* rotation at ξ_- . Thus the local effect of M in the neighbourhood of ξ_- is a rotation of $-(\pi/3)$, and the associated multiplier m_- is $e^{-i(\pi/3)} = (1/m_+)$, as was to be explained.

If we look at [3.31], then we can see the same phenomenon at work in the case of a hyperbolic transformation. In this figure the multiplier associated with ξ_+ is $m = \rho > 1$, and this can now be interpreted as saying that the local effect of M in an infinitesimal neighbourhood of ξ_+ is an expansion centred at that point—we will verify in a moment that the “local expansion factor” is precisely ρ . It is also clear from the figure that the local effect of M in an infinitesimal neighbourhood of ξ_- is a *contraction*, so that the multiplier associated with that point is real and

less than one. However, it is not so clear that this number is precisely $(1/\rho)$, as we know it must be. This too can be demonstrated geometrically, but let us instead content ourselves with showing how our original algebraic argument may be re-interpreted geometrically in terms of the “local effect” of M in the vicinity of each of the fixed points.

Let us write $Z = (z - \xi_+)$ and $W = (w - \xi_+)$ for the complex numbers emanating from ξ_+ connecting that point to z and to its image $w = M(z)$. We have claimed (and partially verified) that if Z is *infinitesimal* then the effect of M is to rotate Z by α and to expand it by ρ : in other words, $W = m Z$. To verify this, note that (3.42) can be rewritten as

$$\frac{W}{Z} = m \left(\frac{w - \xi_-}{z - \xi_-} \right).$$

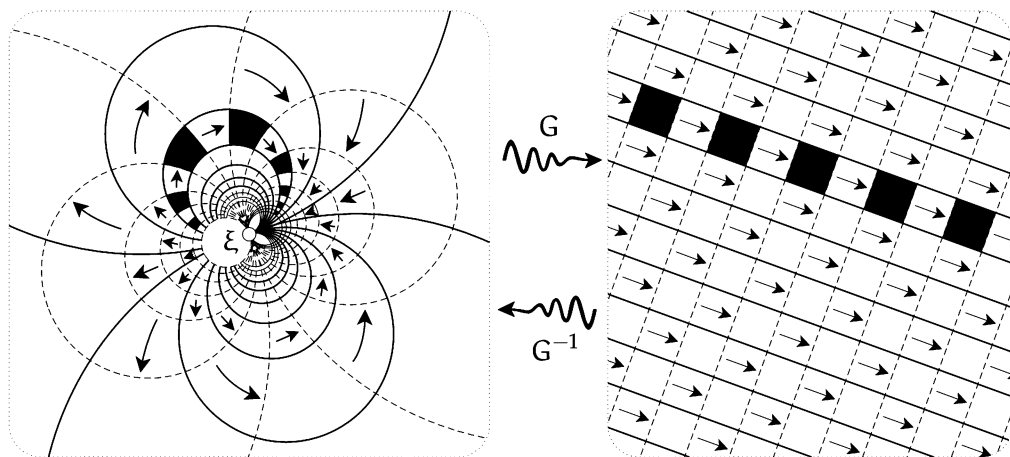
As Z tends to zero, both z and w tend to ξ_+ , and so the fraction on the right is ultimately equal to m . Thus W is ultimately equal to $m Z$, as was to be shown.

After you have read the next chapter, you will be able to look back at what we have just done and recognize it as an example of *differentiating* a complex function.

3.7.4 Parabolic Transformations

We now possess an excellent understanding of Möbius transformations with two fixed points, so all that remains is to treat the case where M has only one fixed point ξ , in which case M is called a *parabolic Möbius transformation*.

Consider the LHS of [3.33], but ignore the arrows for the time being. Here we have drawn two families of circles: the solid ones all pass through the fixed point ξ in one direction, and the dashed ones all pass through ξ in the perpendicular direction. Note that since the two types of circles are orthogonal at ξ , they are also



[3.33] A Parabolic Möbius transformation has a single fixed point, and $\widetilde{M}(\tilde{z}) = \tilde{z} + T$ is a translation, with its single fixed point at ∞ , as illustrated in [3.26d].

(by symmetry) orthogonal at their second intersection point. The RHS illustrates what happens when we send ξ to ∞ by means of the Möbius transformation

$$G(z) = \frac{1}{z - \xi}.$$

Clearly [exercise], the two orthogonal families of circles become two orthogonal families of parallel lines. Conversely, if we apply G^{-1} to any two orthogonal families of lines on the right, then on the left we get two orthogonal families of circles through ξ .

As before, let $\tilde{z} = G(z)$ and $\tilde{w} = G(w)$ be the images on the RHS of z and $w = M(z)$. Thus the Möbius transformation $z \mapsto w = M(z)$ on the LHS induces another Möbius transformation $\tilde{z} \mapsto \tilde{w} = \tilde{M}(\tilde{z})$ on the RHS, where

$$\tilde{M} = G \circ M \circ G^{-1}.$$

Since ∞ is the sole fixed point of $\tilde{M}$, we deduce that $\tilde{M}$ can only be a *translation*:

$$\tilde{M}(\tilde{z}) = \tilde{z} + T.$$

Now suppose that the arrows on the RHS of [3.33] represent the direction of the translation T . As illustrated, we now draw a grid aligned with T , each shaded square being carried into the next by $\tilde{M}$. On the LHS of [3.33] we thus obtain a vivid picture of the action of the original parabolic Möbius transformation M : each solid circle is carried into itself; each dashed circle is carried into another dashed circle; and each shaded region is carried into the next in the direction of the arrows.

If $M(z) = \frac{az+b}{cz+d}$ is *normalized*, then we know from (3.28) that it is parabolic if and only if $(a+d) = \pm 2$, in which case $\xi = (a-d)/2c$. Now let us determine the corresponding translation T in terms of the coefficients. Since $(G \circ M) = (\tilde{M} \circ G)$, the so-called normal form of M is given by

$$\frac{1}{w - \xi} = \frac{1}{z - \xi} + T.$$

Since M maps $z = \infty$ to $w = (a/c)$, we deduce that

$$T = \frac{1}{(a/c) - \xi} = \pm c,$$

where the “ $\pm$ ” is the arbitrarily chosen sign of $(a+d)$.

3.7.5 Computing the Multiplier*

We have seen how the multiplier m determines the character of a Möbius transformation, and we now show how we can determine the character of m directly from the coefficients of $M(z) = \frac{az+b}{cz+d}$.

Suppose we have already calculated the fixed points $\xi_{\pm}$ using (3.28), for example. Since M maps $z = \infty$ to $w = (a/c)$, we deduce from the normal form

(3.42) that the multiplier associated with ξ_+ is

$$m = \frac{a - c \xi_+}{a - c \xi_-}. \quad (3.43)$$

For example, consider complex inversion, $z \mapsto (1/z)$. The fixed points are the solutions of $z = (1/z)$, namely, $\xi_{\pm} = \pm 1$. Thus the multiplier associated with $\xi_+ = 1$ is $m = -1 = e^{i\pi}$, which happens to be the same as the multiplier $(1/m)$ associated with $\xi_- = -1$. Thus complex inversion is elliptic, and an infinitesimal neighbourhood of either fixed point is simply rotated about that point through angle π . Try using a computer to check this prediction.

If desired, we can obtain a completely explicit formula for m by substituting (3.28) into (3.43). If we only want to know the character of the Möbius transformation, then we can proceed as follows.

It turns out—we will prove it in a moment—that m is related to the coefficients of the *normalized* Möbius transformation by the equation,

$$\sqrt{m} + \frac{1}{\sqrt{m}} = a + d. \quad (3.44)$$

Note that the symmetry of this equation implies that if m is a solution, then so is $(1/m)$; this is just as it should be. Without bothering to solve (3.44) for m , we now obtain [exercise] the following algebraic classification: *The normalized Möbius transformation $M(z) = \frac{az+b}{cz+d}$ is*

$$\left. \begin{array}{ll} \text{elliptic,} & \text{iff } (a + d) \text{ is real and } |a + d| < 2; \\ \text{parabolic,} & \text{iff } (a + d) = \pm 2; \\ \text{hyperbolic,} & \text{iff } (a + d) \text{ is real and } |a + d| > 2; \\ \text{loxodromic,} & \text{iff } (a + d) \text{ is complex.} \end{array} \right\} \quad (3.45)$$

Hint: you can get a better feel for this by sketching the graph of $y = x + (1/x)$.

In order to derive (3.44) elegantly, let us use matrices. Rewriting (3.40),

$$[\widetilde{M}] = [F] [M] [F]^{-1} \implies \det[\widetilde{M}] = \det\{[F][F]^{-1}\} \det[M] = \det[M].$$

Thus, regardless of whether or not $[F]$ is normalized, $[M]$ is normalized if and only if $[\widetilde{M}]$ is normalized. Since $\widetilde{M}(z) = mz$, its normalized matrix is [exercise] $[\widetilde{M}] = \begin{bmatrix} \sqrt{m} & 0 \\ 0 & 1/\sqrt{m} \end{bmatrix}$. Recalling (3.36), we deduce that

$$\sqrt{m} + \frac{1}{\sqrt{m}} = \text{tr}\{[F][M][F]^{-1}\} = \text{tr}\{[F]^{-1}[F][M]\} = \text{tr}[M] = a + d,$$

as was to be shown.

3.7.6 Eigenvalue Interpretation of the Multiplier*

If $[M]$ is a linear transformation of $\mathbb{C}^2$, then we saw in (3.33) that its eigenvectors are the homogeneous coordinates of the fixed points of the corresponding

Möbius transformation $M(z)$. We also claimed that if $[M]$ is normalized then the eigenvalues completely determine the character of $M(z)$. We can now be more precise:

If a fixed point of $M(z)$ is represented as an eigenvector (with eigenvalue λ) of the normalized matrix $[M]$, then the multiplier m associated with the fixed point is given by $m = 1/\lambda^2$. (3.46)

Before proving this result, we illustrate it with the example of complex inversion, $z \mapsto (1/z)$. We already know that the fixed points are ± 1 , that the associated multipliers are both given by $m = -1$, and we easily find [exercise] that the normalized matrix is $\begin{bmatrix} 0 & i \\ i & 0 \end{bmatrix}$. If we choose the homogeneous coordinate vector of a finite point z to be $\begin{bmatrix} z \\ 1 \end{bmatrix}$, then the eigenvectors corresponding to the fixed points $z = \pm 1$ are $\begin{bmatrix} \pm 1 \\ 1 \end{bmatrix}$. Since

$$\begin{bmatrix} 0 & i \\ i & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = i \begin{bmatrix} 1 \\ 1 \end{bmatrix} \quad \text{and} \quad \begin{bmatrix} 0 & i \\ i & 0 \end{bmatrix} \begin{bmatrix} -1 \\ 1 \end{bmatrix} = -i \begin{bmatrix} -1 \\ 1 \end{bmatrix},$$

we see that the eigenvalues are given by $\lambda = \pm i$, in agreement with (3.46).

Returning to the general case, comparison of (3.34) and (3.44) reveals that $\sqrt{m}$ and λ satisfy the same quadratic, so we immediately deduce most of (3.46): the two reciprocal values of m are equal to the two reciprocal values of λ^2 . However, this does not tell us which value of λ^2 yields which value of m , nor is this line of attack very illuminating. Here, then, is a more transparent approach.

We begin by recalling a standard result of linear algebra, which is valid for $n \times n$ matrices:

If $\mathbf{e}$ is an eigenvector of $[A]$ with eigenvalue λ , then $\tilde{\mathbf{e}} \equiv [B]\mathbf{e}$ is an eigenvector of $[\tilde{A}] \equiv [B][A][B]^{-1}$, and its eigenvalue is also λ .

This is verified easily:

$$[\tilde{A}]\tilde{\mathbf{e}} = \{[B][A][B]^{-1}\} [B]\mathbf{e} = [B][A]\mathbf{e} = [B]\lambda\mathbf{e} = \lambda\tilde{\mathbf{e}}.$$

Let us return to [3.29], in which the fixed point ξ_+ of M (with associated multiplier m_+) was mapped to the fixed point 0 of $\tilde{M} = (F \circ M \circ F^{-1})$ by means of $z \mapsto \tilde{z} = F(z) = \frac{z - \xi_+}{z - \xi_-}$. In terms of linear transformations of $\mathbb{C}^2$, the eigenvector

$\begin{bmatrix} \xi_+ \\ 1 \end{bmatrix}$ of $[M]$ is being mapped by $[F]$ to the eigenvector $\begin{bmatrix} 0 \\ 1 \end{bmatrix}$ of

$$[\tilde{M}] = [F][M][F]^{-1}.$$

The linear algebra result now tells us that if λ_+ denotes the eigenvalue of $\begin{bmatrix} \xi_+ \\ 1 \end{bmatrix}$, then

$$[\widetilde{M}] \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \lambda_+ \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$

This is true irrespective of whether or not any of the matrices in the above equation are normalized.

Now suppose that $[M]$ is normalized, as demanded in (3.46). Irrespective of whether or not $[F]$ is normalized, we have already noted that $[M]$ is normalized if and only if $[\widetilde{M}]$ is normalized. Since the normalized matrix of $\widetilde{M}(\tilde{z}) = m_+ \tilde{z}$ is given by $[\widetilde{M}] = \begin{bmatrix} \sqrt{m_+} & 0 \\ 0 & 1/\sqrt{m_+} \end{bmatrix}$, we deduce that

$$\lambda_+ \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} \sqrt{m_+} & 0 \\ 0 & 1/\sqrt{m_+} \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \frac{1}{\sqrt{m_+}} \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$

Thus $m_+ = 1/\lambda_+^2$, as was to be shown.

3.8 Decomposition into 2 or 4 Reflections*

3.8.1 Introduction

Recall from (3.4) that the formula for inversion or “reflection” in a circle K has the form

$$\mathcal{J}_K(z) = \frac{A\bar{z} + B}{C\bar{z} + D}.$$

It follows easily that the composition of any two reflections (in circles or lines) is a Möbius transformation. Since the composition of two Möbius transformations is another Möbius transformation, it follows more generally that *the composition of an even number of reflections is a Möbius transformation*.

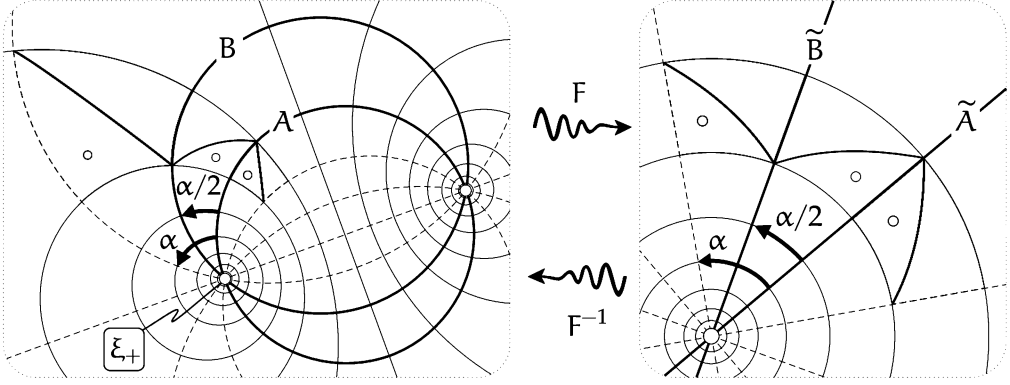
Conversely, in this section we will use the Symmetry Principle [see p. 168] to show that

Every non-loxodromic Möbius transformation can be expressed as the composition of two reflections, and every loxodromic Möbius transformation can be expressed as the composition of four reflections.

In the following, it would be helpful (but not essential) for you to have read the final section of Chapter 1.

3.8.2 Elliptic Case

Consider [3.34], which depicts the same elliptic transformation shown in [3.29] and [3.30]. Recall that the LHS shows a Möbius transformation M such that after



[3.34] Decomposition of an Elliptic Möbius transformation into two reflections. If M is an elliptic Möbius transformation, and the multiplier associated with one of the fixed points ξ_+ is $m = e^{i\alpha}$, then $M = \mathcal{I}_B \circ \mathcal{I}_A$, where A and B are any two circles through the fixed points such that the angle from A to B at ξ_+ is $(\alpha/2)$.

sending ξ_+ and ξ_- to 0 and ∞ by means of $F(z) = (z - \xi_+)/ (z - \xi_-)$, the new transformation on the RHS is a pure rotation $\widetilde{M}(\tilde{z}) = e^{i\alpha} \tilde{z}$. In the illustrated example, $\alpha = (\pi/3)$ and the dark “rectangle” abutting the line $\tilde{A}$ is carried into the dark “rectangle” abutting the line $\tilde{B}$.

As we discussed in Chapter 1 [see p. 42], this origin-centred rotation of α is equivalent to successively reflecting in any two lines containing angle $(\alpha/2)$ at 0, such as the illustrated lines $\tilde{A}$ and $\tilde{B}$. In symbols,

$$\widetilde{M} = \mathfrak{R}_{\tilde{B}} \circ \mathfrak{R}_{\tilde{A}}.$$

In particular, $\mathfrak{R}_{\tilde{A}}$ maps the dark “rectangle” abutting the line $\tilde{A}$ to the light “rectangle”, then $\mathfrak{R}_{\tilde{B}}$ maps this to the dark “rectangle” abutting the line $\tilde{B}$. The figure tries to make this clear by also showing the successive images of both a point and a diagonal circular arc of the original dark “rectangle”.

Now think what this means on the LHS of [3.34]. The Symmetry Principle tells us that if two points are symmetric with respect to the line $\tilde{A}$ then their images under the Möbius transformation F^{-1} are symmetric with respect to the circle $A = F^{-1}(\tilde{A})$ through the fixed points. [Recall that in [3.29] the family of such circles was called $\mathcal{C}_1$.] Thus reflection in $\tilde{A}$ on the RHS becomes reflection (i.e., inversion) in A on the LHS. Of course the same goes for the second reflection in $\tilde{B}$. Thus we have shown the following:

If M is an elliptic Möbius transformation, and the multiplier associated with one of the fixed points ξ_+ is $m = e^{i\alpha}$, then $M = \mathcal{I}_B \circ \mathcal{I}_A$ where A and B are any two circles through the fixed points such that the angle from A to B at ξ_+ is $(\alpha/2)$. (3.47)

3.8.3 Hyperbolic Case

Figure [3.35] (cf. [3.31]) illustrates a similar result in the case of a hyperbolic Möbius transformation. Here the multiplier associated with ξ_+ is a real number $m = \rho$, and the transformation on the RHS is a pure expansion, $\tilde{M}(\tilde{z}) = \rho \tilde{z}$. As with a rotation, an expansion can also be achieved using two reflections: if $\tilde{A}$ and $\tilde{B}$ are any two origin-centred circles such that

$$\frac{r_B}{r_A} = \frac{(\text{radius of } \tilde{B})}{(\text{radius of } \tilde{A})} = \sqrt{\rho}, \quad (3.48)$$

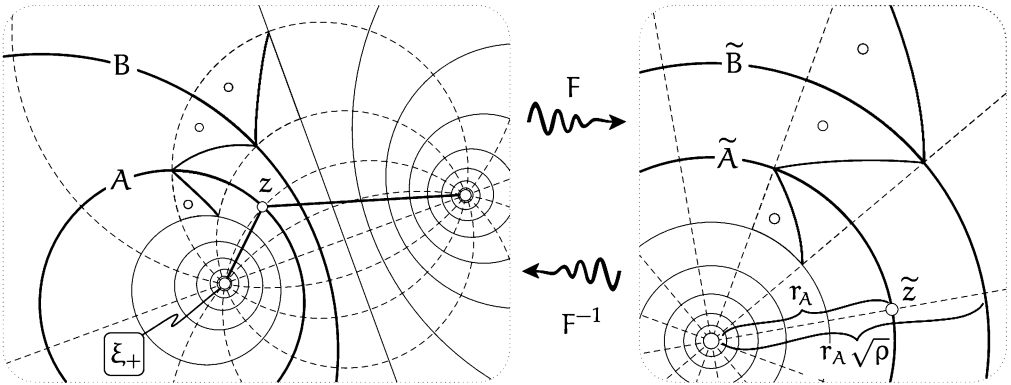
then reflection in $\tilde{A}$ followed by reflection in $\tilde{B}$ yields an origin-centred expansion by ρ . In symbols, this result—which is really the same as (3.8)—says that

$$\tilde{M} = \mathcal{J}_{\tilde{B}} \circ \mathcal{J}_{\tilde{A}}.$$

As in [3.34], the RHS of [3.35] illustrates the successive effect of these two reflections on a dark rectangle abutting $\tilde{A}$. Just as before, the Symmetry Principle applied to F^{-1} tells us that the original Möbius transformation on the LHS can be expressed as

$$M = \mathcal{J}_B \circ \mathcal{J}_A.$$

Recall from [3.29] that A and B belong to the family $\mathcal{C}_2$ of circles orthogonal to the family $\mathcal{C}_1$ of circles through the fixed points. At the time, we pointed out an equivalent property of $\mathcal{C}_2$, namely, that the fixed points $\xi_{\pm}$ are symmetric with respect to each member of $\mathcal{C}_2$; this enables us to explain how it is that $(\mathcal{J}_B \circ \mathcal{J}_A)$ leaves ξ_+ and ξ_- fixed. In the case of [3.34], this was obvious because each reflection separately left those points fixed; in the present case, however, $\mathcal{J}_A$ swaps the points, then $\mathcal{J}_B$ swaps them back again, the net effect being to leave them fixed.



[3.35] Decomposition of a Hyperbolic Möbius transformation into two reflections. If M is a hyperbolic Möbius transformation, and the multiplier associated with one of the fixed points ξ_+ is $m = \rho$, then $M = \mathcal{J}_B \circ \mathcal{J}_A$, where A and B are any two circles of Apollonius with limit points $\xi_{\pm}$ such that $(r_B/r_A) = \sqrt{\rho}$.

In the case of an elliptic transformation, (3.47) describes how to pick out a pair of $\mathcal{C}_1$ circles corresponding to any given angle α . In the present case of a hyperbolic transformation, how are we to pick out a pair of $\mathcal{C}_2$ circles corresponding to any given value of ρ ? The answer depends on a third characterizing property of the $\mathcal{C}_2$ circles: they are the *circles of Apollonius* with *limit points* $\xi_{\pm}$.

This terminology reflects Apollonius' remarkable discovery (c. 250 BCE) that if a point z moves in such a way that the ratio of the distances of z from two fixed points $\xi_{\pm}$ remains constant, then z moves on a *circle*. Figure [3.35] makes this easy to understand. As z travels round A , $\tilde{z} = F(z)$ travels round the origin-centred circle $\tilde{A}$ of radius r_A . But this constant r_A is none other than the ratio of the distances of z from two fixed points $\xi_{\pm}$:

$$r_A = |\tilde{z}| = |F(z)| = \frac{|z - \xi_+|}{|z - \xi_-|}.$$

Note that this also explains the "limit point" terminology: as the ratio r_A tends to 0, the corresponding Apollonian circle A shrinks down towards the limit point ξ_+ ; as r_A tends to infinity, A shrinks down towards the other limit point ξ_- . Another bonus of our discussion is a result that is frequently not mentioned in geometry texts: *the limit points defining a family of Apollonian circles are symmetric with respect to each of these circles*.

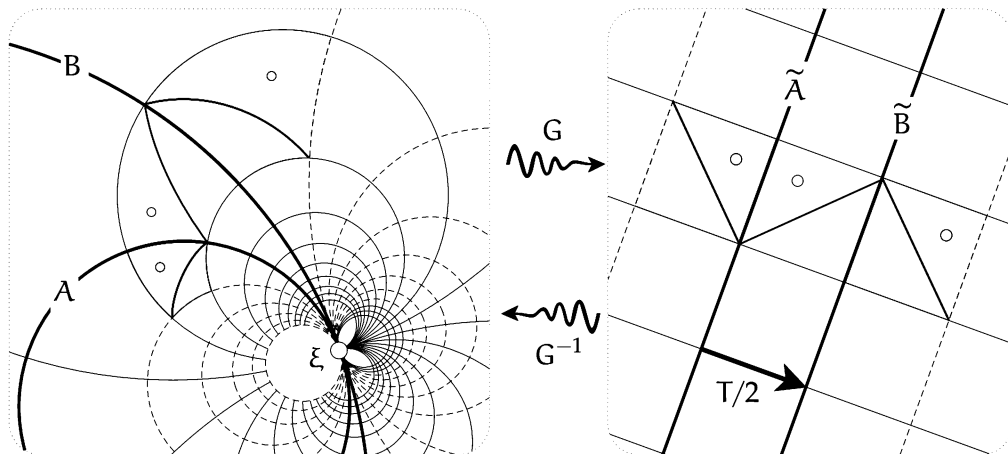
Since the quantities r_A and r_B occurring in (3.48) are now expressible purely in terms of the geometry of the LHS of [3.35], we have solved the problem of picking an appropriate pair of $\mathcal{C}_2$ circles:

If M is a hyperbolic Möbius transformation, and the multiplier associated with one of the fixed points ξ_+ is $m = \rho$, then $M = J_B \circ J_A$, where A and B are any two circles of Apollonius with limit points $\xi_{\pm}$ such that $(r_B/r_A) = \sqrt{\rho}$.

3.8.4 Parabolic Case

Figure [3.36] is a modified copy of [3.33], and it illustrates how the same idea applies to a parabolic transformation. Recall that after we have sent the solitary fixed point ξ to ∞ by means of the Möbius transformation $z \mapsto \tilde{z} = G(z) = 1/(z - \xi)$, the new transformation on the RHS is a translation, $\tilde{M}(\tilde{z}) = \tilde{z} + T$.

As we discussed on p. 42, this translation can be expressed as $\tilde{M} = \mathfrak{R}_{\tilde{B}} \circ \mathfrak{R}_{\tilde{A}}$, where $\tilde{A}$ and $\tilde{B}$ are any two parallel lines such that the perpendicular connecting complex number from $\tilde{A}$ to $\tilde{B}$ is $(T/2)$. Applying the Symmetry Principle to the Möbius transformation G^{-1} , we deduce that (on the LHS)



[3.36] Decomposition of a Parabolic Möbius transformation into two reflections. A parabolic Möbius transformation M with fixed point ξ can be expressed as $M = J_B \circ J_A$, where A and B are circles that touch each other at ξ .

A parabolic Möbius transformation M with fixed point ξ can be expressed as $M = J_B \circ J_A$, where A and B are circles that touch each other at ξ .

3.8.5 Summary

Lest the details obscure the simplicity of what we have discovered, we summarize our results as follows:

A non-loxodromic Möbius transformation M can always be decomposed into two reflections in circles A and B that are orthogonal to the invariant circles of M . Furthermore, M is elliptic, parabolic, or hyperbolic according as A and B intersect, touch, or do not intersect. (3.49)

Recalling (3.41), we also deduce that a loxodromic Möbius transformation M can always be decomposed into four reflections in circles:

$$M = \{J_{B'} \circ J_{A'}\} \circ \{J_B \circ J_A\} = \{J_B \circ J_A\} \circ \{J_{B'} \circ J_{A'}\},$$

where A and B both pass through the fixed points, and where A' and B' are both orthogonal to A and B .

We should stress that these results concern the *least* number of reflections into which a Möbius transformation can be decomposed. Thus if a particular Möbius transformation is expressible as the composition of four reflections, this does *not* necessarily imply that it is loxodromic—one might be able to reduce the number of reflections from four to two. For example, if A and B are lines containing angle $(\pi/12)$ at 0 , and A' and B' are lines containing angle $(\pi/6)$ at 0 , then the

Möbius transformation $(\mathfrak{R}_{B'} \circ \mathfrak{R}_{A'} \circ \mathfrak{R}_B \circ \mathfrak{R}_A)$ represents a rotation of $(\pi/2)$, which can be reduced to two reflections in lines containing angle $(\pi/4)$. As a more extreme example of this idea of redundant reflections, check for yourself that (3.3) represents a decomposition of a general Möbius transformation into *ten* reflections!

3.9 Automorphisms of the Unit Disc*

3.9.1 Counting Degrees of Freedom

An *automorphism* of a region R of the complex plane is a one-to-one, conformal mapping of R to itself. If R is a disc (or a half-plane) then clearly we can map it to itself with a Möbius transformation M , and since M is one-to-one and conformal, it is (by definition) an automorphism. In this subsection we will find all possible Möbius automorphisms of the unit disc. These Möbius transformations are important for at least two reasons: (i) in Chapter 6 we will see that they play a central role in non-Euclidean geometry; (ii) in Chapter 7 we will see that they are the *only* automorphisms of the disc!

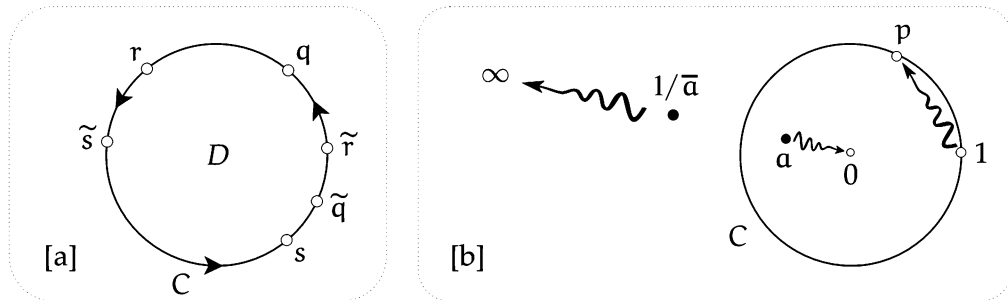
In the following, let C denote the unit circle, let D denote the unit disc (including C), and let $M(z)$ denote a Möbius transformation of D to itself. Before we try to find a formula for the most general M , let us see “how many” such Möbius transformations there are. In other words, how many real numbers (*parameters*) are required to specify a particular M ?

To illustrate how such counting may be done, let us first show that the set of *all* Möbius transformations forms a “six parameter family”. Once we have chosen three points in $\mathbb{C}$, there is a unique Möbius transformation that maps them to three arbitrary image points, and each of these 3 image points $w = u + iv$ requires 2 real numbers (u and v) for its specification. If we think of the three original points as having fixed locations, and the three image points as freely movable, then the total number of parameters needed to specify a particular Möbius transformation is thus $3 \times 2 = 6$. Another suggestive way of describing this fact is to say that the most general Möbius transformation has six *degrees of freedom*.

Returning to the original problem, it is clear that we will lose some of these six degrees of freedom when we impose the condition that $M(z)$ map D to itself. In fact we lose half of them:

$$\text{Möbius automorphisms of } D \text{ have three degrees of freedom.} \quad (3.50)$$

Figure [3.37a] gives one way of seeing this. Here q, r, s may be viewed as having fixed locations on C , while $\tilde{q}, \tilde{r}, \tilde{s}$ are thought of as freely movable. Provided (as illustrated) that $\tilde{q}, \tilde{r}, \tilde{s}$ induce the same orientation of C as q, r, s , we know from



[3.37] [a] Möbius automorphisms of D have three real degrees of freedom. Think of q, r, s as fixed on C , but $\tilde{q}, \tilde{r}, \tilde{s}$ as freely movable, their locations specified by three angles. Then we know from (3.31) that the unique Möbius automorphism of D mapping q, r, s to $\tilde{q}, \tilde{r}, \tilde{s}$ is given by $z \mapsto \tilde{z} = M(z)$, where $[\tilde{z}, \tilde{q}, \tilde{r}, \tilde{s}] = [z, q, r, s]$.

[b] Finding the most general Möbius automorphism via the Symmetry Principle. If a Möbius automorphism of D sends a to 0 , then the Symmetry Principle tells us that it sends $(1/\bar{a})$ to ∞ . It then follows easily [see text] that the most general Möbius automorphism of D is $M_a^\phi(z) = e^{i\phi} \left[\frac{z-a}{\bar{a}z-1} \right]$, which does indeed have three real degrees of freedom.

(3.31) that the unique Möbius automorphism of D mapping q, r, s to $\tilde{q}, \tilde{r}, \tilde{s}$ is given by $z \mapsto \tilde{z} = M(z)$, where

$$[\tilde{z}, \tilde{q}, \tilde{r}, \tilde{s}] = [z, q, r, s].$$

Since *three* real numbers are needed to specify $\tilde{q}, \tilde{r}, \tilde{s}$ —their angles, for example—this establishes (3.50).

3.9.2 Finding the Formula via the Symmetry Principle

According to (3.50), the specification of a particular M requires three bits of information. However, we are not obliged to give this information in the form of three points on C —any data that are equivalent to three real numbers will do equally well. A particularly useful alternative of this kind is shown in [3.37b]. We specify which point a inside D is to be mapped to the origin, and we also specify which point p on C is to be the image of the point 1 (or of some other definite point on C). Choosing a uses up two degrees of freedom; choosing p uses up the third and last degree of freedom.

Before pursuing this, we note another consequence of (3.50): we cannot generally find a Möbius automorphism that simultaneously sends the interior point a to 0 and sends another interior point to some other interior point. These requirements amount to *four* conditions on M , while (3.50) tells us that only *three* such conditions can be accommodated. It is very much as if we were seeking to draw a circle through four arbitrary points—it can't be done! However, suppose in this analogy that we are lucky, and that the four points just happen to be concyclic, then the

circle that passes through them is *unique*. By the same token,

If two Möbius automorphisms M and N map two interior points to the same image points, then $M = N$. (3.51)

Returning to [3.37b], note that since C is mapped to itself by M , the Symmetry Principle tells us that if a pair of points are symmetric with respect to C , then so are their images. Now we apply this to the symmetric pair of points, a and $(1/\bar{a})$ shown in [3.37b]. Since a is mapped to 0, $(1/\bar{a})$ must be mapped to the reflection of 0 in C , namely, ∞ . Thus M must have the form

$$M(z) = k \left(\frac{z - a}{\bar{a}z - 1} \right),$$

where k is a constant. Finally, we require that $p = M(1)$ be a point on C , so

$$1 = |p| = |k| \frac{|1 - a|}{|\bar{a} - 1|} = |k| \implies k = e^{i\phi}.$$

Thus the choice of p is equivalent to the choice of ϕ . Using the angle ϕ and the point a to label the transformation, we have discovered that the most general Möbius automorphism of D is

$$M_a^\phi(z) = e^{i\phi} \left(\frac{z - a}{\bar{a}z - 1} \right). \quad (3.52)$$

Note that $M_0^\phi(z) = -e^{i\phi}z = e^{i(\pi+\phi)}z$ simply rotates D about its centre 0 through angle $(\pi + \phi)$. The general Möbius automorphism M_a^ϕ may be interpreted as M_a^0 followed by a rotation of ϕ , and from this point of view the really interesting part of the transformation is M_a^0 , which we will now abbreviate to M_a . This is the same M_a whose properties you were asked to investigate algebraically in Chapter 2, Ex. 3.

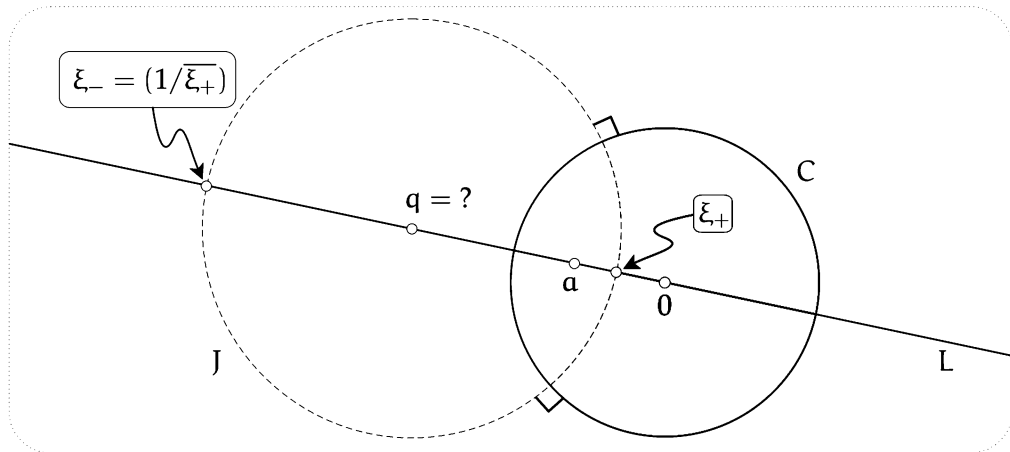
3.9.3 Interpreting the Simplest Formula Geometrically*

To find the geometric meaning of

$$M_a(z) = \frac{z - a}{\bar{a}z - 1}, \quad (3.53)$$

we could simply apply our whole arsenal of classification techniques. We ask that you try this yourself in Ex. 26.

Here we will instead attempt to make sense of M_a “with our bare hands”, as it were. This is probably more illuminating, and it certainly provides better geometric sport! Begin by noting that M_a has the property that it *swaps* a and 0: not only is $M_a(a) = 0$, but also $M_a(0) = a$. According to (3.51), this is the *only* Möbius automorphism with this property, so if we can geometrically construct a Möbius automorphism that swaps a and 0, then it must be M_a .



[3.38] Geometric meaning of the most general Möbius automorphism. *The most general Möbius automorphism $M_a^\Phi(z)$ (above) is the composition of a rotation with the fundamental mapping $M_a(z) = \frac{z-a}{\bar{a}z-1}$. To discover the geometric meaning of $M_a(z)$, we first observe that M_a swaps a and 0 . The Symmetry Principle then guides us [see text] to the answer: $M_a = \mathfrak{R}_L \circ \mathcal{I}_J$, the order of the reflections being immaterial. The fact that M_a swaps a and 0 can now be recognized as a special case of the fact that M_a is involutory: $(M_a \circ M_a) = \mathcal{E}$, and every pair of points $\{z, M_a(z)\}$ is swapped by M_a .*

As was explained earlier in [3.6] on page 145, the reflection $\mathcal{I}_J$ in any circle J orthogonal to C will map D to itself, the two regions into which D is divided by J being swapped. See [3.38]. At this point the obvious thing to do is to find the circle J such that $\mathcal{I}_J$ swaps a and 0 . Clearly the centre q of J must lie on the line L through a and 0 , but where?

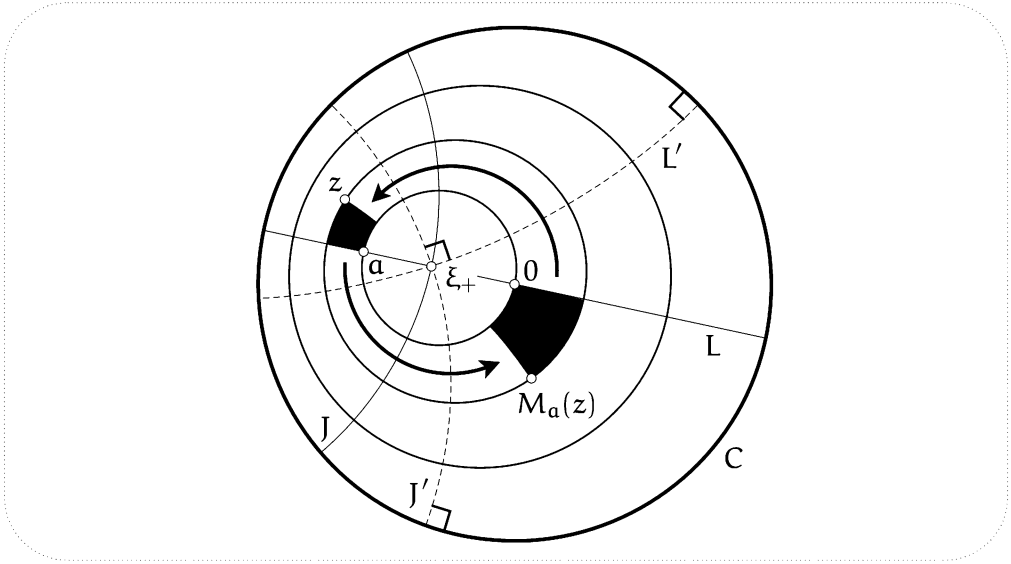
We can answer this question with the same symmetry argument that we used earlier. Since a and $(1/\bar{a})$ are symmetric with respect to C , their images under $\mathcal{I}_J$ are symmetric with respect to $\mathcal{I}_J(C) = C$. Because we want $\mathcal{I}_J(a) = 0$, we deduce that $\mathcal{I}_J(1/\bar{a}) = \infty$. But the point that is mapped to infinity by $\mathcal{I}_J$ is the centre of J , so $q = (1/\bar{a})$.

Of course $\mathcal{I}_J$ is an anticonformal mapping; to obtain a conformal Möbius automorphism we must compose it with another reflection. However, we have already successfully swapped a and 0 , so this second reflection must leave these points fixed. The obvious (and only) choice is thus reflection in L . Here, then, is our geometric interpretation of M_a :

$$M_a = \mathfrak{R}_L \circ \mathcal{I}_J.$$

Incidentally, observe [exercise] that the order of these reflections doesn't matter: we may also write $M_a = \mathcal{I}_J \circ \mathfrak{R}_L$.

Clearly the fixed points $\xi_\pm$ are the intersection points of J and L , and so they are symmetric with respect to C . Since the reflections occur in orthogonal circles



[3.39] The *true* meaning of M_a awaits in Chapter 6: it is a half-turn of the hyperbolic plane about ξ_+ . We can also express M_a as $(J_{L'} \circ J_{J'})$, where J' and L' are any two orthogonal circles through ξ_+ that are orthogonal to C . Here we illustrate some of the invariant circles, together with the effect of M_a on a “square”. The deeper meaning of all this will become clear once we have encountered hyperbolic geometry in Chapter 6. We will then recognize that for inhabitants of the hyperbolic plane, M_a is nothing more than an ordinary rotation of the plane about ξ_+ through angle π ! See Section 6.3.11.

through these points, M_a is elliptic, and the multipliers associated with $\xi_{\pm}$ are both given by $m = e^{i\pi} = -1$. Thus the effect of M_a on an infinitesimal neighbourhood of the interior fixed point ξ_+ is a rotation of π . The fact that M_a swaps a and 0 can now be recognized as a special case of the fact that M_a is *involutory*: $(M_a \circ M_a) = \mathcal{E}$, and *every* pair of points $z, M_a(z)$ is swapped by M_a . Finally, note that we can also express M_a as $(J_{L'} \circ J_{J'})$, where J' and L' are any two orthogonal circles through ξ_+ that are orthogonal to C . All this is illustrated in [3.39], which also shows some of the invariant circles, together with the effect of M_a on a “square”.

The deeper meaning of all this will become clear once we have encountered Hyperbolic Geometry in Chapter 6. We will then recognize that for inhabitants of the hyperbolic plane, M_a is *nothing more than an ordinary rotation of the plane about ξ_+ through angle π* ! See Section 6.3.11.

While we will return to the geometry of the general Möbius automorphisms M_a^ϕ in Chapter 6, we remark here that they can only be elliptic, parabolic, or hyperbolic. This is because (by construction) they leave C invariant, while a loxodromic Möbius transformation has no invariant circles. To be more precise, in Chapter 6 we will use the above interpretation of M_a to show geometrically that

If we define $\Phi \equiv 2 \cos^{-1} |a|$, then M_a^Φ is

- (i) elliptic if $|\phi| < \Phi$,
 - (ii) parabolic if $|\phi| = \Phi$,
 - (iii) hyperbolic if $|\phi| > \Phi$.
- (3.54)

For an algebraic proof, see Ex. 27.

3.9.4 Introduction to Riemann's Mapping Theorem

Riemann's doctoral thesis of 1851 contained many profound new results, one of the most famous being the following, which is now called *Riemann's Mapping Theorem*:

Any simply connected region R (other than the entire plane) may be mapped one-to-one and conformally to any other such region S .

(3.55)

In Chapter 12 we shall discuss this in detail, but for the time being we merely wish to point out some connections between Riemann's result and what we have learnt concerning automorphisms of the disc.

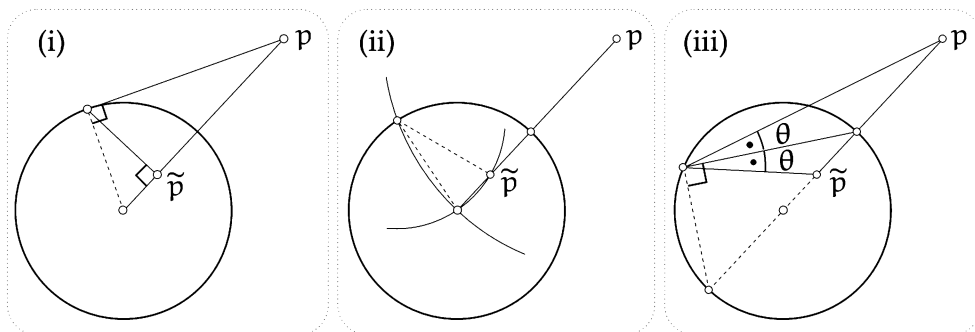
First note that to establish (3.55) in general, it is sufficient to establish it in the special case that S is the unit disc D . For if F_R is a one-to-one conformal mapping from R to D , and F_S is likewise a one-to-one conformal mapping of S to D , then $F_S^{-1} \circ F_R$ is a one-to-one conformal mapping of R to S , as required.

If M is an arbitrary automorphism of D , then $M \circ F_R$ is clearly another one-to-one conformal mapping from R to D . In fact *every* such mapping must be of this form. For if $\tilde{F}_R$ were any other such mapping, then $\tilde{F}_R \circ F_R^{-1}$ would be some automorphism M of D , in which case $\tilde{F}_R = M \circ F_R$.

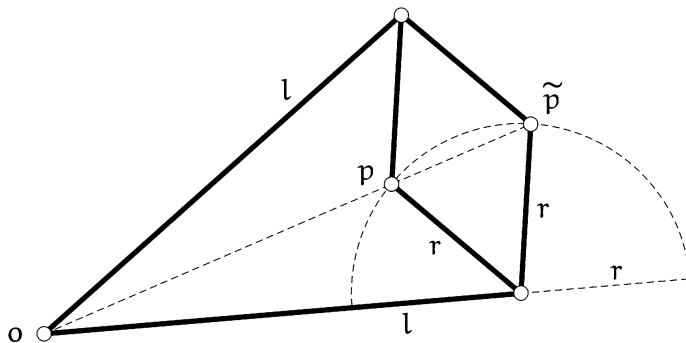
Thus the number of one-to-one conformal mappings from R to S is equal to the number from R to D , which in turn is equal to the number of automorphisms of D . As we have already said, in Chapter 7 we will show that these automorphisms are the Möbius transformations M_a^Φ , which form a 3-parameter family. Thus (3.55) in fact implies that *the one-to-one conformal mappings from R to S form a three-parameter family*.

3.10 Exercises

- 1 In each of the figures below, show that p and $\tilde{p}$ are symmetric with respect to the circle. The dashed lines are not strictly part of the constructions, rather they are intended to be helpful or suggestive.



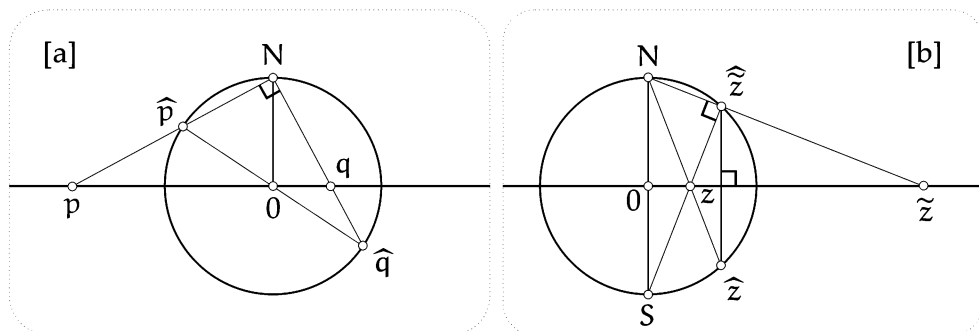
- 2 In 1864 a French officer named Peaucellier caused a sensation by discovering a simple mechanism (*Peaucellier's linkage*) for transforming linear motion (say of a piston) into circular motion (say of a wheel). The figure below shows six rods hinged at the white dots, and anchored at o . Two of the rods have length l , and the other four have length r . With the assistance of the dashed circle, show that $\tilde{p} = \mathcal{I}_K(p)$, where K is the circle of radius $\sqrt{l^2 - r^2}$ centred at o . Construct this mechanism—perhaps using strips of fairly stiff cardboard for rods, and drawing pins for hinges—and use it to verify properties of inversion. In particular, try moving p along a line.



- 3 Let S be a sphere, and let p be a point not on S . Explain why $\mathcal{I}_S(p)$ may be constructed as the second intersection point of any three spheres that pass through p and are orthogonal to S . Explain the preservation of three-dimensional symmetry in terms of this construction.
- 4 Deduce (3.23), p. 167 directly from (3.18), p. 162.
- 5 Consider the following two-stage mapping: first stereographically project $\mathbb{C}$ onto the Riemann sphere Σ in the usual way; now stereographically project Σ

back to $\mathbb{C}$, but from the south pole instead of the north pole. The net effect of this is some complex mapping $z \mapsto f(z)$ of $\mathbb{C}$ to itself. What is f ?

- 6 Both figures below show vertical cross sections of the Riemann sphere.
- In figure [a], show that the triangles $p0N$ and $N0q$ are similar. Deduce (3.23).
 - Figure [b] is a modified copy of [3.21b]. Show that the triangles $z0N$ and $N0\tilde{z}$ are similar. Deduce (3.18).



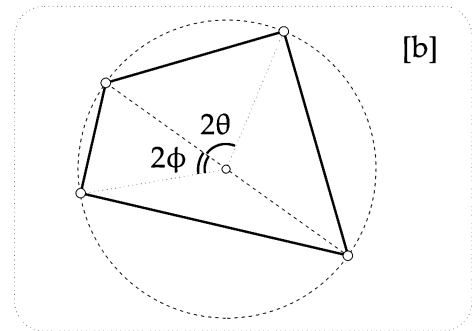
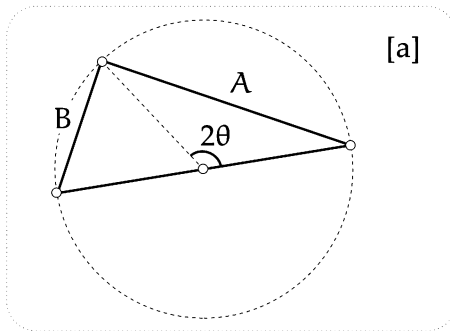
- 7
- Use a computer to draw the images in $\mathbb{C}$ of several origin-centred circles under the exponential mapping, $z \mapsto e^z$. Explain the obvious symmetry of these image curves with respect to the real axis.
 - Now use the computer to draw these same image curves on the Riemann sphere, instead of in $\mathbb{C}$. Note the surprising new symmetry!
 - Use (3.19) to explain this extra symmetry.
- 8 This exercise continues the discussion of (3.2), p. 138. If a point p in space emits a flash of light, we claimed that each of the light rays could be represented by a *complex number*. Here is one, indirect method of establishing this correspondence. Once again, we choose units of space and time so that the speed of light is 1. After one unit of time, the expanding sphere of light emitted by p —made up of particles of light called *photons*—forms a unit sphere. Thus each photon may be identified with a point on the Riemann sphere, and hence, via stereographic projection, with a complex number. Indeed, if the photon has spherical polar coordinates (ϕ, θ) , then (3.22) tells us that the corresponding complex number is $z = \cot(\phi/2) e^{i\theta}$.

Sir Roger Penrose (see Penrose and Rindler (1984, p. 13)) discovered the following remarkable method of passing from a light ray to the associated complex number *directly*, without the assistance of the Riemann sphere. Imagine that p is one unit vertically above the origin of the (horizontal) complex plane. At the instant that p emits its flash, let $\mathbb{C}$ begin to travel straight up (in the direction $\phi = 0$) at the speed of light ($= 1$) towards p . Decompose the velocity of the

photon F emitted by p in the direction (ϕ, θ) into components perpendicular and parallel to $\mathbb{C}$. Hence find the time at which F hits $\mathbb{C}$. Deduce that F hits $\mathbb{C}$ at the point $z = \cot(\phi/2) e^{i\theta}$. Amazingly, we see that *Penrose's construction is equivalent to stereographic projection!*

- 9 In order to analyse astronomical data, Ptolemy required accurate trigonometric tables, which he constructed using the addition formulae for sine and cosine. The figures below explain how he discovered these key addition formulae. Both the circles have unit radius.

- (i) In figure [a], show that $A = 2 \sin \theta$ and $B = 2 \cos \theta$.
 (ii) In figure [b], apply Ptolemy's Theorem to the illustrated quadrilateral, and deduce that $\sin(\theta + \phi) = \sin \theta \cos \phi + \sin \phi \cos \theta$.



- 10 The aim of this question is to understand the following result:

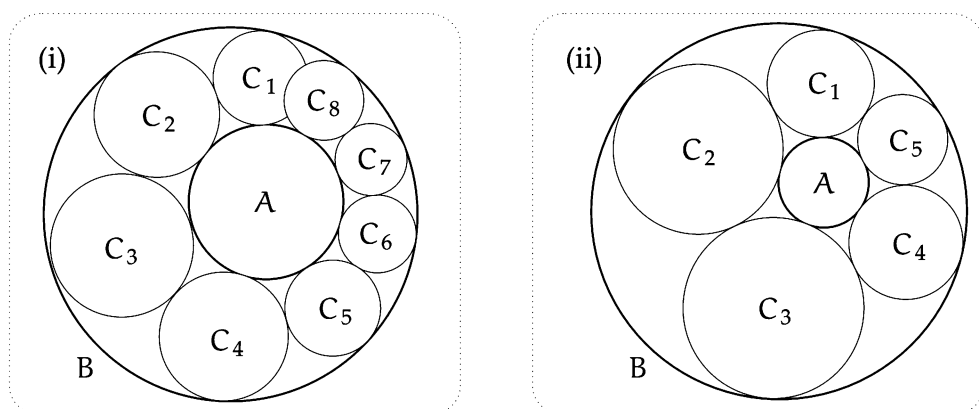
Any two non-intersecting, non-concentric circles can be mapped to concentric circles by means of a suitable Möbius transformation.

- (i) If A and B are the two circles in question, show that there exists a pair of points $\xi_{\pm}$ that are symmetric with respect to both A and B .
 (ii) Deduce that if $F(z) = (z - \xi_+)/ (z - \xi_-)$, then $F(A)$ and $F(B)$ are concentric circles, as was desired.
- 11 This exercise yields a more intuitive proof of the result of the previous exercise. Using different colours for each, draw two non-intersecting, non-concentric circles, A and B , then draw the line L through their centres. Label as p and q the intersection points of B with L .
- (i) Using corresponding colours, draw a fresh picture showing the images $\tilde{A}$, $\tilde{B}$, $\tilde{L}$, $\tilde{q}$ of A , B , L , q under inversion in any circle centred at p . To get you started, note that $\tilde{L} = L$.
 (ii) Now add to your figure by drawing the circle K , centred at $\tilde{q}$, that cuts $\tilde{A}$ at right angles, and let g and h be the intersection points of K and L .

- (iii) Now draw a new picture showing the images K', L', h' of K, L, h under inversion in any circle centred at g .
- (iv) By appealing to the anticonformal nature of inversion, deduce that $\tilde{A}', \tilde{B}'$ are concentric circles centred at h' .

Since the composition of two inversions is a Möbius transformation, you have proved the result of the previous exercise.

- 12 Figure (i) below shows two non-intersecting, non-concentric circles A and B , together with a chain of circles $C_1, C_2, \dots$ that touch one another successively, and that all touch A and B . As you would expect, the chain fails to “close up”: C_8 overlaps C_1 instead of touching it. Figure (ii) shows that this failure to close is not inevitable. Given a different pair A, B , it is possible to obtain a closed chain where C_n touches C_1 . Here $n = 5$, but by considering the case where A, B are concentric, you can easily see that any value of n is possible, given the right A and B .

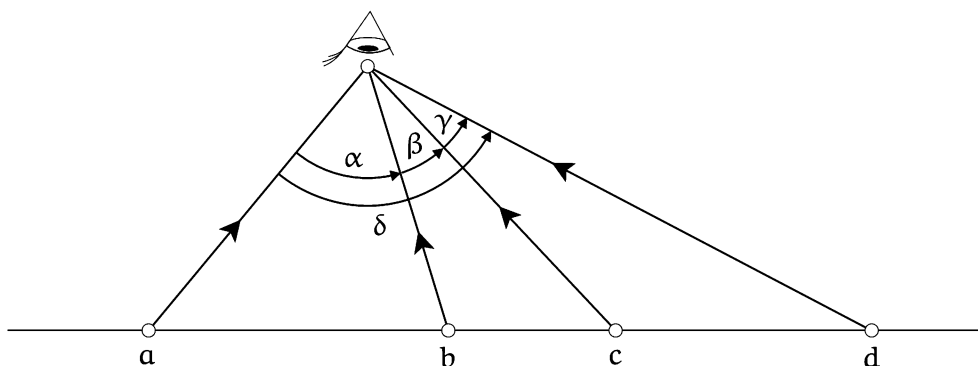


Steiner discovered, very surprisingly, that if the chain closes for one choice of C_1 , then it closes for *every* choice of C_1 , and the resulting chain always contains the same number of touching circles. Explain this using the result of Ex. 10.

- 13 (i) Let P be a sphere resting on the flat surface Q of a table. Let $S_1, S_2, \dots$ be a string of spheres touching one another successively and all the same size as P . If each S -sphere touches both P and Q , show that S_6 touches S_1 , so that we have a closed “necklace” of six spheres around P .
- (ii) Let A, B, C be three spheres (not necessarily of equal size) all touching one another. As in the previous part, let $S_1, S_2, \dots$ be a string of spheres (now of unequal size) touching one another successively, and all touching A, B, C . Astonishingly (cf. previous exercise), S_6 will always touch S_1 , forming a closed “necklace” of six spheres interlocked with A, B, C . Prove this by first applying an inversion centred at the point of contact of A and B , then appealing to part (i).

The chain of six spheres in part (ii) is called *Soddy's Hexlet*, after the amateur mathematician Frederick Soddy—fellow graduate of Merton College, Oxford!—who discovered it (*without inversion!*). For further information on Soddy's Hexlet, see [Ogilvy \(1990\)](#). Soddy's full time job was chemistry—in 1921 he won the Nobel Prize for his discovery of isotopes!

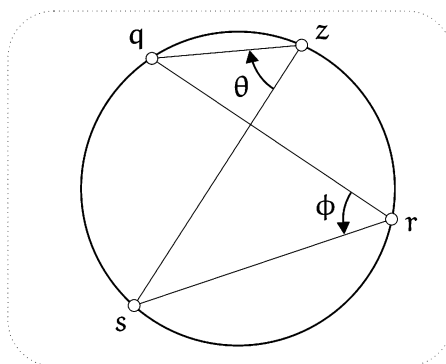
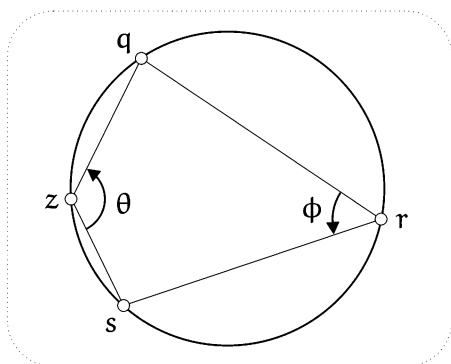
- 14 The figure below shows four collinear points a, b, c, d , together with the (necessarily coplanar) light rays from those points to an observer. Imagine that the collinear points lie in the complex plane, and that the observer is above the plane looking down. Show that the cross-ratio $[a, b, c, d]$ can be expressed purely in terms of the directions of these light rays; more precisely, show that



$$[a, b, c, d] = -\frac{\sin \alpha \sin \gamma}{\sin \beta \sin \delta}.$$

Suppose the observer now does a *perspective drawing* on a glass “canvas plane” C (arbitrarily positioned between himself and $\mathbb{C}$). That is, for each point p in $\mathbb{C}$ he draws a point p' where the light ray from p to his eye hits C . Use the above result to show that although angles and distances are both distorted in his drawing, *cross-ratios of collinear points are preserved*: $[a', b', c', d'] = [a, b, c, d]$.

- 15 Show that in both of the figures below, $\text{Arg } [z, q, r, s] = \theta + \phi$. Hence deduce (3.32), p. 156.



16 As in figure [3.28], think of the cross-ratio $[z, q, r, s]$ as a Möbius transformation.

- (i) Explain geometrically why permuting q, r, s , in $[z, q, r, s]$ yields six different Möbius transformations.
- (ii) If $I(z)$ is the Möbius transformation that leaves 1 fixed and that swaps 0 with ∞ , explain geometrically why $I \circ [z, q, r, s] = [z, s, r, q]$.
- (iii) If $J(z)$ is the Möbius transformation that sends 0, 1, ∞ to 1, ∞ , 0, respectively, explain geometrically why $J \circ [z, q, r, s] = [z, s, q, r]$.
- (iv) Employing the abbreviation $\chi \equiv [z, q, r, s]$, explain why the six Möbius transformations in part (i) can be expressed as

$$\chi, \quad I \circ \chi, \quad J \circ \chi, \quad I \circ J \circ \chi, \quad J \circ I \circ \chi, \quad I \circ J \circ I \circ \chi.$$

(v) Show that $I(z) = (1/z)$ and $J(z) = 1/(1 - z)$.

(vi) Deduce that the six possible values of the cross-ratio are

$$\chi, \quad \frac{1}{\chi}, \quad \frac{1}{1-\chi}, \quad 1-\chi, \quad \frac{\chi}{\chi-1}, \quad \frac{\chi-1}{\chi}.$$

17 Show geometrically that if a and c lie on a circle K , and b and d are symmetric with respect to K , then the point $[a, b, c, d]$ lies on the unit circle. [Hints: Draw the two circles through a, b, d and through b, c, d . Now think of $[z, b, c, d]$ as a Möbius transformation.]

18 The *curvature* κ of a circle is defined to be the reciprocal of its radius. Let $M(z) = \frac{az+b}{cz+d}$ be normalized. Use (3.3) to show geometrically that M maps the real line to a circle of curvature

$$\kappa = \left| 2c^2 \operatorname{Im} \left(\frac{d}{c} \right) \right|.$$

19 Let $M(z) = \frac{az+b}{cz+d}$ be normalized.

- (i) Using (3.3), draw diagrams to illustrate the successive effects of these transformations on a family of *concentric* circles. Note that the image circles are generally *not* concentric.
- (ii) Deduce that the image circles *are* concentric if and only if the original family of circles are centred at $q = -(d/c)$. Write down the centre of the image circles in this case. [Note that this is not the image of the centre of the original circles: $M(q)$ is the point at infinity!]
- (iii) Hence show geometrically that the circle I_M with equation $|cz + d| = 1$ is mapped by M to a circle of equal size. Furthermore, show that *each arc* of I_M is mapped to an image arc of equal size. For this reason, I_M is called the *isometric circle* of M .

For applications of the isometric circle, see Ford (1929) and Katok (1992).

- 20 (i) Show that every Möbius transformation of the form

$$M(z) = \frac{pz + q}{\bar{q}z + \bar{p}} \quad \text{where } |p| > |q|$$

can be rewritten in the form

$$M(z) = e^{i\theta} \left(\frac{z - a}{\bar{a}z - 1} \right), \quad \text{where } |a| < 1.$$

[Notice that the converse is also true. In other words, the two sets of functions are the same.]

- (ii) Use the matrix representation of the first equation to show that this set of Möbius transformations forms a group under the operation of composition.
- (iii) Use the disc-automorphism interpretation of these transformations to give a geometric explanation of the fact that they form a group.
- 21 (i) Use the matrix representation to show algebraically that the set of Möbius transformations

$$R(z) = \frac{az + b}{-\bar{b}z + \bar{a}} \quad \text{with} \quad |a|^2 + |b|^2 = 1$$

forms a group under the operation of composition.

- (ii) Using the interpretation of these functions given on page 184, explain part (i) geometrically.
- 22 Let H be the rectangular hyperbola with Cartesian equation $x^2 - y^2 = 1$. Show that $z \mapsto w = z^2$ maps H to the line $\operatorname{Re}(w) = 1$. What is the image of this line under complex inversion, $w \mapsto (1/w)$? Referring back to figure [2.9], p. 69, deduce that complex inversion maps H to a *lemniscate*!
- [Hint: Think of complex inversion as $z \mapsto \sqrt{(1/z^2)}$.]
- 23 From the simple fact that $z \mapsto (1/z)$ is involutory, deduce that it is elliptic, with multiplier -1 .
- 24 (i) Use the Symmetry Principle to show that the most general Möbius transformation of the upper half-plane to the unit disc has the form

$$M(z) = e^{i\theta} \left(\frac{z - a}{z - \bar{a}} \right),$$

where $\operatorname{Im} a > 0$.

- (ii) The most general Möbius transformation back *from* the unit disc *to* the upper half-plane will therefore be the inverse of $M(z)$. Let's call this inverse $N(z)$. Use the matrix form of M to show that

$$N(z) = M^{-1}(z) = \frac{\bar{a}z - a e^{i\theta}}{z - e^{i\theta}}.$$

- (iii) Explain why the Symmetry Principle implies that $N(1/\bar{z}) = \overline{N(z)}$.
 - (iv) Show by direct calculation that the formula for N in part (ii) does indeed satisfy the equation in part (iii).
- 25** Let $M(z)$ be the general Möbius automorphism of the upper half-plane.
- (i) Observing that M maps the real axis into itself, use (3.30) to show that the coefficients of M are *real*.
 - (ii) By considering $\text{Im}[M(i)]$, deduce that the only restriction on these real coefficients is that they have positive determinant: $(ad - bc) > 0$.
 - (iii) Explain (both algebraically and geometrically) why these Möbius transformations form a group under composition.
 - (iv) How many degrees of freedom does M have? Why does this make sense?
- 26** Reconsider (3.53), p. 201.
- (i) Use (3.45), p. 192 to show that M_a is elliptic.
 - (ii) Use (3.44), p. 192 to show that both multipliers are given by $m = -1$.
 - (iii) Calculate the matrix product $[M_a][M_a]$, and thereby verify that M_a is involutory.
 - (iv) Use (3.28), p. 172 to calculate the fixed points of M_a .
 - (v) Show that the result of the previous part is in accord with figure [3.38].
- 27** Use (3.45), p. 192 to verify (3.54), p. 204.